



Automatic Water-Level Motor-Pump Controller Using NE555 Timer and Conductive Probes

Design, Calculation, Scheme, Simulation,
Troubleshooting, and Academic Questions

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Khazar, May 2026



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Abstract

This report presents the design and simulation of an automatic water-level motor-pump controller using conductive probes, an NE555 timer integrated circuit, BC547 transistor switching stages, relay-based motor control, diode protection, and LED indication. The system is designed to reduce manual pump operation by detecting low and high water-level conditions inside a tank and automatically switching the pump according to the required control state. When the water level is below the LOW probe, the controller activates the relay and turns the motor pump on. When the water reaches the HIGH probe, the controller deactivates the relay and turns the pump off to prevent overflow. The project is verified through Proteus simulation, where probe states, relay action, LED indication, motor response, and circuit behavior are examined under different water-level conditions. The report also includes component calculations, scheme explanation, simulation analysis, troubleshooting procedures, limitations, and academic defense questions. The main contribution of the project is a clear, low-cost, hardware-based control design that demonstrates how sensing, electronic decision-making, switching, protection, and actuation work together in an automatic water-level control system.

Keywords: Automatic water-level controller, NE555 timer, conductive probes, relay switching, BC547 transistor, Proteus simulation.

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1

Description

1.1 Introduction

Water storage systems are widely used in domestic, institutional, agricultural, and small industrial environments where water must be collected, stored, and supplied according to demand. In many such systems, a motor pump is used to fill an overhead or storage tank from a lower reservoir or water source. When this process is controlled manually, the user must continuously observe the tank level and operate the pump at the correct time. This method is simple, but it is unreliable because it depends entirely on human attention.

The purpose of this project is to design and simulate an automatic water-level motor-pump controller that can switch a pump ON and OFF according to the level of water inside a tank. The system uses conductive water-level probes to detect the water level, an NE555 timer integrated circuit as the central control element, BC547 transistors for signal switching and relay driving, a relay for motor control, a diode for inductive protection, an LED for indication, and a DC motor pump as the controlled load.

The system is designed around two main water-level conditions. When the water level is low, the pump should turn ON and begin filling the tank. When the water reaches the high-level probe, the pump should turn OFF to prevent overflow. Therefore, the project performs automatic control by converting the physical condition of water level into an electrical switching action.

The project is simulated in Proteus in order to verify the operating logic before physical construction. The simulation demonstrates the relationship between the sensor probes, control circuit, relay operation, and motor pump response. The central engineering idea is that water-level detection must not remain only a sensing event; it must become a controlled electrical decision that safely operates the pump.

In formal terms, the project can be described as a closed threshold-control system. The controlled variable is the water level. The actuator is the motor pump. The sensing elements are the HIGH, LOW, and REF probes. The control unit is the NE555-based

circuit. The switching interface is the transistor and relay stage. The output action is the automatic starting and stopping of the pump.

The basic logical behavior of the system can be expressed as follows:

$$P_{next} = \begin{cases} 1, & \text{if the water level is below the LOW probe} \\ 0, & \text{if the water level reaches the HIGH probe} \\ P_{previous}, & \text{if the water level is between LOW and HIGH} \end{cases}$$

where ($P = 1$) represents pump ON and ($P = 0$) represents pump OFF.

This expression shows that the system should not switch randomly at every small change in water level. Instead, it should preserve a stable state between the lower and upper sensing points. This is important because a water-level controller must avoid rapid switching, relay chattering, unnecessary pump stress, and unstable operation.

1.2 Background of the Study

Water management is an important practical issue because stored water is often supplied intermittently, consumed irregularly, and controlled by pumps that require electrical energy. In many buildings and small facilities, water tanks are filled manually by switching a pump ON when the tank becomes low and switching it OFF when the tank becomes full. Although this method is common, it creates several risks.

The first risk is overflow. If the pump remains ON after the tank is full, water is wasted and the surrounding area may be damaged. The second risk is unnecessary energy consumption, because the pump may continue running after its useful task has already been completed. The third risk is user inconvenience, since manual supervision requires attention and time. The fourth risk is possible pump damage, especially when the pump is operated without proper monitoring.

Automatic water-level control systems are developed to solve these problems by detecting water-level conditions and controlling the pump without continuous human intervention. Recent academic discussion on automatic water-level control identifies reduced manual monitoring, reduced human error, lower water waste, and improved energy use as major motivations for automation. (rsisinternational.org)

Different technical approaches can be used for automatic water-level control. Some systems use float switches. Some use ultrasonic sensors. Some use pressure sensors. Some use microcontrollers. Some use programmable logic controllers. The present project uses conductive probes and a discrete electronic control circuit. This approach is suitable for a low-cost educational design because it shows how sensing, switching, control, and actuation can be achieved without software.

The NE555 timer is an important component in this project because it can operate as a control and switching device when its trigger, threshold, and reset behavior are correctly used. According to the Texas Instruments NE555 datasheet, the device contains internal comparators, a flip-flop, and an output stage, and it responds to trigger and

threshold voltage levels that are related to the supply voltage. (ti.com) This internal structure makes the NE555 useful not only for timing applications, but also for simple control applications where a stable switching decision is required.

The relay is used because the control circuit cannot directly switch the motor pump in a safe and reliable manner. The relay allows a low-power electronic circuit to control a higher-current load through an electromagnetic switching mechanism. Since a relay coil is an inductive load, a flyback diode is required to suppress the reverse voltage generated when the relay coil is switched OFF. Application notes on inductive load switching describe the flyback diode as a protection device that clamps the transient voltage produced by an inductive coil. (support.automationdirect.com)

The background of this study therefore combines three main areas: water-level automation, NE555-based electronic control, and relay-based motor switching. The project is academically important because it connects these areas into one complete system.

1.3 Problem Statement

Manual water-pump control is inefficient and unreliable because it depends on continuous human observation. When a user forgets to switch the pump OFF, the tank may overflow. When a user forgets to switch the pump ON, the tank may become empty. When the pump operates longer than necessary, electrical energy is wasted. When the circuit does not include proper switching and protection, electrical components may become damaged.

The central problem addressed by this project is the absence of an automatic, low-cost, and understandable control system that can start and stop a pump according to actual tank water level. A reliable system must detect low water level, activate the pump, continue filling the tank until the high-level point is reached, and then deactivate the pump without requiring manual action.

The technical problem can be divided into four smaller problems.

First, the water level must be sensed. This requires a method for detecting whether water has reached specific points inside the tank.

Second, the sensor condition must be converted into a usable electrical signal. Since water-level probes produce simple conductive states, the circuit must interpret those states correctly.

Third, the control circuit must make a stable decision. The pump should not turn ON and OFF repeatedly because of small disturbances, incomplete contact, or water movement.

Fourth, the motor pump must be switched safely. The NE555 and transistor stages cannot be treated as direct motor drivers. A relay and protective components are required to separate the control circuit from the motor-load circuit.

Therefore, the problem is not only to make a pump turn ON and OFF. The real engineering problem is to create a complete chain of operation:

Water Level → Probe State
→ Electrical Signal → Control Decision
→ Relay Action → Motor Pump Operation

If any part of this chain is missing or weak, the system becomes incomplete. For this reason, the project must be described as a full control system rather than as a simple timer circuit.

1.4 Aim of the Project

The aim of this project is to design and simulate an automatic water-level motor-pump controller that uses conductive probes, an NE555 timer integrated circuit, BC547 transistors, a relay switching stage, and an indicator circuit to control a DC motor pump according to the water level in a tank.

The project aims to demonstrate that a pump can be controlled automatically by detecting two water-level thresholds. The lower threshold is used to identify when water is low and pumping is required. The upper threshold is used to identify when the tank is full and pumping must stop.

The project also aims to show that a discrete electronic design can perform useful automatic control without using a microcontroller. This is important because it makes the operation visible at the component level. Each part of the system has a clear function: the probes sense the water, the NE555 processes the control state, the transistor drives the relay, the relay switches the motor pump, and the LED indicates the operating condition.

The project further aims to verify the circuit behavior through Proteus simulation before hardware implementation. Simulation is used to observe whether the system responds correctly when the water level changes from low to high and from high to low.

1.5 Objectives of the Project

The objectives of the project are as follows.

The first objective is to design a water-level sensing arrangement using three probes named HIGH, LOW, and REF. The REF probe provides the common reference path, the LOW probe identifies the lower threshold, and the HIGH probe identifies the upper threshold.

The second objective is to use the NE555 timer as the main control device for interpreting the level-sensing conditions and producing the required output state.

The third objective is to use BC547 transistors as switching elements so that weak control signals can operate the relay stage properly.

The fourth objective is to use a relay to switch the motor pump, since the motor requires a separate load path from the low-power control circuit.

The fifth objective is to use a diode across the relay coil in order to protect the transistor from the reverse voltage generated when the relay is switched OFF.

The sixth objective is to provide visual indication of the system state using an LED indicator.

The seventh objective is to simulate the complete circuit in Proteus and verify that the pump turns ON when the water level is low and turns OFF when the high-level condition is reached.

The eighth objective is to identify important limitations of the design, including probe corrosion, sensor reliability, relay current rating, motor current rating, and the difference between simulation behavior and real hardware behavior.

The ninth objective is to prepare a report structure that explains the project in a clear academic order, beginning with description, then calculation, scheme, simulation, troubleshooting, and questions.

The tenth objective is to ensure that the system is not presented as a random connection of components, but as a logically complete control system whose operation can be explained, tested, and defended.

1.6 Scope of the Project

The scope of this project is limited to the design and simulation of an automatic water-level motor-pump controller using discrete electronic components. The project focuses on the sensing of tank water level, the interpretation of sensor conditions, the control of a relay, and the switching of a DC motor pump.

The project includes the design of a conductive probe-based sensing section. It includes an NE555-based control section. It includes BC547 transistor switching stages. It includes relay operation for motor control. It includes LED indication. It includes Proteus simulation of the circuit behavior. It also includes theoretical explanation of how the system should operate under different water-level conditions.

The project does not include full physical hardware construction unless a separate hardware phase is added. It does not include printed circuit board design unless a layout chapter is later prepared. It does not include enclosure design, waterproofing, long-term corrosion testing, or certification for real electrical installation. It also does not include advanced functions such as wireless monitoring, mobile application control, liquid flow measurement, water-quality measurement, or internet-based data logging.

The present design is most suitable for a low-voltage DC demonstration system. If the system is later used with an AC mains pump, additional safety measures must be included. These measures would include electrical isolation, fuse protection, proper relay contact selection, insulated wiring, enclosure protection, grounding, and compliance with relevant electrical safety standards.

The scope of this report is therefore academic and simulation-oriented. The report explains the working principle, component function, system logic, and expected behavior. It does not claim that simulation alone proves complete real-world safety.

1.7 Importance and Significance of the Project

The project is important because it addresses a real and common problem in water storage systems. Manual pump operation can waste water, waste electricity, and create inconvenience for users. An automatic water-level controller reduces the need for continuous monitoring and makes pump operation more consistent.

The project is significant from an engineering education perspective because it demonstrates how basic electronic components can be combined to create an automatic control system. The design uses sensing, signal conditioning, decision-making, switching, protection, and actuation. These are fundamental concepts in electronics and control systems.

The project is also significant because it explains automation without depending on software. In many modern projects, a microcontroller is immediately used for control. Although microcontrollers are powerful, they can hide the electrical logic behind code. In this project, the operation is visible through component behavior. The student can understand how a sensor input affects a transistor, how the transistor affects the relay, and how the relay affects the motor.

The project has practical significance because it can help reduce overflow and unnecessary pump operation. When the pump stops automatically at the high-water level, water loss is reduced. When the pump operates only when needed, energy use is reduced. Studies on automatic water-level control describe reduced manual monitoring, reduced human error, and improved resource use as important advantages of such systems. (rsisinternational.org)

The project also has diagnostic significance because it allows important engineering questions to be studied. These include whether the transistor can safely drive the relay, whether the relay can safely switch the motor, whether the sensor probes give reliable readings, whether the circuit is protected from inductive voltage spikes, and whether the simulation result is enough for practical implementation.

Therefore, the significance of the project is not only that it turns a pump ON and OFF. Its real value is that it teaches the complete relationship between physical sensing, electronic control, switching safety, and automatic system behavior.

1.8 Literature Review

Automatic water-level control has been studied and implemented in different forms because water storage systems require reliable supervision. The general purpose of these systems is to prevent overflow, reduce water wastage, protect the pump, reduce manual effort, and improve the efficiency of water use.

In the literature, automatic water-level controllers are commonly designed around threshold detection. A low-level condition is used to activate filling, while a high-level condition is used to stop filling. This principle matches the logic of the present project. The system does not need to measure the exact height of water continuously. It only needs to know whether water has reached important decision points.

Recent work on automatic water-level control emphasizes that automated systems are relevant for homes and small establishments because they reduce the need for manual monitoring and help prevent unnecessary pump operation. (rsisinternational.org) Other project literature on automatic water-level controllers similarly identifies water waste, power waste, and manual supervision as central concerns. (Teras)

The present project belongs to the category of discrete electronic water-level control. Unlike microcontroller-based systems, it does not require programming. Its logic is implemented through probes, transistor switching, NE555 behavior, and relay actuation. This makes the design suitable for academic explanation because each physical and electrical state can be traced directly.

The literature also supports the need for proper component-level design. The NE555 is not merely a symbolic block; it has defined voltage thresholds, reset behavior, and output limitations. The BC547 is also not a universal power driver; it has a limited collector current rating. A common BC547 datasheet lists a maximum DC collector current of 100 mA and collector power dissipation of 500 mW, so relay-coil current must be verified before hardware construction. (SparkFun)

Relay protection is another important part of the literature foundation. When current through an inductive coil is interrupted, a reverse voltage spike can occur. A fly-back diode provides a safe current path and suppresses the transient voltage. Inductive load protection references describe this as a necessary protection method for relay coils and other inductive loads. (support.automationdirect.com)

The literature therefore confirms the relevance of this project, but it also reveals the design requirements that must be respected. A proper academic report must not only state that the circuit works. It must explain why the sensing method is chosen, why the NE555 is suitable, why a relay is required, why a transistor driver is needed, why diode protection is necessary, and why simulation must be followed by real hardware verification.

1.8.1 Manual Water-Level Control

Manual water-level control is the simplest method of operating a pump. In this method, the user observes the tank level and switches the pump ON or OFF manually. Although the method requires few components, it is not reliable for continuous use because it depends on human attention.

The main weakness of manual control is delay. The user may not switch the pump ON when the tank becomes low. The user may also forget to switch the pump OFF when the tank is full. This causes either water shortage or overflow. Both conditions

are undesirable.

Manual control also wastes energy when the pump continues to operate after the tank has reached the required level. In addition, manual control creates inconvenience because the user must check the tank repeatedly. This is especially problematic when the tank is located on a roof or in a place that is not easy to observe.

From a control-system perspective, manual water-level control is an open human-dependent process. It does not contain an automatic feedback mechanism. The tank level changes, but the pump does not respond unless a human operator notices the condition and acts.

The limitation of manual control justifies the need for an automatic system. In this project, automatic control replaces human observation with probe-based sensing and replaces manual switching with relay-based pump control.

1.8.2 Automatic Water-Level Control Systems

Automatic water-level control systems use sensors and control circuits to operate a pump according to tank water level. Their main purpose is to maintain water within a desired range without continuous human supervision.

A typical automatic water-level controller has four main parts. The first part is the sensing unit, which detects the water level. The second part is the control unit, which interprets the sensor condition. The third part is the switching unit, which controls the pump. The fourth part is the load, which is the motor pump.

In the present project, these four parts are clearly represented. The sensing unit consists of the HIGH, LOW, and REF probes. The control unit is the NE555-based circuit. The switching unit consists of BC547 transistors and the relay. The load is the DC motor pump.

Automatic systems are superior to manual systems because they reduce human error and provide repeatable operation. They can also reduce water wastage because the pump can be switched OFF automatically when the tank is full. Recent literature on water-level control identifies reduced manual monitoring and reduced resource wastage as important advantages of automation. (rsisinternational.org)

The system in this project uses threshold-based automatic control. It does not attempt to measure every possible water height. Instead, it uses important decision points. This is sufficient for pump control because the pump usually needs only two commands: start filling and stop filling.

1.8.3 Conductive Probe-Based Water-Level Detection

Conductive probe-based sensing uses the electrical conductivity of water to detect whether water has reached a certain level. In this method, metal probes are placed at selected heights inside the tank. When water touches a probe and also contacts the reference probe, a conductive path is created. The circuit can then detect that the water has reached that level.

In this project, three probes are used. The REF probe is the reference electrode. The LOW probe represents the lower water-level threshold. The HIGH probe represents the upper water-level threshold.

The probe conditions may be defined as binary variables:

$$L = \begin{cases} 1, & \text{if water touches the LOW probe} \\ 0, & \text{if water does not touch the LOW probe} \end{cases}$$

$$H = \begin{cases} 1, & \text{if water touches the HIGH probe} \\ 0, & \text{if water does not touch the HIGH probe} \end{cases}$$

The expected control logic can then be stated as follows:

If $L = 0$ and $H = 0$, the pump should be ON.

If $H = 1$, the pump should be OFF.

If $L = 1$ and $H = 0$, the pump should keep its previous state.

This middle condition is important because it prevents unnecessary switching while the water is between the lower and upper probes.

Conductive probes are simple and low-cost, but they also have limitations. Their performance depends on water conductivity. Very pure water may not conduct enough current for reliable detection. Dirty water, mineral deposits, corrosion, and electrolysis may change the sensor behavior over time. Therefore, the report must not present conductive probes as perfect sensors. They are suitable for a low-cost design, but they require careful current limiting, corrosion awareness, and practical testing.

1.8.4 NE555 Timer-Based Control Circuits

The NE555 timer integrated circuit is widely used in timing, pulse generation, oscillation, switching, and simple control circuits. In this project, the NE555 is not used mainly as an oscillator. It is used as a control element that responds to input conditions and produces an output state for the relay-driver stage.

The NE555 contains internal comparators and a flip-flop structure. Its trigger and threshold behavior are connected to fractions of the supply voltage. The Texas Instruments NE555 datasheet describes the trigger and threshold operation and identifies the reset input as a control input that can force the output into a defined state. (ti.com)

This makes the NE555 suitable for a water-level controller because the system requires a stable decision. The pump should start when the water is low, but it should not stop immediately when the LOW probe becomes wet. It should continue until the HIGH probe is reached. This requires memory of the previous state, and the NE555

internal latch behavior can support this type of control when connected correctly.

The NE555 output is used to control the transistor driver stage. The output does not directly operate the motor pump. Instead, it provides the signal that causes Q1 to energize or de-energize the relay coil.

The use of the NE555 is academically useful because it allows the project to demonstrate hardware-based control. The report should clearly explain the relationship between NE555 pins and project function, especially the trigger input, reset input, output pin, supply pin, and ground pin.

1.8.5 Relay-Based Motor Switching

A relay is an electromechanical switching device that allows a low-power control circuit to switch a higher-current load. In this project, the relay is used to switch the DC motor pump. This is necessary because the motor pump requires more current than the NE555 output can safely provide.

The relay has two functional sides. The first side is the coil side. When current flows through the relay coil, the relay contact changes position. The second side is the contact side. This side completes or interrupts the motor pump circuit.

The relay provides separation between the control circuit and the motor-load circuit. The control circuit operates at a lower power level, while the motor circuit may operate at a higher current. This separation improves the safety and clarity of the design.

However, the relay coil is an inductive load. When the transistor switches OFF and current through the relay coil is interrupted, the collapsing magnetic field can produce a reverse voltage spike. This spike can damage the transistor if it is not controlled. For this reason, a flyback diode is connected across the relay coil. Inductive load protection references describe the flyback diode as a method for clamping the reverse voltage generated by coils. (support.automationdirect.com)

The relay contact must also be selected carefully. It must be rated for the voltage and current of the motor pump. The report should state that relay contact rating must be greater than the motor running current and should also tolerate motor starting current.

The basic relay-coil current relation is:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

This relation belongs mainly to the calculation chapter, but it should be introduced in the description chapter to show that relay selection is not arbitrary.

1.8.6 Comparison with Microcontroller-Based Systems

Microcontroller-based water-level control systems use programmable devices such as Arduino, PIC, AVR, or other controllers to read sensors and control output devices.

These systems can support advanced features such as displays, alarms, wireless communication, data logging, calibration, and software-based control logic.

The present project does not use a microcontroller. It uses a discrete electronic design based on probes, an NE555 timer, transistor switches, a relay, and a motor pump. This makes the system simpler in terms of programming because no code is required. It also makes the electrical behavior easier to observe and explain at component level.

A microcontroller-based system has advantages when the design requires advanced features. It can process multiple inputs, use complex logic, display information, store data, and communicate with other systems. However, it also introduces software dependency, firmware errors, programming requirements, and the need for stable digital input conditioning.

The NE555-based system has advantages when the design goal is low cost, direct operation, simple maintenance, and educational clarity. It is especially suitable for demonstrating basic automation principles.

The comparison can be summarized as follows:

Table 1.1: *Comparison Between NE555-Based and Microcontroller-Based Systems*

Criterion	NE555-Based System	Microcontroller-Based System
Programming required	No	Yes
Circuit simplicity	High	Medium
Feature expansion	Limited	High
Cost	Low	Medium
Educational visibility	High	Medium
Calibration flexibility	Low	High
Display and communication	Not included	Easy to add
Fault diagnosis	Mostly hardware-based	Hardware and software-based

The correct academic position is not that one approach is absolutely superior. The correct position is that each approach has a different purpose. The present project is strongest as a low-cost hardware-control demonstration.

1.8.7 Summary of Literature Gap

The reviewed background shows that automatic water-level control is a useful and practical field of study. Existing systems often focus on either manual improvement, microcontroller automation, or general pump control. However, many basic designs do not clearly explain the complete engineering chain from water detection to motor switching. Some designs also present the circuit function without sufficiently discussing component limits, relay protection, sensor reliability, and simulation limitations.

The gap addressed by this project is the need for a clear, component-level, simulation-based academic explanation of an automatic water-level motor-pump controller. The project does not only show that a pump can be switched. It explains how water-level

sensing, NE555 control behavior, transistor switching, relay actuation, and motor operation form one complete system.

The project also emphasizes that the design must be evaluated structurally. The probe section must be understood as the sensing layer. The NE555 section must be understood as the control layer. The transistor and relay section must be understood as the switching layer. The motor pump must be understood as the actuator layer. The diode and supply elements must be understood as the protection and stability layer.

Therefore, the literature gap is not the absence of water-level controllers in general. The gap is the need for a rigorously organized, formally explained, and academically defensible report that connects every component to the root function of automatic water-level control.

1.9 Components Used

The circuit consists of sensing, control, switching, indication, protection, and power-supply components. Each component has a specific function in the system. No component should be treated as decorative or accidental. A valid academic report must explain why each component is present and what would fail if it were removed.

The main components used in the project are the NE555 timer IC, BC547 transistors, relay module or relay coil, HIGH probe, LOW probe, REF probe, DC motor pump, diodes, resistors, LED indicator, and power supplies.

The component relationship can be expressed as a functional chain:

Probes → NE555 Control
→ Transistor Driver → Relay
→ Motor Pump

The protection and indication components support this chain:

Diode → Protection

LED → Status Indication

Resistors → Current Limiting and Biasing

Power Supplies → Energy Source for Control and Load

Each component is described in the following subsections.

1.9.1 NE555 Timer IC

The NE555 timer IC is the central control component of the project. It receives the electrical effect of the sensing circuit and produces an output signal that controls the relay-driver transistor.

In this project, the NE555 is used as a control and latch-like switching element rather than as a simple timing oscillator. The reason is that water-level control requires a stable output state. The pump should remain ON while the water rises from the LOW probe toward the HIGH probe. The pump should remain OFF after the HIGH probe is reached until the water falls below the lower level again.

The NE555 is suitable for this role because its internal structure includes comparators and a flip-flop. The trigger and threshold functions allow the output to respond to defined voltage conditions. The reset pin allows the circuit to force the output into a known state when required. The NE555 datasheet explains that its internal operation is based on threshold and trigger levels related to the supply voltage. (ti.com)

In the report, the NE555 must be described by function, not only by name. Its purpose is to convert sensor-related input conditions into a controlled output state. The output of the NE555 is then used to drive the transistor relay stage.

The important NE555 pins in this project are ground, trigger, output, reset, control voltage, threshold, discharge, and VCC. The exact connection of each pin must be explained in the Scheme chapter.

1.9.2 BC547 Transistors

The BC547 is an NPN bipolar junction transistor used for switching and signal control. In this project, BC547 transistors are used because the NE555 and sensing section cannot be expected to directly handle every current requirement in the circuit.

One BC547 transistor may be used as a relay driver. Its function is to switch current through the relay coil when the NE555 output provides the required signal. Another BC547 may be used in the sensing or reset path, depending on the circuit connection.

The transistor works as an electronic switch. When sufficient base current flows, the transistor allows collector current to pass through the relay coil or associated circuit section. When base current is absent, the transistor switches OFF.

The BC547 must be used within its electrical limits. A commonly cited BC547 datasheet lists a maximum DC collector current of 100 mA and collector power dissipation of 500 mW. (SparkFun) Therefore, the relay coil current must be lower than the safe collector-current limit of the transistor. It is not enough to say that the relay works in simulation. The real coil current must be calculated or measured.

The relay-driver relation can be introduced as:

$$I_C \approx I_{coil}$$

where (I_C) is the collector current of the transistor and (I_{coil}) is the relay coil cur-

rent.

For safe switching:

$$I_C \leq I_{C(max)}$$

The detailed calculation belongs to Chapter 2, but this description chapter must state the design requirement clearly.

1.9.3 Relay Module or Relay Coil

The relay is the output switching component that controls the motor pump. It allows the low-voltage control circuit to operate the motor-load circuit without connecting the motor directly to the NE555 output.

The relay coil is energized by the transistor driver. When the coil is energized, the relay contact changes state and completes the motor circuit. When the coil is de-energized, the contact returns to its normal state and the motor circuit is opened.

The relay is necessary because the motor pump requires more current than the NE555 can safely supply. The relay also separates the control side from the motor side. This makes the design clearer and safer.

The relay has two main ratings that must be considered. The first is the coil rating. The coil voltage must match the control circuit supply. The second is the contact rating. The contact must be able to handle the voltage and current required by the motor pump.

The relay coil current is determined by:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

where (V_{coil}) is relay coil voltage and (R_{coil}) is relay coil resistance.

The motor-contact requirement can be expressed as:

$$I_{contact(rated)} > I_{motor(running)}$$

A stronger design also considers motor starting current:

$$I_{contact(rated)} \geq I_{motor(start)}$$

The relay must therefore be selected according to both coil compatibility and motor-load compatibility.

1.9.4 Water-Level Probes: HIGH, LOW, REF

The water-level probes are the sensing elements of the project. They detect the level of water inside the tank by using the conductive property of water.

The REF probe acts as the reference electrode. It provides the common conductive path needed for level detection. Without the REF probe, the circuit would not have

a reliable reference path for detecting whether water has touched the LOW or HIGH probes.

The LOW probe is placed at the lower desired water level. When the water falls below this point, the circuit should interpret the tank as needing water. This condition should lead to pump activation.

The HIGH probe is placed at the upper desired water level. When water reaches this probe, the circuit should stop the pump in order to prevent overflow.

The correct physical order of the probes is:

REF probe extends lowest

LOW probe is placed at the minimum control level

HIGH probe is placed at the maximum control level

The HIGH probe must be above the LOW probe. The LOW probe must be above the lowest reference contact point. This arrangement allows the circuit to distinguish between empty, filling, full, and draining states.

The sensor logic can be described in this table:

Table 1.2: *Water-Level Probe Logic*

Water condition	LOW probe	HIGH probe	Required pump state
Water below LOW	Dry	Dry	ON
Water reaches LOW but not HIGH	Wet	Dry	Hold previous state
Water reaches HIGH	Wet	Wet	OFF
Water falls below HIGH but remains above LOW	Wet	Dry	Hold previous state
Water falls below LOW again	Dry	Dry	ON

This table is essential because it shows that the project is not a single-threshold design. It is a two-threshold control system.

1.9.5 DC Motor Pump

The DC motor pump is the output load of the system. Its function is to fill the tank when the controller activates it. The motor pump is controlled through the relay contact, not directly through the NE555 or transistor signal path.

The motor pump converts electrical energy into mechanical pumping action. When the relay contact closes, the motor receives supply voltage and begins running. When the relay contact opens, the motor supply path is interrupted and the pump stops.

The motor pump must be selected according to its rated voltage and current. If the pump is rated for 12 V DC, the motor supply must provide 12 V with enough current capacity. The relay contact must also be rated for the motor current.

The motor power can be expressed as:

$$P_{motor} = V_{motor} \times I_{motor}$$

where (P_{motor}) is motor power, (V_{motor}) is motor voltage, and (I_{motor}) is motor current.

The motor may draw higher current at starting than during normal operation. This starting current must be considered when selecting the relay and power supply. A simulation model may not fully represent real motor inrush current, so hardware verification remains necessary.

1.9.6 Diodes

Diodes are used in the project for protection and current-direction control. The most important diode in the circuit is the flyback diode connected across the relay coil.

When the relay coil is energized, magnetic energy is stored in the coil. When the transistor switches OFF, the current through the coil cannot stop instantly. The collapsing magnetic field can generate a high reverse voltage. This voltage may damage the transistor if it is not controlled.

The flyback diode provides a safe path for the coil current after the transistor turns OFF. It clamps the reverse voltage and protects the transistor. Inductive load protection references identify the flyback diode as a standard method for reducing voltage transients in coil circuits. (support.automationdirect.com)

The diode must be connected in the correct polarity. During normal relay operation, it must be reverse-biased. During relay turn-off, it becomes forward-biased and conducts the coil discharge current.

The diode-current requirement can be written as:

$$I_{diode} \geq I_{coil}$$

This means that the diode should be able to carry at least the relay coil current during the discharge period.

If the motor is a DC motor, additional protection may also be required at the motor side, because motors are also inductive and can generate electrical noise. This point should be discussed in the limitations and troubleshooting sections.

1.9.7 Resistors

Resistors are used to control current, set bias conditions, protect components, and define input states. In this project, resistors may be used as transistor base resistors, LED

current-limiting resistors, pull-up resistors, pull-down resistors, and probe-current limiting resistors.

A transistor base resistor limits the current entering the base of the transistor. Without a proper base resistor, excessive current may flow into the base, causing unnecessary power loss or component stress.

The base current relation is:

$$I_B = \frac{V_{drive} - V_{BE}}{R_B}$$

where (I_B) is base current, (V_{drive}) is the driving voltage, (V_{BE}) is the base-emitter voltage, and (R_B) is the base resistor.

The LED resistor limits current through the LED. Without this resistor, the LED may draw excessive current and fail.

The LED resistor relation is:

$$R_{LED} = \frac{V_{supply} - V_F}{I_{LED}}$$

where (V_F) is LED forward voltage and (I_{LED}) is the desired LED current.

Pull-up and pull-down resistors are used to prevent floating inputs. A floating input can produce unpredictable circuit behavior. In water-level sensing circuits, stable biasing is especially important because water conductivity may vary.

Probe-related resistors should be selected carefully. A very low resistor value in the probe path may cause excessive current through water and may accelerate corrosion. A higher resistor value is usually more appropriate for sensing.

1.9.8 LED Indicator

The LED indicator provides a visual signal that shows the operating state of the circuit. It may indicate that the relay is active, that the pump is ON, or that the control output is in a particular state, depending on its exact connection.

The LED is not part of the pumping mechanism, but it is important for observation, testing, and troubleshooting. In simulation, the LED helps confirm whether the control output has changed. In hardware, it helps the user or technician quickly understand the system state.

The LED must always be connected with a current-limiting resistor. The current through the LED should be selected according to the LED rating and the supply voltage.

The basic LED current relation is:

$$I_{LED} = \frac{V_{supply} - V_F}{R_{LED}}$$

The LED should not be connected directly across the supply without a resistor. If the resistor value is too small, the LED current may become too high.

In the report, the LED should be described as an indicator and diagnostic component, not as a control component. The system should still operate even if the indicator LED is removed, provided the circuit is otherwise correctly connected.

1.9.9 Power Supplies

The project requires electrical power for both the control circuit and the motor pump. These two parts may use different voltage levels. For example, the control circuit may use a low-voltage supply suitable for the NE555 and relay coil, while the motor pump may use a separate 12 V DC supply.

The power supply system must be clearly described because supply confusion can damage components. The NE555 has a valid supply-voltage range defined by its datasheet. The relay coil must receive its rated voltage. The motor pump must receive its rated operating voltage.

The control supply provides power to the NE555, transistors, sensing circuit, relay coil, and LED indicator. The motor supply provides power to the pump through the relay contact.

The supply relationship can be stated as:

$$V_{control} \rightarrow \text{NE555, sensors, transistors, relay coil, LED}$$

$$V_{motor} \rightarrow \text{relay contact and motor pump}$$

The grounds must be arranged carefully. Control current and motor current should not create unwanted voltage drops in the sensing section. If the motor creates noise, it can disturb the NE555 and cause false triggering. For this reason, decoupling capacitors and good wiring practices are important.

The total supply-current requirement can be expressed as:

$$I_{total} = I_{NE555} + I_{relay} + I_{LED} + I_{sensor} + I_{transistor}$$

The power supply should provide more current than the minimum calculated requirement in order to maintain stable operation.

1.10 General Working Principle

The general working principle of the project is based on automatic switching according to water-level thresholds. The circuit observes the water level through the probes and uses the NE555-based control circuit to determine whether the pump should be ON or OFF.

When the tank water level is below the LOW probe, the circuit detects a low-water condition. Under this condition, the control circuit produces an output that activates

the relay-driver transistor. The transistor allows current to flow through the relay coil. The relay contact closes the motor circuit, and the pump turns ON.

As the pump fills the tank, the water level rises. When water reaches the LOW probe but has not yet reached the HIGH probe, the pump should continue running. This prevents premature stopping and ensures that the tank continues to fill.

When water reaches the HIGH probe, the circuit detects that the tank has reached the full level. The control circuit then changes state and switches the relay-driver transistor OFF. The relay coil is de-energized, the relay contact opens, and the pump turns OFF.

After the pump stops, the water level may gradually fall as water is used. If water falls below the HIGH probe but remains above the LOW probe, the pump should not immediately restart. The system should remain OFF until the water falls below the LOW probe again. This behavior prevents frequent switching and gives the system stable two-threshold operation.

The operating sequence can be summarized as:

Low water → Pump ON

Water rising → Pump remains ON

High water → Pump OFF

Water falling but above LOW → Pump remains OFF

Water below LOW again → Pump ON

This is the essential control identity of the project.

1.11 Functional Block Explanation

The complete system can be divided into functional blocks. Each block performs one necessary task in the control chain.

The first block is the water tank and probe block. This block represents the physical water level and the conductive probes placed inside the tank. The probes convert water position into electrical contact conditions.

The second block is the sensing and input-conditioning block. This block interprets the probe states and produces the electrical conditions required by the NE555 control circuit.

The third block is the NE555 control block. This block receives the input conditions and produces a stable output state. It determines whether the relay-driver transistor

should be ON or OFF.

The fourth block is the transistor relay-driver block. This block amplifies the control action because the NE555 output alone should not be treated as a direct relay or motor power path in a robust design.

The fifth block is the relay switching block. This block uses electromagnetic switching to connect or disconnect the motor pump from its supply.

The sixth block is the motor pump block. This block performs the physical action of pumping water into the tank.

The seventh block is the protection block. This includes the flyback diode across the relay coil and any additional protective components needed for safe operation.

The eighth block is the indication block. This includes the LED indicator, which helps show the operating state of the system.

The ninth block is the power-supply block. This provides energy to the control circuit and motor circuit.

The complete block-level flow is:

```
Tank Water Level → Probes
→ Input Circuit → NE555
→ BC547 Driver → Relay
→ Motor Pump
```

This block explanation is important because it prevents the report from becoming only a list of components. It shows the system as a complete engineering structure.

1.12 Expected Output of the Project

The expected output of the project is a simulated automatic water-level control system that operates the pump according to the level of water in the tank.

When the tank is below the LOW level, the pump is expected to turn ON. The relay should energize, the motor pump should receive power, and the LED indicator should show the corresponding operating state if it is connected to represent pump activity.

When the water level rises and reaches the LOW probe only, the pump should continue running. The system should not stop at the LOW probe because the tank is not yet full.

When the water reaches the HIGH probe, the pump is expected to turn OFF. The relay should de-energize, the motor pump supply path should open, and the indicator should change according to the circuit design.

When the water level falls from HIGH to a point between HIGH and LOW, the pump should remain OFF. This confirms that the system has stable two-threshold behavior. The pump should restart only when the water level falls below the LOW probe.

The expected output can be represented as a state table:

Table 1.3: *Expected Output State Table*

System condition	Required output
Tank below LOW	Pump ON
Tank between LOW and HIGH while filling	Pump ON
Tank reaches HIGH	Pump OFF
Tank between HIGH and LOW while draining	Pump OFF
Tank below LOW after draining	Pump ON

The expected academic output is also important. The project should produce a complete report containing description, calculations, scheme, simulation evidence, troubleshooting, and questions. Chapter 1 provides the descriptive foundation for all later chapters.

1.13 Limitations of the Project

The project has several limitations that must be stated honestly.

The first limitation is that the system is simulated in Proteus. Simulation is useful for testing logic, but it does not prove complete real-world reliability. Real components have tolerances, temperature effects, contact resistance, wiring noise, relay delay, motor inrush current, and physical wear.

The second limitation is that conductive probes depend on water conductivity. If the water is very pure, the sensor path may not conduct enough current. If the water contains minerals or dirt, deposits may form on the probes. This can affect long-term accuracy.

The third limitation is probe corrosion. If DC current passes continuously through water and metal probes, electrochemical effects may occur. Over time, this can damage the probes or change their electrical behavior.

The fourth limitation is that dry-run protection is not fully included unless an additional source-water sensor is added. The project detects the level in the controlled tank, but it may not detect whether the source reservoir contains enough water for pumping.

The fifth limitation is relay contact wear. A relay is mechanical, so repeated switching can wear the contacts. If the motor current is high, contact arcing may occur.

The sixth limitation is motor-side electrical noise. DC motors can generate electrical noise and transient voltages. The relay coil diode protects the relay-driver transistor, but additional protection may be needed for the motor side.

The seventh limitation is current-rating uncertainty. The BC547 transistor can only drive limited current. The relay coil current must be checked against transistor rating. The BC547 datasheet lists a maximum DC collector current of 100 mA, so the relay coil must remain within a safe range. (SparkFun)

The eighth limitation is that the circuit is not automatically suitable for AC mains pumps. If an AC pump is used, electrical isolation, relay contact rating, fuse protection, enclosure safety, and wiring standards become critical.

The ninth limitation is that the project does not include advanced monitoring features such as display, alarm, remote control, flow sensing, water-quality sensing, or data logging.

The tenth limitation is that the project uses threshold detection rather than continuous level measurement. It can decide when the tank is low or full, but it does not measure the exact water height at every moment.

These limitations do not make the project weak. They define the real boundary of the design. A strong academic report must state them clearly because honest limitation analysis is part of engineering correctness.

1.14 Chapter Summary

This chapter described the automatic water-level motor-pump controller and established the foundation for the complete academic report. The project was introduced as a threshold-based automatic control system that uses HIGH, LOW, and REF probes to detect water level and uses an NE555-based circuit to control a relay-driven motor pump.

The chapter explained the background of manual water-pump control and identified the problems of overflow, water wastage, energy wastage, user inconvenience, and unreliable human supervision. It stated the aim of the project as the design and simulation of a low-cost automatic controller using discrete electronic components.

The objectives of the project were presented, including probe-based sensing, NE555 control, transistor switching, relay operation, diode protection, LED indication, and Proteus simulation. The scope of the project was also defined, with attention to what the project includes and what it does not include.

The literature review showed that automatic water-level control is a practical and academically relevant field. It also showed that the present project belongs to the discrete electronic control category rather than the microcontroller category. The review established the importance of proper component selection, transistor-current verification, relay protection, and recognition of simulation limitations.

The components used in the project were described individually. The NE555 was identified as the control element. The BC547 transistors were identified as switching devices. The relay was identified as the motor-switching interface. The probes were identified as the sensing elements. The motor pump was identified as the actuator. The diode, resistors, LED, and power supplies were explained according to their necessary functions.

The chapter also presented the general working principle, functional block explanation, expected output, and limitations of the project. The system was shown to operate according to a stable two-threshold logic:

$$P_{next} = \begin{cases} 1, & \text{if water is below LOW} \\ 0, & \text{if water reaches HIGH} \\ P_{previous}, & \text{if water is between LOW and HIGH} \end{cases}$$

This chapter therefore proves the structural identity of the project. It establishes that the project is not merely a collection of electronic components, but a complete automatic water-level control system. The next chapter should provide the mathematical calculations required to verify component values, current limits, resistor selection, relay compatibility, motor-load safety, and power-supply capacity.

2

Calculation

2.1 Introduction

This chapter presents the mathematical verification of the automatic water level motor pump controller. The purpose of this chapter is to prove that the circuit is not only logically functional in simulation, but also electrically justified through voltage, current, resistance, power, and protection calculations. Since the project uses an NE555 timer integrated circuit, BC547 transistor stages, a relay, water level probes, an LED indicator, and a motor pump, each of these elements must be checked against its electrical operating requirements.

The calculation chapter is necessary because a circuit diagram alone cannot prove design correctness. A schematic may show how components are connected, but calculation proves whether each component can safely perform its assigned function. In this project, the NE555 must operate within its valid supply range, the transistor must safely drive the relay coil, the relay must safely switch the motor pump, the LED resistor must limit current correctly, the sensing resistors must prevent excessive current through the water probe path, and the power supply must provide sufficient current with proper safety margin.

The central calculation logic of this chapter follows the actual control chain of the system. First, the supply voltages are defined. Second, the NE555 operating conditions are verified. Third, the relay coil current is calculated. Fourth, the Q1 relay driver transistor is examined. Fifth, the Q2 sensor transistor is checked. Sixth, the LED current limiting resistor is verified. Seventh, the pull up, pull down, and probe resistors are evaluated. Eighth, the relay flyback diode is justified. Ninth, the motor pump current and relay contact rating are calculated. Finally, the total current consumption, power supply capacity, protection margin, and final component values are summarized.

Therefore, this chapter acts as the mathematical foundation of the whole report. If the description chapter explains what the project does, this chapter proves whether the electrical design is reasonable, safe, and defensible.

2.2 Design Assumptions

Before any calculation is performed, the design assumptions must be clearly stated. These assumptions define the numerical basis of the project and prevent uncertainty in the analysis. In an academic engineering report, assumptions are not weaknesses when they are stated clearly. They become weaknesses only when they remain hidden or when they are used without verification.

For this project, the control circuit is assumed to operate from a low voltage direct current supply, while the motor pump is assumed to operate from a separate direct current supply. The relay is used to separate the low power control circuit from the higher current motor circuit. The transistor stages are assumed to operate as switches rather than as linear amplifiers. The LED is assumed to operate as a visual indicator. The water level probes are assumed to provide a conductive path through water when submerged.

The main assumed design values used in this chapter are as follows. The control supply voltage is taken as 5 V. The motor supply voltage is taken as 12 V. The relay coil voltage is taken as 5 V. The LED forward voltage is taken as approximately 2 V for a green indicator LED. The base emitter voltage of each BC547 transistor is taken as approximately 0.7 V when the transistor is conducting.

These values are suitable for calculation because they match the common operating conditions of an NE555 based relay control circuit. However, for real hardware implementation, the exact relay coil resistance, motor current, transistor datasheet limits, and pump specifications must be measured or taken from the exact component datasheets.

2.2.1 Control Supply Voltage Assumption

The control section of the circuit is assumed to operate at:

$$V_{CC} = 5V$$

This voltage supplies the NE555 timer, the transistor control stages, the relay coil if a 5 V relay is used, and the LED indicator if it is connected to the control supply. A 5 V supply is suitable for this project because it is low enough to be safe for the electronic control circuit and high enough to operate common low voltage components.

The control supply must be stable because the NE555 uses internal voltage thresholds that depend directly on the supply voltage. If the control supply fluctuates, the trigger and reset behavior of the NE555 may also become unstable. Therefore, the 5 V control supply must not be treated as a random power source. It must be regulated, filtered, and capable of delivering the relay coil current without excessive voltage drop.

The control supply assumption is therefore:

$$V_{control} = 5V$$

This value is used throughout the NE555 calculation, transistor base current calculation, LED resistor calculation, and sensing resistor analysis.

2.2.2 Motor Supply Voltage Assumption

The motor pump section is assumed to operate at:

$$V_{motor} = 12V$$

This voltage is separate from the control supply because the motor pump requires more current than the NE555 and sensing circuit. The relay contact is used to connect and disconnect this 12 V supply to the motor pump. The NE555 does not directly drive the motor. The transistor does not directly carry the motor current. The relay contact carries the motor current.

This separation is important because motor loads can produce electrical noise, voltage spikes, and high starting current. If the motor supply were directly mixed with the NE555 input pins, the control circuit could become unstable or damaged. Therefore, the motor supply must be treated as a load supply, while the 5 V supply must be treated as the control supply.

The motor supply assumption is therefore:

$$V_{pump} = 12V$$

This value is used in the motor current calculation, motor power calculation, and relay contact rating verification.

2.2.3 Relay Coil Voltage Assumption

The relay coil is assumed to be rated for:

$$V_{coil} = 5V$$

This means the relay coil is designed to energize when approximately 5 V is applied across it. The relay coil is controlled by transistor Q1. When Q1 is switched on, current flows through the relay coil and the relay contact changes state. When Q1 is switched off, the coil de energizes and the relay contact returns to its normal position.

The relay coil voltage must match the control supply voltage. If a 5 V relay coil is used with a 5 V supply, the relay can energize correctly. If a 12 V relay coil were accidentally used with a 5 V supply, the relay might not activate reliably. If a 5 V relay coil were connected directly to 12 V, the relay coil could overheat and fail.

The relay coil voltage assumption is therefore:

$$V_{relay} = 5V$$

The exact relay coil current cannot be known only from the relay voltage. The coil resistance must also be known. Therefore, the relay current calculation must use the relay coil resistance.

2.2.4 LED Forward Voltage Assumption

The LED is assumed to be a green indicator LED with approximate forward voltage:

$$V_F = 2V$$

The exact forward voltage depends on the LED color, model, and current. A green LED commonly has a forward voltage near 2 V, although the exact value may vary. The LED cannot be connected directly to the supply because it would draw excessive current. Therefore, a resistor must be placed in series with the LED to limit current.

The LED calculation is based on the difference between the supply voltage and the LED forward voltage. If the LED is connected to the 5 V supply, the resistor must drop the remaining voltage after the LED forward voltage. If the LED is connected to the 12 V supply, the resistor must be much larger because the remaining voltage is higher.

The LED forward voltage assumption is therefore:

$$V_{LED} = 2V$$

This value is used to calculate the current limiting resistor for safe LED operation.

2.2.5 Transistor Base Emitter Voltage Assumption

The BC547 transistor stages are assumed to have an approximate base emitter voltage of:

$$V_{BE} = 0.7V$$

This value is used when calculating the transistor base current. In a silicon bipolar junction transistor, the base emitter junction behaves similarly to a forward biased diode when the transistor is conducting. Therefore, approximately 0.7 V is dropped between the base and emitter.

For transistor switching calculations, the base current is found by subtracting the base emitter voltage from the driving voltage and dividing by the base resistor. The basic equation is:

$$I_B = \frac{V_{drive} - V_{BE}}{R_B}$$

This assumption is used for both Q1 and Q2. Q1 is the relay driver transistor. Q2 is the sensing or input transistor. Both must be checked so that the base current is sufficient for switching but not excessively high.

2.3 NE555 Supply Verification

The NE555 timer integrated circuit must operate within a valid supply voltage range. In this project, the assumed control supply voltage is:

$$V_{CC} = 5V$$

The NE555 is commonly used in low voltage timing and switching circuits, and a 5 V supply is suitable for many NE555 variants. Therefore, the supply condition for the NE555 is accepted as valid when:

$$V_{CC} = 5V$$

Since the NE555 control thresholds are internally related to the supply voltage, the trigger and threshold reference levels depend on this 5 V value. The internal trigger reference is approximately one third of the supply voltage, and the threshold reference is approximately two thirds of the supply voltage.

The trigger reference voltage is:

$$V_{TRIG} = \frac{1}{3}V_{CC}$$

Substituting the assumed supply voltage:

$$V_{TRIG} = \frac{1}{3} \times 5$$

$$V_{TRIG} = 1.67V$$

The threshold reference voltage is:

$$V_{THRESH} = \frac{2}{3}V_{CC}$$

Substituting the assumed supply voltage:

$$V_{THRESH} = \frac{2}{3} \times 5$$

$$V_{THRESH} = 3.33V$$

Therefore, when the NE555 is powered by 5 V, its main internal reference levels are approximately 1.67 V and 3.33 V. These values are important because the water level sensing circuit must drive the NE555 inputs into clearly defined voltage regions. A vague input voltage near the switching boundary may create unstable behavior. A correct design must produce input levels that are clearly below or clearly above the required threshold.

The NE555 supply verification confirms that the control voltage is suitable and that the internal switching levels can be calculated from the selected supply voltage.

2.4 NE555 Trigger and Reset Voltage Analysis

The NE555 performs control action according to the voltage conditions at its input pins. In this project, the NE555 is not used mainly as an oscillator. It is used as a control element that changes output state according to the water level probe signals.

The trigger input is responsible for setting the output state when the input voltage falls below the trigger reference level. With a 5 V supply, the trigger reference is:

$$V_{TRIG} = 1.67V$$

The trigger condition is therefore:

$$V_{pin2} < 1.67V$$

When this condition occurs, the NE555 output is driven high, provided the reset input is not forcing the output low. In the water level controller, this condition corresponds to the pump start command when the tank level is below the required lower level.

The reset input has priority over ordinary triggering behavior. When the reset pin is driven low, the output is forced low. In this project, the reset action is used to stop the pump when the high level condition is detected. For normal operation, the reset pin must remain at a valid high level. For stopping action, the reset pin must be pulled low by the sensing circuit.

The reset condition may be expressed as:

$$V_{RESET} = LOW \Rightarrow V_{OUT} = LOW$$

The normal operation condition may be expressed as:

$$V_{RESET} = HIGH \Rightarrow \text{NE555 responds to trigger control}$$

Therefore, the NE555 output logic in this project can be summarized as follows. When the tank is low, the trigger condition causes the output to become high. When the tank reaches the high probe, the reset condition forces the output low. This output then controls Q1, which energizes or de energizes the relay.

The NE555 output voltage can be represented as:

$$V_{OUT} \approx V_{CC}$$

when the output is high, and:

$$V_{OUT} \approx 0V$$

when the output is low.

Since the assumed supply is 5 V, the approximate output states are:

$$V_{OUT,HIGH} \approx 5V$$

$$V_{OUT,LOW} \approx 0V$$

This analysis proves that the NE555 can provide a clear switching signal to the transistor stage, provided that the trigger and reset inputs are driven by stable and correctly limited sensor circuits.

2.5 Relay Coil Current Calculation

The relay coil current is one of the most important calculations in the project because transistor Q1 must carry this current. The relay coil current is determined by the coil voltage and coil resistance.

The general equation is:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

where:

$$I_{coil}$$

is the relay coil current,

$$V_{coil}$$

is the relay coil voltage, and:

$$R_{coil}$$

is the relay coil resistance.

Using the assumed relay coil voltage:

$$V_{coil} = 5V$$

the equation becomes:

$$I_{coil} = \frac{5}{R_{coil}}$$

The exact current depends on the real relay coil resistance. If the relay coil resistance is measured as $100\ \Omega$, then:

$$I_{coil} = \frac{5}{100}$$

$$I_{coil} = 0.05A$$

$$I_{coil} = 50mA$$

If the relay coil resistance is measured as $70\ \Omega$, then:

$$I_{coil} = \frac{5}{70}$$

$$I_{coil} = 0.071A$$

$$I_{coil} = 71mA$$

The relay coil current must remain within the safe collector current limit of the transistor. Since Q1 is a BC547 transistor, the coil current must not be assumed safe without calculation. If the relay coil current is too high, Q1 may overheat or fail. Therefore, the relay coil current is acceptable only when:

$$I_{coil} < I_{C,max}$$

For a safe design, the coil current should be comfortably below the maximum collector current rating of Q1, not merely equal to it.

This calculation proves that the relay coil must be selected together with the transistor, not independently from it.

2.6 Q1 Relay Driver Transistor Calculation

Q1 is the relay driver transistor. Its purpose is to allow the NE555 output to control the relay coil. The relay coil requires more current than a logic input, so Q1 acts as a switching current path for the relay. When the NE555 output is high, base current flows into Q1. Q1 then saturates and allows collector current to flow through the relay coil. When the NE555 output is low, Q1 switches off and the relay coil de-energizes.

The Q1 calculation must prove five things. First, the transistor collector current must match the relay coil current. Second, the base current must be sufficient to saturate the transistor. Third, the base resistor must be correctly selected. Fourth, transistor saturation must be verified. Fifth, transistor power dissipation must remain safe.

2.6.1 Required Collector Current

The collector current of Q1 is equal to the relay coil current because Q1 is placed in the relay coil current path. Therefore:

$$I_{C,Q1} = I_{coil}$$

If the relay coil current is 50 mA, then:

$$I_{C,Q1} = 50mA$$

If the relay coil current is 71 mA, then:

$$I_{C,Q1} = 71mA$$

The required collector current is acceptable only if it is lower than the maximum collector current allowed for Q1. Therefore:

$$I_{C,Q1} < I_{C,max}$$

This condition is essential because simulation may show relay operation even when the real transistor is operating too close to its limit. A mathematically safe design must ensure that Q1 does not exceed its current rating.

2.6.2 Required Base Current

For reliable switching, Q1 should be driven into saturation. In saturation, the transistor behaves like a closed switch with a small collector emitter voltage drop. To force saturation, the design should not rely on the transistor maximum current gain. Instead, a conservative forced gain is used.

A common conservative switching assumption is:

$$\beta_{forced} = 10$$

The required base current is:

$$I_{B,Q1} = \frac{I_{C,Q1}}{\beta_{forced}}$$

Since:

$$I_{C,Q1} = I_{coil}$$

then:

$$I_{B,Q1} = \frac{I_{coil}}{10}$$

If:

$$I_{coil} = 50mA$$

then:

$$I_{B,Q1} = \frac{50mA}{10}$$

$$I_{B,Q1} = 5mA$$

If:

$$I_{coil} = 71mA$$

then:

$$I_{B,Q1} = \frac{71mA}{10}$$

$$I_{B,Q1} = 7.1mA$$

Therefore, the required base current depends directly on the relay coil current. The higher the relay coil current, the higher the required transistor base current.

2.6.3 Base Resistor Calculation

The base resistor R1 limits current flowing from the NE555 output into the base of Q1. The base resistor is calculated using:

$$R_1 = \frac{V_{OUT} - V_{BE}}{I_{B,Q1}}$$

Using:

$$V_{OUT} = 5V$$

and:

$$V_{BE} = 0.7V$$

the equation becomes:

$$R_1 = \frac{5 - 0.7}{I_{B,Q1}}$$

$$R_1 = \frac{4.3}{I_{B,Q1}}$$

If:

$$I_{B,Q1} = 5mA$$

then:

$$R_1 = \frac{4.3}{0.005}$$

$$R_1 = 860\Omega$$

The nearest standard value may be selected as:

$$R_1 = 820\Omega$$

or:

$$R_1 = 1k\Omega$$

If:

$$I_{B,Q1} = 7.1mA$$

then:

$$R_1 = \frac{4.3}{0.0071}$$

$$R_1 = 606\Omega$$

The nearest standard value may be:

$$R_1 = 560\Omega$$

or:

$$R_1 = 680\Omega$$

If the schematic uses:

$$R_1 = 120\Omega$$

then the base current becomes:

$$I_{B,Q1} = \frac{5 - 0.7}{120}$$

$$I_{B,Q1} = \frac{4.3}{120}$$

$$I_{B,Q1} = 0.0358A$$

$$I_{B,Q1} = 35.8mA$$

This base current is much higher than necessary for many small relay coils. It may still make the transistor switch in simulation, but it is not an efficient or elegant academic design unless the NE555 output current capacity and transistor base current limit are explicitly verified. Therefore, R1 should preferably be recalculated from the actual relay coil current.

2.6.4 Transistor Saturation Verification

Q1 is considered properly saturated when the available base current is at least equal to the required base current.

The saturation condition is:

$$I_{B,available} \geq I_{B,required}$$

The available base current is:

$$I_{B,available} = \frac{V_{OUT} - V_{BE}}{R_1}$$

The required base current is:

$$I_{B,required} = \frac{I_{coil}}{10}$$

Therefore, the saturation condition becomes:

$$\frac{V_{OUT} - V_{BE}}{R_1} \geq \frac{I_{coil}}{10}$$

Substituting:

$$V_{OUT} = 5V$$

and:

$$V_{BE} = 0.7V$$

gives:

$$\frac{4.3}{R_1} \geq \frac{I_{coil}}{10}$$

This equation is the formal verification condition for Q1. If this inequality is true, Q1 has sufficient base drive to saturate. If this inequality is false, Q1 may operate in the

active region, causing excessive collector emitter voltage drop, relay weakness, heating, or unstable switching.

2.6.5 Transistor Power Dissipation

When Q1 is saturated, the power dissipated in the transistor is calculated using:

$$P_{Q1} = V_{CE,sat} \times I_{C,Q1}$$

where:

$$V_{CE,sat}$$

is the collector emitter saturation voltage, and:

$$I_{C,Q1}$$

is the collector current through Q1.

Since:

$$I_{C,Q1} = I_{coil}$$

the equation becomes:

$$P_{Q1} = V_{CE,sat} \times I_{coil}$$

Assuming:

$$V_{CE,sat} = 0.2V$$

and:

$$I_{coil} = 50mA$$

then:

$$P_{Q1} = 0.2 \times 0.05$$

$$P_{Q1} = 0.01W$$

$$P_{Q1} = 10mW$$

If:

$$I_{coil} = 71mA$$

then:

$$P_{Q1} = 0.2 \times 0.071$$

$$P_{Q1} = 0.0142W$$

$$P_{Q1} = 14.2mW$$

These values are low and acceptable for a small signal transistor if the relay coil current is within the transistor rating. However, if the relay coil current is much higher, Q1 may no longer be safe. Therefore, the power dissipation calculation must always be tied to the actual relay coil current.

2.7 Q2 Sensor Input Transistor Calculation

Q2 is associated with the sensing and input control part of the circuit. Its function is to translate the water probe condition into a defined electrical signal for the NE555 control input. Since water level probes do not provide a strong digital signal by themselves, Q2 helps convert the conductive water path into a switching action.

The calculation of Q2 is different from Q1 because Q2 does not drive the relay coil. Q2 normally handles a much smaller signal current. Therefore, the purpose of Q2 calculation is to verify that the base current is limited, the collector current is reasonable, and the transistor switches clearly when the water probe condition changes.

2.7.1 Base Current Calculation

If Q2 uses an input resistor R4 connected to a 5 V control supply, the base current can be estimated as:

$$I_{B,Q2} = \frac{V_{CC} - V_{BE}}{R_4}$$

Using:

$$V_{CC} = 5V$$

$$V_{BE} = 0.7V$$

and:

$$R_4 = 10k\Omega$$

the base current is:

$$I_{B,Q2} = \frac{5 - 0.7}{10000}$$

$$I_{B,Q2} = \frac{4.3}{10000}$$

$$I_{B,Q2} = 0.00043A$$

$$I_{B,Q2} = 0.43mA$$

This base current is suitable for a sensor switching transistor because it is small enough to avoid excessive current but large enough to create a switching action in a small signal transistor.

2.7.2 Collector Current Calculation

The collector current of Q2 depends on the resistor connected to its collector and the voltage applied through that resistor. If Q2 uses a 10 kΩ pull up resistor, the collector current can be estimated as:

$$I_{C,Q2} = \frac{V_{CC}}{R_C}$$

If:

$$R_C = 10k\Omega$$

then:

$$I_{C,Q2} = \frac{5}{10000}$$

$$I_{C,Q2} = 0.0005A$$

$$I_{C,Q2} = 0.5mA$$

This is a very small collector current and is safe for a BC547 transistor. It also means that Q2 is used as a signal switching component rather than a power switching component.

2.7.3 Switching Condition Verification

For Q2 to switch properly, the available base current must be sufficient to control the collector current. The condition is:

$$I_{B,Q2} \times \beta_{forced} \geq I_{C,Q2}$$

Using a conservative forced gain:

$$\beta_{forced} = 10$$

and:

$$I_{B,Q2} = 0.43mA$$

then the supported collector current is:

$$I_{C,supported} = 0.43mA \times 10$$

$$I_{C,supported} = 4.3mA$$

Since the estimated collector current is:

$$I_{C,Q2} = 0.5mA$$

the condition becomes:

$$4.3mA \geq 0.5mA$$

This condition is true. Therefore, Q2 has sufficient base drive to switch the collector node reliably under the assumed resistor values.

2.8 LED Resistor Calculation

The LED indicator must be protected by a current limiting resistor. Without a resistor, the LED would draw excessive current and could be damaged. The resistor value depends on the supply voltage, the LED forward voltage, and the desired LED current.

The general equation is:

$$R_{LED} = \frac{V_{supply} - V_F}{I_{LED}}$$

where:

$$R_{LED}$$

is the LED series resistor,

$$V_{supply}$$

is the voltage applied to the LED resistor branch,

$$V_F$$

is the LED forward voltage, and:

$$I_{LED}$$

is the desired LED current.

2.8.1 LED Current Selection

A suitable indicator LED current is commonly selected between 5 mA and 10 mA for clear visual indication with low power consumption. For this project, a safe design current is selected as:

$$I_{LED} = 10mA$$

This value provides sufficient brightness for an indicator while keeping the current moderate. If lower power consumption is required, the current may be reduced to:

$$I_{LED} = 5mA$$

For the main calculation, the selected value is:

$$I_{LED} = 10mA$$

2.8.2 Resistor Value for 5 V Supply

If the LED is connected to the 5 V control supply, the resistor is:

$$R_{LED} = \frac{5 - 2}{0.01}$$

$$R_{LED} = \frac{3}{0.01}$$

$$R_{LED} = 300\Omega$$

The nearest standard value is:

$$R_{LED} = 330\Omega$$

Therefore, if the LED is powered from 5 V, a 330 Ω resistor is academically and practically suitable for approximately 10 mA operation.

If the schematic uses:

$$R_2 = 120\Omega$$

then the LED current at 5 V is:

$$I_{LED} = \frac{5 - 2}{120}$$

$$I_{LED} = \frac{3}{120}$$

$$I_{LED} = 0.025A$$

$$I_{LED} = 25mA$$

This current is higher than the preferred 10 mA indicator current. It may be acceptable for some LEDs for a limited condition, but it is not the most conservative academic design. A 330 Ω resistor is safer and more efficient for 5 V operation.

2.8.3 Resistor Value for 12 V Supply

If the LED is connected to the 12 V motor supply, the resistor value must be much larger. The calculation is:

$$R_{LED} = \frac{12 - 2}{0.01}$$

$$R_{LED} = \frac{10}{0.01}$$

$$R_{LED} = 1000\Omega$$

Therefore:

$$R_{LED} = 1k\Omega$$

If a 120 Ω resistor were used with a 12 V supply, the current would be:

$$I_{LED} = \frac{12 - 2}{120}$$

$$I_{LED} = \frac{10}{120}$$

$$I_{LED} = 0.083A$$

$$I_{LED} = 83mA$$

This value is excessive for a normal indicator LED. Therefore, R2 must be selected

according to the actual LED supply voltage.

2.8.4 Safety Check for LED Current

The LED current is safe when:

$$I_{LED} \leq I_{LED,max}$$

For a conservative indicator design, the preferred condition is:

$$I_{LED} \leq 10mA$$

If the LED is powered from 5 V and R2 is 330 Ω , then:

$$I_{LED} = \frac{5 - 2}{330}$$

$$I_{LED} = \frac{3}{330}$$

$$I_{LED} = 0.00909A$$

$$I_{LED} = 9.09mA$$

This is safe.

If the LED is powered from 12 V and R2 is 1 k Ω , then:

$$I_{LED} = \frac{12 - 2}{1000}$$

$$I_{LED} = 0.01A$$

$$I_{LED} = 10mA$$

This is also safe.

Therefore, the LED resistor must be chosen from the actual supply connection. The recommended values are 330 Ω for 5 V and 1 k Ω for 12 V.

2.9 Pull Up and Pull Down Resistor Calculation

Pull up and pull down resistors are used to define stable logic states. Without them, input nodes may float. A floating node can randomly behave as high or low due to noise, leakage current, or nearby switching effects. In this project, stable sensing and stable NE555 input behavior are essential, so the pull up, input, and probe related resistors must be verified.

2.9.1 R3 Pull Up Calculation

If R3 is used as a pull up resistor connected to the 5 V control supply, its current is:

$$I_{R3} = \frac{V_{CC}}{R_3}$$

Assuming:

$$R_3 = 10k\Omega$$

then:

$$I_{R3} = \frac{5}{10000}$$

$$I_{R3} = 0.0005A$$

$$I_{R3} = 0.5mA$$

This current is suitable for a pull up resistor. It is low enough to avoid wasting power and high enough to define a stable logic level. Therefore, a 10 k Ω pull up resistor is reasonable for the input control part of the circuit.

The power dissipated by R3 is:

$$P_{R3} = V_{CC} \times I_{R3}$$

$$P_{R3} = 5 \times 0.0005$$

$$P_{R3} = 0.0025W$$

$$P_{R3} = 2.5mW$$

This is far below the rating of a common quarter watt resistor.

2.9.2 R4 Input Resistor Calculation

If R4 is used to limit current into the base of Q2 or into an input path, its current is calculated as:

$$I_{R4} = \frac{V_{CC} - V_{BE}}{R_4}$$

Using:

$$V_{CC} = 5V$$

$$V_{BE} = 0.7V$$

$$R_4 = 10k\Omega$$

the current is:

$$I_{R4} = \frac{5 - 0.7}{10000}$$

$$I_{R4} = 0.43mA$$

This value is suitable for a transistor input path. It limits current enough to protect the transistor while still allowing reliable switching.

The power dissipated in R4 is:

$$P_{R4} = I_{R4}^2 \times R_4$$

$$P_{R4} = (0.00043)^2 \times 10000$$

$$P_{R4} = 0.00185W$$

$$P_{R4} = 1.85mW$$

This is safe for a standard resistor.

2.9.3 R5 Probe Resistor Verification

R5 is especially important because it may be connected to the water probe path. The probe path should use low current because water sensing does not require high current. Excessive current through water probes can increase corrosion, waste power, and create unsafe or unstable sensing behavior.

If R5 were:

$$R_5 = 10\Omega$$

then the current from a 5 V supply would be:

$$I_{R5} = \frac{5}{10}$$

$$I_{R5} = 0.5A$$

This value is extremely high for a water level sensing path and would be unacceptable for a probe resistor.

If R5 were:

$$R_5 = 10k\Omega$$

then the current would be:

$$I_{R5} = \frac{5}{10000}$$

$$I_{R5} = 0.0005A$$

$$I_{R5} = 0.5mA$$

This value is suitable for a sensing path.

Therefore, R5 must be verified carefully. If the schematic label truly shows 10 Ω , the value should be corrected. If the intended value is 10 k Ω , the schematic should write it clearly as 10 k Ω . In a water probe sensing circuit, R5 should normally be a high value resistor, not a low value resistor.

The formal acceptance condition for the probe current is:

$$I_{probe} \leq I_{safe,sense}$$

A suitable sensing current should remain small, preferably in the microampere to low milliampere range, depending on the sensing method and probe material.

2.10 Relay Flyback Diode Calculation

The relay coil is an inductive component. When current flows through the coil, energy is stored in its magnetic field. When Q1 switches off, the magnetic field collapses and the coil attempts to keep current flowing. This can produce a high reverse voltage across the transistor. The flyback diode provides a safe path for this current and protects Q1 from voltage stress.

2.10.1 Relay Coil Back EMF Explanation

The energy stored in the relay coil is expressed as:

$$E = \frac{1}{2}LI^2$$

where:

$$E$$

is the stored magnetic energy,

$$L$$

is the relay coil inductance, and:

$$I$$

is the relay coil current.

When Q1 turns off, this stored energy must be released. Without a diode, the coil voltage may rise sharply until it finds a path through the transistor or surrounding circuit. This may damage Q1 or disturb the NE555 operation.

The flyback diode is connected across the relay coil in reverse bias during normal operation. When the transistor turns off, the coil voltage reverses and the diode becomes forward biased. The coil current then circulates through the diode until the stored energy is dissipated safely.

2.10.2 Diode Current Rating

The flyback diode must be able to carry at least the relay coil current immediately after Q1 turns off. Therefore, the minimum diode current condition is:

$$I_D \geq I_{coil}$$

If:

$$I_{coil} = 50mA$$

then:

$$I_D \geq 50mA$$

If:

$$I_{coil} = 71mA$$

then:

$$I_D \geq 71mA$$

A common rectifier diode with a current rating much higher than the relay coil current is suitable for ordinary relay protection. The diode reverse voltage rating must also be higher than the supply voltage. Since the relay coil supply is 5 V, even a modest diode reverse voltage rating is electrically sufficient. However, a diode with a higher voltage rating provides stronger general design margin.

2.10.3 Diode Polarity Verification

The relay flyback diode must be connected in the correct direction. During normal relay operation, the diode must not conduct. Therefore, its cathode must be connected to the positive side of the relay coil, and its anode must be connected to the transistor side of the coil.

The normal condition is:

$$V_{cathode} > V_{anode}$$

In this condition, the diode is reverse biased and does not conduct.

When Q1 turns off, the relay coil reverses its voltage. The diode becomes forward biased and conducts the coil current. The protection condition is therefore:

$$V_{anode} > V_{cathode}$$

during coil turn off.

If the diode is connected in the wrong direction, it will conduct immediately when the relay is powered and may short the supply through the coil path. Therefore, diode polarity is not optional. It is a critical safety and functionality condition.

2.11 Motor Pump Current Calculation

The motor pump is the main load of the project. It is switched by the relay contact, not directly by the NE555 or transistor. The motor current determines the relay contact rating, the required supply capacity, the fuse rating, and the protection requirements.

The motor calculation must consider both running current and starting current. A motor may draw a much higher current when it starts than when it is already rotating. Therefore, using only the running current may produce an unsafe relay selection.

2.11.1 Running Current

The running current of the motor is calculated from its rated power and voltage:

$$I_{run} = \frac{P_{motor}}{V_{motor}}$$

where:

$$I_{run}$$

is the running current,

$$P_{motor}$$

is the motor power, and:

$$V_{motor}$$

is the motor supply voltage.

If the motor is rated at 12 V and 6 W, then:

$$I_{run} = \frac{6}{12}$$

$$I_{run} = 0.5A$$

If the motor is rated at 12 V and 12 W, then:

$$I_{run} = \frac{12}{12}$$

$$I_{run} = 1A$$

Therefore, the motor current cannot be guessed. It must be taken from the motor rating or measured during operation.

2.11.2 Starting Inrush Current

The starting current of a direct current motor can be several times higher than its running current. The inrush current may be estimated as:

$$I_{start} = k \times I_{run}$$

where:

$$k$$

is the starting current multiplier.

For a conservative academic estimate, the multiplier may be taken as:

$$k = 3$$

Therefore:

$$I_{start} = 3I_{run}$$

If:

$$I_{run} = 0.5A$$

then:

$$I_{start} = 3 \times 0.5$$

$$I_{start} = 1.5A$$

If:

$$I_{run} = 1A$$

then:

$$I_{start} = 3 \times 1$$

$$I_{start} = 3A$$

The relay contact and power supply should be able to tolerate this starting current.

2.11.3 Relay Contact Current Rating

The relay contact must be rated higher than the motor current. The minimum condition is:

$$I_{contact} \geq I_{run}$$

However, a stronger condition is needed for motor switching:

$$I_{contact} \geq I_{start}$$

For a more reliable design, a safety margin should be added:

$$I_{contact} \geq 1.5I_{start}$$

If:

$$I_{start} = 1.5A$$

then:

$$I_{contact} \geq 1.5 \times 1.5$$

$$I_{contact} \geq 2.25A$$

If:

$$I_{start} = 3A$$

then:

$$I_{contact} \geq 1.5 \times 3$$

$$I_{contact} \geq 4.5A$$

Therefore, a relay contact rated only slightly above the running current is not sufficient for a motor load. The contact rating must account for starting current and switching stress.

2.11.4 Motor Power Calculation

Motor power is calculated using:

$$P_{motor} = V_{motor} \times I_{motor}$$

If:

$$V_{motor} = 12V$$

and:

$$I_{motor} = 0.5A$$

then:

$$P_{motor} = 12 \times 0.5$$

$$P_{motor} = 6W$$

If:

$$I_{motor} = 1A$$

then:

$$P_{motor} = 12 \times 1$$

$$P_{motor} = 12W$$

This calculation is required because the motor supply must be selected according to actual motor power, not only voltage.

2.12 Total Current Consumption

The total current consumption of the control circuit includes the NE555 current, relay coil current, LED current, sensor current, and transistor base currents. The total control current can be expressed as:

$$I_{control,total} = I_{555} + I_{coil} + I_{LED} + I_{sensor} + I_{B,Q1} + I_{B,Q2}$$

If the following values are assumed:

$$I_{555} = 10mA$$

$$I_{coil} = 50mA$$

$$I_{LED} = 10mA$$

$$I_{sensor} = 0.5mA$$

$$I_{B,Q1} = 5mA$$

$$I_{B,Q2} = 0.43mA$$

then:

$$I_{control,total} = 10 + 50 + 10 + 0.5 + 5 + 0.43$$

$$I_{control,total} = 75.93mA$$

Therefore, the control supply must provide more than 75.93 mA. However, a supply should not be selected exactly equal to the calculated current. A safety margin must be added.

The motor supply current is separate and is calculated from the motor running and starting current. The total system current cannot be treated as one value unless both supplies are provided from the same source.

2.13 Power Supply Capacity Calculation

The control supply capacity should include a safety margin. A common design condition is:

$$I_{supply,control} \geq 1.5 \times I_{control,total}$$

Using:

$$I_{control,total} = 75.93mA$$

then:

$$I_{supply,control} \geq 1.5 \times 75.93$$

$$I_{supply,control} \geq 113.895mA$$

Therefore, a 5 V supply rated at 200 mA or higher would be acceptable for the assumed control load. A 5 V supply rated at 500 mA would provide stronger margin.

For the motor supply, the condition should be based on starting current:

$$I_{supply,motor} \geq I_{start}$$

For safer operation:

$$I_{supply,motor} \geq 1.5I_{start}$$

If:

$$I_{start} = 1.5A$$

then:

$$I_{supply,motor} \geq 2.25A$$

If:

$$I_{start} = 3A$$

then:

$$I_{supply,motor} \geq 4.5A$$

Therefore, the 12 V motor supply must be selected according to the real motor starting current, not merely the motor rated voltage.

2.14 Protection Margin Calculation

Protection margin is the difference between the component rating and the actual operating demand. A design without margin may work temporarily but fail under toler-

ance, temperature, aging, or load variation.

The general protection margin can be written as:

$$\text{Margin} = \frac{\text{Rating} - \text{Operating Value}}{\text{Operating Value}} \times 100$$

For the relay contact, if the contact is rated at 5 A and the motor starting current is 3 A, the margin is:

$$\text{Margin} = \frac{5 - 3}{3} \times 100$$

$$\text{Margin} = \frac{2}{3} \times 100$$

$$\text{Margin} = 66.7$$

For the control supply, if the supply is rated at 500 mA and the calculated control current is 75.93 mA, then:

$$\text{Margin} = \frac{500 - 75.93}{75.93} \times 100$$

$$\text{Margin} = 558.5$$

For the LED resistor, if the selected LED current is 9.09 mA and the safe target is 10 mA, then the design is within the preferred current range.

For Q1, if the relay coil current is 50 mA and the transistor maximum collector current is 100 mA, then:

$$\text{Margin} = \frac{100 - 50}{50} \times 100$$

$$\text{Margin} = 100$$

This margin is acceptable. If the relay coil current is 90 mA, then:

$$\text{Margin} = \frac{100 - 90}{90} \times 100$$

$$\text{Margin} = 11.1$$

This margin is weak and not recommended.

Therefore, every component should be selected with enough margin. The relay, transistor, resistor, diode, and power supply must not operate at their absolute limits.

2.15 Final Component Value Table

The final component values should be recorded after calculation so that the reader can see the complete design decision in one place. The table must include the component, its function, the calculated or assumed value, and the design judgment.

This table shows that the final design is not based on arbitrary component placement. Each component value is connected to a voltage, current, power, or protection requirement.

2.16 Chapter Summary

This chapter presented the complete calculation basis of the automatic water level motor pump controller. The control supply voltage was assumed to be 5 V, and the motor supply voltage was assumed to be 12 V. The NE555 supply condition was verified, and its internal trigger and threshold levels were calculated as 1.67 V and 3.33 V respectively. These values define the voltage boundaries required for reliable control operation.

The relay coil current was expressed as a function of relay coil resistance, proving that the relay cannot be selected by voltage alone. The Q1 relay driver transistor was analyzed through collector current, base current, base resistor selection, saturation verification, and power dissipation. The calculation showed that the Q1 base resistor should be chosen from the actual relay coil current and that a very low base resistor such as 120 Ω may create unnecessarily high base current.

The Q2 sensor input transistor was also analyzed. Its base current and collector current were shown to be suitable under 10 k Ω resistor assumptions. The LED resistor calculation proved that the resistor value depends on whether the LED is connected to 5 V or 12 V. A 330 Ω resistor is suitable for a 5 V LED branch, while a 1 k Ω resistor is suitable for a 12 V LED branch. The pull up, input, and probe resistor analysis showed that 10 k Ω values are reasonable for low current sensing, while a 10 Ω probe resistor would be unsuitable.

The relay flyback diode was justified by the inductive behavior of the relay coil. Its current rating must be greater than the relay coil current, and its polarity must be correct. The motor pump calculation showed that the relay contact must be selected according to motor starting current, not only running current. The total current consumption and power supply capacity calculations proved that both the control supply and motor supply require safety margin.

Overall, this chapter proves that the project must be defended through measurable electrical conditions. The circuit is valid only when the NE555 receives stable supply voltage, the relay coil current is safe for Q1, the relay contact can switch the motor starting current, the LED current is properly limited, the probe path current remains low, and the power supplies provide sufficient margin. Therefore, the mathematical analysis confirms the technical foundation of the automatic water level motor pump

Table 2.1: Final Component Values and Design Judgments

Component	Function	Calculated or selected value	Design judgment
Control supply	Powers NE555 and control circuit	5 V	Suitable for low voltage control
Motor supply	Powers motor pump	12 V	Suitable if matched to pump rating
NE555 trigger level	Internal low trigger reference	1.67 V	Based on one third of 5 V
NE555 threshold level	Internal upper reference	3.33 V	Based on two thirds of 5 V
Relay coil	Electromagnetic actuator	5 V coil assumed	Coil resistance must be verified
Relay coil current	Current through relay coil	$(I_{coil} = \frac{5}{R_{coil}})$	Must remain safe for Q1
Q1	Relay driver transistor	BC547 assumed	Acceptable only if coil current is within rating
Q1 base resistor	Limits base current	Prefer 560 Ω to 1 k Ω depending on relay current	120 Ω may be excessive
Q2	Sensor input transistor	BC547 assumed	Suitable for low current signal switching
R3	Pull up resistor	10 k Ω assumed	Suitable, gives 0.5 mA
R4	Input resistor	10 k Ω assumed	Suitable, gives about 0.43 mA base current
R5	Probe resistor	Prefer high value such as 10 k Ω	10 Ω would be unsuitable
LED resistor at 5 V	Limits LED current	330 Ω	Gives about 9.09 mA
LED resistor at 12 V	Limits LED current	1 k Ω	Gives about 10 mA
Flyback diode	Protects Q1 from relay coil back EMF	Diode current rating greater than relay coil current	Required
Relay contact	Switches motor supply	Must exceed starting current	Must be rated for motor load
Motor pump	Water pumping load	12 V pump assumed	Running and starting current must be known
Control supply current	Total control current	About 75.93 mA under sample assumptions	Supply should exceed 113.895 mA
Recommended control supply	Safe control supply capacity	5 V, 200 mA minimum under sample assumptions	5 V, 500 mA gives stronger margin
Recommended motor supply	Safe motor supply capacity	At least 1.5 times motor starting current	Must be selected from real pump current

controller and prepares the report for the scheme, simulation, and troubleshooting chapters.

3

Scheme

3.1 Introduction

This chapter presents the complete scheme of the automatic water level motor pump controller. The purpose of the scheme chapter is to transform the project from a general idea into a clear electrical structure. A water level controller cannot be understood only by naming its components. It must be explained as a connected system in which sensing, decision making, switching, actuation, indication, and protection operate together.

The circuit is designed around a conductive water level sensing arrangement, an NE555 timer integrated circuit used as the control core, transistor based relay driving, relay based motor switching, and indicator output. The water probes detect the physical water level inside the tank. The NE555 interprets the electrical condition produced by the probes and generates a control signal. The transistor driver strengthens this control signal so that the relay coil can be energized. The relay then switches the motor pump circuit. The LED indicator provides a visible status signal. Protection components reduce the risk of damage caused by inductive voltage spikes, supply disturbance, reverse polarity, and excessive current.

The scheme is therefore not a random collection of components. It is a structured control system. Each part has a defined position, a defined electrical function, and a defined relationship with the other parts. The correct interpretation of this chapter is essential because all later simulation, troubleshooting, and defense questions depend on the accuracy of the scheme.

The main design principle of the circuit is threshold based automatic control. The pump must turn on when the tank requires water and must turn off when the tank reaches the full level. This behavior is achieved by using three sensing probes, namely the reference probe, the low level probe, and the high level probe. These probes form the input condition of the system. The relay controlled motor pump forms the output condition of the system. The NE555 and transistor sections form the internal control

path between sensing and actuation.

The complete scheme can be summarized as follows:

Water Level → Probe Conductivity
→ Control Signal → Relay Drive
→ Pump Switching

This chapter explains every part of this chain in a formal engineering manner.

3.2 Overall System Block Diagram

The overall system block diagram represents the complete working structure of the automatic water level motor pump controller. The purpose of the block diagram is to show the functional relationship among the major sections before studying the detailed circuit diagram. It gives a higher level view of how the system receives input, processes the input, and produces output.

The complete system consists of seven main blocks:

Water level sensing block

NE555 control block

Transistor driver block

Relay switching block

Motor pump block

LED indicator block

Power supply block

The functional flow of the system is:

Water Tank → Level Probes
→ NE555 Control Circuit
→ Transistor Driver → Relay
→ Motor Pump

The water tank provides the physical condition that must be controlled. The level probes convert the water level into an electrical condition. The NE555 control section receives this electrical condition and determines whether the output should represent pump activation or pump deactivation. The transistor driver receives the NE555 output and supplies sufficient current to energize the relay coil. The relay uses its contact mechanism to connect or disconnect the motor pump from its supply. The motor pump then fills the tank when required. The LED indicator shows the operating status of the control circuit.

The system block diagram must show that the sensing side and the motor side are not the same section. The sensing and NE555 sections belong to the low power control domain. The motor pump belongs to the load domain. The relay provides the

separation between the control circuit and the motor circuit.

A correct block diagram for this project should therefore contain the following structure:

Water Tank → REF, LOW, HIGH Probes
 → NE555 Control Section
 → BC547 Relay Driver → Relay Contact
 → Motor Pump

The LED indicator must be shown as a parallel status output from the control or relay driver section. The power supply block must feed the control section and the motor section according to their voltage requirements. If the control circuit uses 5 volts and the motor pump uses 12 volts, the diagram must clearly show these as separate voltage domains.

The block diagram also establishes the main control condition:

If water level is below LOW, pump state becomes ON

If water level reaches HIGH, pump state becomes OFF

This is the fundamental logic of the project.

3.3 Complete Circuit Diagram

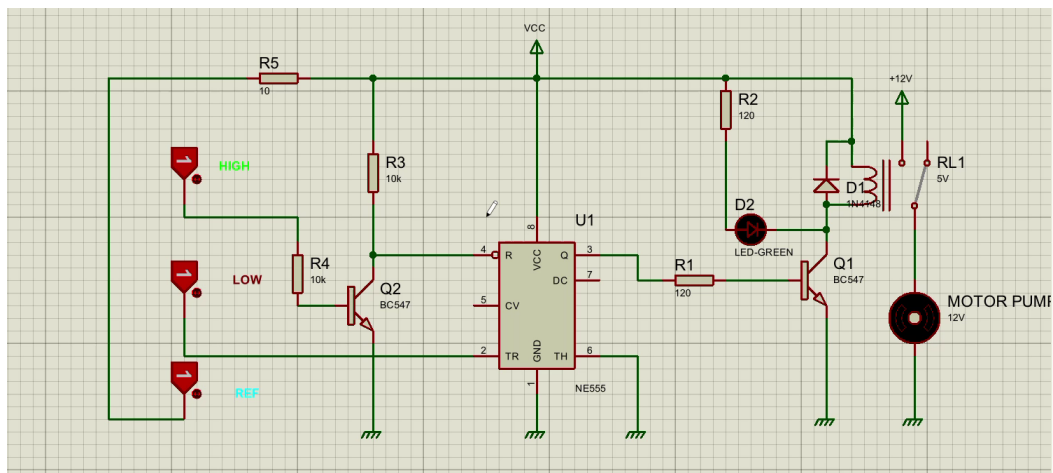


Figure 3.1: Complete Circuit Diagram from Proteus Simulation

The complete circuit diagram shows the actual electrical realization of the system. It contains all components, connections, supply terminals, sensing probes, relay connections, motor connections, transistor connections, and protection elements. The circuit diagram is the most important visual evidence in Chapter 3 because it proves how the block diagram is implemented at component level.

The complete circuit diagram must include the NE555 timer integrated circuit as the central control element. Pin 1 of the NE555 must be connected to ground. Pin 8 must be connected to the control supply. Pin 3 must be used as the output pin that controls the relay driver transistor. Pin 2 and pin 4 must be connected according to the probe based start and stop logic. Pin 5 should be treated as the control voltage pin and must either be properly filtered or left in a defined condition. Pin 6 and pin 7 must be connected according to the selected NE555 operating arrangement. No NE555 pin should be left unexplained in the report.

The complete circuit diagram must also show the transistor driver stage. The relay cannot be treated as a direct logic load without verification. The transistor driver receives the control signal and provides the coil current required by the relay. In this project, the BC547 transistor performs switching action. The transistor base receives current through a base resistor. The emitter is connected to ground in a low side switching arrangement. The collector is connected to one side of the relay coil. The other side of the relay coil is connected to the positive control supply.

The relay coil must have a diode connected across it. The diode must be reverse biased during normal relay operation. When the relay coil is switched off, the diode becomes forward biased and provides a safe path for the collapsing coil current. This prevents a high reverse voltage from damaging the transistor.

The motor pump must be connected to the relay contact side, not directly to the NE555. The relay contact acts as an electrically controlled switch. When the relay coil is energized, the contact closes and the motor pump receives supply voltage. When the relay coil is de energized, the contact opens and the motor pump turns off.

The conductive probes must be shown as three separate sensing electrodes:

REF probe LOW probe HIGH probe

The REF probe acts as the common reference electrode in the water. The LOW probe identifies the lower switching threshold. The HIGH probe identifies the upper switching threshold. The circuit diagram must clearly show which probe is connected to the trigger path and which probe is connected to the reset or stop path.

The complete circuit diagram must therefore prove the following structural facts:

The sensing section produces the input condition. The NE555 section produces the control decision. The transistor section provides relay coil current. The relay section switches the motor pump. The diode protects the relay driver. The LED indicates the control state. The power supply provides correct voltage to each part.

If any of these relationships is missing from the diagram, the scheme becomes incomplete.

3.4 Circuit Section Division

The circuit is divided into functional sections so that each part can be explained independently before the full operation is analyzed. This division is necessary because the circuit performs several different engineering functions at the same time. It senses wa-

ter level, interprets electrical states, switches a relay, controls a motor, displays status, and protects sensitive components.

The major circuit sections are:

Water level sensing section NE555 control section Transistor driver section Relay switching section Motor pump section Indicator section Power supply section

This division prevents confusion between input, control, drive, load, indication, and supply. It also supports troubleshooting because a fault can be located section by section.

The root function of the entire circuit is automatic pump control. Each section contributes to that root function in a different way. The sensing section answers the question: what is the water level? The NE555 control section answers the question: should the pump be on or off? The transistor driver section answers the question: can the control signal energize the relay coil? The relay switching section answers the question: is the pump supply connected? The motor pump section performs the physical water filling action. The indicator section communicates circuit status. The power supply section provides energy to all sections.

This section division is therefore both electrical and logical.

3.4.1 Water Level Sensing Section

The water level sensing section is the input section of the system. Its purpose is to convert the physical water level into an electrical condition that the control circuit can use. This section uses conductive probes placed inside the water tank. Since ordinary water usually contains dissolved minerals and impurities, it can provide a conductive path between probes. When water touches a probe, the electrical condition of that probe changes. When water does not touch a probe, the electrical path remains open.

The sensing section contains three probes:

REF probe LOW probe HIGH probe

The REF probe provides the common reference path. The LOW probe represents the minimum water threshold. The HIGH probe represents the maximum water threshold.

The sensing section must distinguish between three important tank conditions. In the first condition, the water is below the LOW probe. This means the tank requires filling, so the pump should turn on. In the second condition, the water is between the LOW and HIGH probes. This means the tank is in an intermediate state, so the previous pump state should be maintained. In the third condition, the water reaches the HIGH probe. This means the tank is full, so the pump should turn off.

The basic sensing logic can be expressed as:

Water below LOW $\Rightarrow$ LOW probe dry and HIGH probe dry

Water between LOW and HIGH $\Rightarrow$ LOW probe wet and HIGH probe dry

Water reaches HIGH $\Rightarrow$ LOW probe wet and HIGH probe wet

The sensing section must be designed with high enough resistance so that the current through the water is small. The water sensing path should not allow unnecessary current because excessive current may accelerate probe corrosion and may create unsafe or unstable sensing behavior.

The sensing current can be represented as:

$$I_{sense} = \frac{V_{supply}}{R_{sense}}$$

For safe low current sensing, the resistor used in the probe path should be large enough to keep (I_{sense}) small. If a sensing resistor is too small, the probe current becomes excessive. Therefore, the sensing resistor must be selected using electrical reasoning, not visual guesswork.

3.4.2 NE555 Control Section

The NE555 control section is the decision making section of the circuit. Although the NE555 is commonly known as a timer, in this project it is used as a control element that responds to trigger and reset conditions. The NE555 receives input conditions from the sensing section and generates an output state that controls the relay driver.

The control behavior depends on the internal comparator and latch structure of the NE555. The trigger input can set the output, while the reset input can force the output into the off state. This makes the NE555 suitable for a simple automatic pump controller because the pump requires two different switching thresholds: a lower threshold for starting and an upper threshold for stopping.

The NE555 output at pin 3 is the main control output. When the output is high, the transistor driver can be activated and the relay can be energized. When the output is low, the transistor driver turns off and the relay de energizes.

The control relation can be written as:

$$V_{OUT} = HIGH \Rightarrow \text{Relay energized} \Rightarrow \text{Pump ON}$$

$$\begin{aligned} V_{OUT} = LOW &\Rightarrow \text{Relay de energized} \\ &\Rightarrow \text{Pump OFF} \end{aligned}$$

The NE555 section must be supplied with a stable control voltage. An unstable supply can cause false triggering, relay chatter, or incorrect output behavior. For that reason, a decoupling capacitor should be connected near the NE555 supply pins. The

capacitor reduces sudden supply disturbance caused by relay switching and other transient events.

The NE555 control section must also have defined input conditions. Floating trigger or reset pins are not acceptable in a reliable control system because they can produce unpredictable output. Pull up or pull down resistors must be used wherever necessary to maintain a defined logic state.

In this project, the NE555 is the structural center of the scheme. However, it does not directly perform sensing, heavy current switching, or motor actuation. It only decides and provides a control signal. This distinction is important because it prevents the report from incorrectly presenting the NE555 as a direct motor controller.

3.4.3 Transistor Driver Section

The transistor driver section is the current amplification and switching section between the NE555 output and the relay coil. The NE555 output can provide a control voltage, but the relay coil requires current. The transistor driver allows a small control signal to switch a larger relay coil current.

In this project, a BC547 transistor is used as a switching transistor. The transistor operates in saturation when it is fully on and in cutoff when it is off. When the NE555 output becomes high, current flows into the base of the transistor through the base resistor. This causes the transistor to conduct between collector and emitter. The relay coil then receives current and becomes energized. When the NE555 output becomes low, the base current stops. The transistor enters cutoff, relay coil current stops, and the relay returns to its inactive state.

The base current is determined by:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

where (I_B) is the transistor base current, (V_{OUT}) is the NE555 output voltage, (V_{BE}) is the base emitter voltage of the transistor, and (R_B) is the base resistor.

The collector current is approximately equal to the relay coil current:

$$I_C \approx I_{coil}$$

For reliable saturation, the base current must be sufficient. A conservative switching design commonly uses forced current gain:

$$I_B \geq \frac{I_C}{10}$$

Therefore, the base resistor should satisfy:

$$R_B \leq \frac{V_{OUT} - V_{BE}}{I_C/10}$$

At the same time, (R_B) must not be so small that it wastes output current or over-

loads the NE555 output. Thus, the transistor driver scheme must balance two conditions:

I_B must be large enough for saturation

I_B must be small enough to protect the control output

The transistor driver section must also include the relay flyback diode because the relay coil is an inductive load. Without the diode, the transistor may be exposed to a high voltage spike when the coil current is interrupted.

3.4.4 Relay Switching Section

The relay switching section provides electrical switching between the control circuit and the motor pump circuit. The relay has two main parts: the coil and the contact. The coil belongs to the control side. The contact belongs to the motor switching side.

When current flows through the relay coil, a magnetic field is produced. This magnetic field moves the internal contact. When the contact closes, the motor pump receives supply voltage and starts running. When the coil current stops, the contact returns to its original position and disconnects the motor pump.

The relay switching relation is:

$$I_{coil} > I_{operate} \Rightarrow \text{Relay contact closed}$$

$$I_{coil} < I_{release} \Rightarrow \text{Relay contact open}$$

The relay is important because the motor pump may require more current than the NE555 and transistor signal path can safely provide. The relay also allows the motor supply voltage to be different from the control supply voltage. For example, the control circuit may operate at 5 volts while the motor pump operates at 12 volts. The relay contact can switch the 12 volt motor circuit without forcing the NE555 to directly handle motor current.

The relay section must be drawn in a way that clearly separates coil terminals from contact terminals. Confusing the coil side and contact side is a serious schematic error. The coil side is connected to the transistor driver. The contact side is connected in series with the motor pump and motor supply.

The relay contact rating must be greater than the motor current. If the motor pump has running current (I_{run}) and starting current (I_{start}), the relay contact should satisfy:

$$I_{contact} \geq I_{start}$$

If only the running current is considered and the starting current is ignored, the relay contact may fail in real hardware because motor starting current can be significantly

larger than running current.

3.4.5 Motor Pump Section

The motor pump section is the output load section of the system. Its purpose is to perform the physical action required by the controller: moving water into the tank. The motor pump receives supply voltage only when the relay contact is closed.

The motor pump is not controlled directly by the probes and is not controlled directly by the NE555. It is controlled through the relay contact. This is essential because the motor is a load device that may draw significant current and may generate electrical noise. Direct connection of a motor to a sensitive control integrated circuit would be unsafe and structurally incorrect.

The motor pump state can be expressed as:

$$\begin{aligned} \text{Relay contact closed} &\Rightarrow V_{motor} = V_{supply} \\ &\Rightarrow \text{Pump ON} \end{aligned}$$

$$\begin{aligned} \text{Relay contact open} &\Rightarrow V_{motor} = 0 \\ &\Rightarrow \text{Pump OFF} \end{aligned}$$

The motor power can be calculated as:

$$P_{motor} = V_{motor} I_{motor}$$

where (P_{motor}) is motor power, (V_{motor}) is motor supply voltage, and (I_{motor}) is motor current.

The motor section must be protected because a motor is an inductive and mechanical load. It can produce voltage spikes, current surges, and noise during switching. If the pump is a direct current motor, a diode, transient suppressor, or suitable suppression network should be considered across the motor terminals. If the pump is an alternating current pump, the safety requirements become much stricter and must include isolation, enclosure, fuse, and correct relay contact rating.

In this project scheme, the motor pump is treated as the final controlled device. Its behavior confirms whether the entire control chain has worked correctly.

3.4.6 Indicator Section

The indicator section provides visual feedback about the circuit state. The LED indicator allows the user to observe whether the control circuit is in an active or inactive condition. In many simple automatic pump controllers, the LED is used to indicate relay activation, pump activation, or output state.

The LED must always be connected with a current limiting resistor. An LED cannot be connected directly across a supply because excessive current can damage it. The LED resistor is selected by:

$$R_{LED} = \frac{V_{supply} - V_F}{I_{LED}}$$

where (R_{LED}) is the LED resistor, (V_{supply}) is the supply voltage applied to the LED branch, (V_F) is the forward voltage of the LED, and (I_{LED}) is the selected LED current.

For a typical low power indicator, (I_{LED}) may be selected around 5 milliamperes to 10 milliamperes. If the supply is 5 volts and the LED forward voltage is approximately 2 volts, then:

$$R_{LED} = \frac{5 - 2}{0.01} = 300\Omega$$

The nearest common resistor value is:

$$R_{LED} = 330\Omega$$

If the LED is supplied from 12 volts, then:

$$R_{LED} = \frac{12 - 2}{0.01} = 1000\Omega$$

Therefore, the LED resistor must be selected according to the actual supply voltage used in the LED branch. The indicator section must not be treated as electrically unimportant. Incorrect LED resistor selection can cause excessive current, overheating, or LED failure.

The indicator section supports the usability of the system because it makes the circuit state visible during testing, simulation, and troubleshooting.

3.4.7 Power Supply Section

The power supply section provides electrical energy to the control circuit, the relay coil, and the motor pump. This section must be clearly described because the project uses both low power control components and a higher current motor load.

The control circuit requires a stable supply for the NE555, sensing circuit, transistor driver, relay coil, and LED indicator. The motor pump may require a separate supply, commonly 12 volts in the simulated design. If the control supply and motor supply are different, the scheme must show the difference clearly.

The total control supply current can be estimated as:

$$I_{control} = I_{NE555} + I_{relay} + I_{LED} + I_{sensor} + I_{base}$$

The power supply must provide more than the calculated current. A practical margin is required:

$$I_{supply} \geq 1.5I_{control}$$

The motor supply current must be selected according to the motor running current

and starting current:

$$I_{motor\ supply} \geq I_{start}$$

The ground connection must also be treated carefully. The control ground and motor return path should not be arranged in a way that allows motor current noise to disturb the NE555 input section. In practical hardware, high current motor paths should be physically separated from sensitive sensing and control paths as much as possible.

The power supply section must therefore provide correct voltage, sufficient current, stable operation, and safe grounding.

3.5 Water Level Probe Arrangement

The water level probe arrangement defines the physical placement of the sensing electrodes inside the water tank. The operation of the complete controller depends on this arrangement. If the probes are placed incorrectly, the circuit may switch at the wrong water levels.

The probe arrangement contains three electrodes:

REF probe LOW probe HIGH probe

The REF probe is the common reference electrode and must be positioned so that it remains in contact with water across the normal operating range of the system. The LOW probe is placed at the minimum acceptable water level. The HIGH probe is placed at the maximum desired water level.

The water level state is determined by whether water provides a conductive path between the REF probe and the other probes. When water touches a probe, that probe becomes electrically connected through the water path. When water does not touch a probe, that connection is open.

The arrangement must support automatic hysteresis. Hysteresis means that the pump does not start and stop at the same level. Instead, the pump starts at the lower threshold and stops at the upper threshold. This prevents rapid switching and makes the system more stable.

The physical order of the probes from bottom to top must be:

REF probe positioned for reference contact LOW probe positioned at the lower switching level HIGH probe positioned at the upper switching level

The HIGH probe must not be placed below the LOW probe. The LOW probe must not be placed above the HIGH probe. Such an arrangement would invert the control logic and make the scheme invalid.

3.5.1 REF Probe Position

The REF probe is the reference electrode of the water sensing system. Its function is to provide a common electrical path through the water. Without the REF probe, the

LOW and HIGH probes cannot produce a reliable conductive sensing condition.

The REF probe should be placed at a low position in the tank, preferably below the LOW probe. This ensures that the reference electrode remains in contact with water during normal operation. If the REF probe becomes dry while the system is still expected to sense water level, the sensing circuit may lose its reference and produce an incorrect state.

The REF probe position must satisfy the following condition:

$$h_{REF} < h_{LOW} < h_{HIGH}$$

where (h_{REF}) is the height of the reference probe, (h_{LOW}) is the height of the low level probe, and (h_{HIGH}) is the height of the high level probe.

This height relationship is essential because the REF probe must support both low level and high level sensing. The reference probe does not represent the tank full condition. It represents the electrical baseline for the water sensing system.

The REF probe should be made from corrosion resistant conductive material. Stainless steel is more suitable than reactive metals because it resists corrosion better in water. The probe surface should be clean enough to maintain stable conduction. If corrosion or mineral deposits form on the REF probe, the sensing resistance may increase and cause unreliable operation.

3.5.2 LOW Probe Position

The LOW probe is the lower threshold electrode. It defines the level at which the system recognizes that the tank has reached a low water condition. Its main purpose is to trigger pump activation when the water level has fallen below the acceptable minimum level.

The LOW probe must be placed at the minimum safe water level selected for the tank. If it is placed too high, the pump will turn on too often and reduce system efficiency. If it is placed too low, the tank may become nearly empty before the pump starts. Therefore, the LOW probe height must be selected according to the required water reserve.

The LOW probe height must satisfy:

$$h_{REF} < h_{LOW} < h_{HIGH}$$

When the water level falls below (h_{LOW}), the LOW probe becomes dry. This condition indicates that water should be pumped into the tank. The controller should then activate the relay and start the motor pump.

The LOW probe is also part of the hold behavior. Once the pump starts, the water level rises and eventually touches the LOW probe again. However, the pump should not immediately stop at the LOW probe. It should continue running until the water reaches the HIGH probe. This behavior is necessary for stable automatic control.

The LOW probe therefore performs two related functions. It identifies the lower boundary of the water level and participates in the start condition of the pump control sequence.

3.5.3 HIGH Probe Position

The HIGH probe is the upper threshold electrode. It defines the level at which the system recognizes that the tank is full. Its main purpose is to stop the pump before overflow occurs.

The HIGH probe must be placed at the maximum safe water level. It should be below the physical overflow point of the tank. If it is placed too high, water may overflow before the pump stops. If it is placed too low, the tank will not be filled to the desired capacity.

The HIGH probe height must satisfy:

$$h_{HIGH} > h_{LOW}$$

The maximum water level must also satisfy:

$$h_{HIGH} < h_{overflow}$$

where ($h_{overflow}$) is the height at which water would overflow from the tank.

When water reaches the HIGH probe, a conductive path is formed through the water between HIGH and REF. This condition must cause the control circuit to turn the pump off. The HIGH probe is therefore the stop threshold of the system.

The HIGH probe is the most important probe for overflow prevention. Its correct placement directly determines whether the system can stop the motor pump before the tank overfills.

3.5.4 Probe Working Logic

The probe working logic describes how the physical water level produces electrical states. The sensing logic depends on whether the LOW and HIGH probes are wet or dry relative to the REF probe.

The basic logic is:

LOW dry and HIGH dry $\Rightarrow$ Tank low
 $\Rightarrow$ Pump ON

LOW wet and HIGH dry
 $\Rightarrow$ Tank filling or intermediate
 $\Rightarrow$ Previous pump state maintained

LOW wet and HIGH wet $\Rightarrow$ Tank full
 $\Rightarrow$ Pump OFF

The most important feature of this logic is state memory. The system should not turn the pump off immediately when water reaches the LOW probe during filling. It should remain on until water reaches the HIGH probe. Similarly, when water falls below HIGH but remains above LOW, the pump should not immediately start. It should remain off until the water falls below LOW again.

This produces two threshold control:

Start threshold = h_{LOW}

Stop threshold = h_{HIGH}

The difference between these heights is the control band:

$$\Delta h = h_{HIGH} - h_{LOW}$$

A larger (Δh) reduces frequent switching but allows a wider water level range. A smaller (Δh) keeps water level more constant but may increase switching frequency. Therefore, probe spacing must be selected according to the desired system behavior.

A formal probe state table is shown below.

Table 3.1: Probe State and Pump Control Logic

Water condition	REF probe	LOW probe	HIGH probe	Controller state	Pump state
Tank below low level	Wet if water remains at reference	Dry	Dry	Start condition	ON
Water reaches low level while filling	Wet	Wet	Dry	Hold condition	ON
Water reaches high level	Wet	Wet	Wet	Stop condition	OFF
Water falls below high but above low	Wet	Wet	Dry	Hold condition	OFF
Water falls below low again	Wet or nearly dry depending on tank design	Dry	Dry	Start condition	ON

The probe working logic is valid only if the REF probe remains effective and the water provides sufficient conductivity. If the water is too pure or the probes are corroded, the sensing resistance may become too high and the controller may not detect the correct state.

3.6 NE555 Pin Connection Scheme

The NE555 pin connection scheme explains the role of every pin of the NE555 integrated circuit in the project. This is necessary because the NE555 is the control center of the circuit. A complete academic report must not simply show the NE555 in a schematic. It must explain why each pin is connected as shown.

The NE555 has eight pins:

Pin 1: Ground Pin 2: Trigger Pin 3: Output Pin 4: Reset Pin 5: Control voltage Pin 6: Threshold Pin 7: Discharge Pin 8: VCC

The control logic of the water level controller depends mainly on pin 2, pin 3, and pin 4. Pin 2 is related to the start condition. Pin 4 is related to the stop condition. Pin 3 provides the output signal that drives the relay transistor. Pin 1 and pin 8 provide the supply connection. Pin 5 improves stability if filtered. Pin 6 and pin 7 must be connected according to the selected control configuration.

The NE555 output state can be expressed generally as:

$$\text{Trigger condition active} \Rightarrow V_{OUT} = HIGH$$

$$\text{Reset condition active} \Rightarrow V_{OUT} = LOW$$

This is the control principle used to start and stop the pump.

3.6.1 Pin 1: Ground

Pin 1 is the ground pin of the NE555. It must be connected to the common reference of the control circuit. All control voltages are measured relative to this ground point. The transistor emitter, sensor reference network, and NE555 ground must share a defined reference so that the circuit can interpret voltages correctly.

If pin 1 is not connected properly, the NE555 cannot operate correctly. A weak or noisy ground connection can cause unstable switching, false triggering, or relay chatter. The ground connection must therefore be low resistance and physically reliable.

In the scheme, pin 1 must be connected to the control ground. If the motor has a separate high current return path, that path should not disturb the NE555 ground. Sensitive control ground should be protected from motor current noise.

The ground condition can be represented as:

$$V_{GND} = 0V$$

All NE555 input and output voltages are interpreted relative to this value.

3.6.2 Pin 2: Trigger

Pin 2 is the trigger input of the NE555. It is responsible for setting the output high when the trigger condition becomes active. In the water level controller, the trigger function is associated with the low water condition. When the water level falls below the LOW probe, the circuit should activate the pump.

The trigger input is sensitive to voltage level. In a standard NE555 configuration, the trigger comparator responds when the trigger voltage falls below a fraction of the supply voltage. This makes the trigger pin suitable for detecting a low level condition produced by the sensing network.

The trigger condition can be represented as:

$$V_{TRIG} < \frac{1}{3}V_{CC} \Rightarrow V_{OUT} = HIGH$$

In the scheme, pin 2 must be connected so that the low water condition produces the correct trigger voltage. If pin 2 is left floating, it may randomly trigger the NE555. Therefore, pin 2 must have a defined electrical path through the sensing network and resistors.

Pin 2 is structurally connected to the pump start behavior. If pin 2 does not receive the correct signal when the tank becomes low, the pump will not start.

3.6.3 Pin 3: Output

Pin 3 is the output pin of the NE555. It provides the control signal that drives the relay transistor. When the output is high, the transistor base receives current through the base resistor. The transistor saturates, the relay coil energizes, and the motor pump turns on. When the output is low, transistor base current stops, the relay coil de-energizes, and the motor pump turns off.

The output relation is:

$$\begin{aligned} V_{OUT} = HIGH &\Rightarrow I_B > 0 \Rightarrow I_C > 0 \\ &\Rightarrow \text{Relay ON} \Rightarrow \text{Pump ON} \end{aligned}$$

$$\begin{aligned} V_{OUT} = LOW &\Rightarrow I_B = 0 \Rightarrow I_C = 0 \\ &\Rightarrow \text{Relay OFF} \Rightarrow \text{Pump OFF} \end{aligned}$$

Pin 3 should not be connected directly to the motor pump. It should only drive the transistor base through an appropriate resistor. This protects the NE555 from excessive load current.

The output voltage must also be high enough to provide base current to the transistor. The base current is calculated by:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

Thus, pin 3 is the electrical bridge between the NE555 control logic and the relay driver stage.

3.6.4 Pin 4: Reset

Pin 4 is the reset input of the NE555. It is used to force the output into the low state. In the water level controller, the reset function is associated with the high water condition. When water reaches the HIGH probe, the circuit must stop the pump.

The reset condition can be represented as:

$$\begin{aligned} \text{HIGH probe wet} &\Rightarrow \text{Reset condition active} \\ &\Rightarrow V_{OUT} = \text{LOW} \end{aligned}$$

In a standard NE555, the reset input is active when pulled low. Therefore, the circuit must be arranged so that the HIGH probe condition produces the correct reset action. If the reset pin is not used properly, the pump may continue running even when the tank is full.

Pin 4 must never be left floating. A floating reset pin can cause unpredictable output behavior. If the reset function is controlled by the water sensing network, a pull up resistor is usually required to keep the reset input high during normal operation and allow it to be pulled to the reset condition when the HIGH probe is activated.

The reset input is structurally more important than the trigger input for safety because it stops the pump and prevents overflow. Therefore, pin 4 must be treated as a critical control input.

3.6.5 Pin 5: Control Voltage

Pin 5 is the control voltage pin of the NE555. It gives access to the internal reference voltage network. In many circuits, pin 5 is not used as an active control input. However, it should still be stabilized because noise on pin 5 can affect the internal threshold levels of the NE555.

A small capacitor can be connected from pin 5 to ground to reduce noise sensitivity. A common value is:

$$C_{control} = 0.01\mu F$$

The purpose of this capacitor is not to control the water level logic directly. Its purpose is to improve stability by filtering unwanted noise.

If pin 5 is left without a capacitor, the circuit may still work in simulation, but the real hardware may become more sensitive to supply noise and relay switching disturbance. Therefore, the scheme should show pin 5 either connected to a capacitor or explicitly described as unused but stable according to the selected design.

Pin 5 must not be ignored in the academic explanation because it affects the internal comparator reference structure of the NE555.

3.6.6 Pin 6: Threshold

Pin 6 is the threshold input of the NE555. In many timing circuits, it is used to reset the internal latch when the timing capacitor reaches a defined voltage. In this water level controller, its exact role depends on the selected NE555 control arrangement.

If pin 6 is connected to a fixed level, timing node, or ground, the report must explain why that connection is required. If pin 6 is tied to a stable condition so that the circuit behaves as a latch controlled mainly by trigger and reset, this must be stated clearly.

The standard threshold relation is:

$$V_{THRESH} > \frac{2}{3}V_{CC} \Rightarrow V_{OUT} = LOW$$

If the project does not use pin 6 as the primary stop input, then the scheme must ensure that pin 6 does not accidentally force an unwanted output state. Floating threshold input is not acceptable because it can cause unpredictable behavior.

Pin 6 must therefore be connected to a defined voltage condition. Its connection must support the intended pump control logic and must not conflict with the reset function.

3.6.7 Pin 7: Discharge

Pin 7 is the discharge pin of the NE555. In normal timing circuits, it discharges an external capacitor when the internal discharge transistor turns on. In this project, if no timing capacitor is used, pin 7 may not perform the standard timing discharge role. However, it must still be shown correctly in the circuit diagram.

If pin 7 is unused, the report must state that it is not part of the active control path. If it is connected to any resistor or capacitor network, the function of that network must be explained.

The discharge pin should not be connected randomly. Any connection to pin 7 changes the behavior of the internal transistor and may influence circuit operation. Therefore, the scheme must treat pin 7 as a defined pin, not as an invisible pin.

In the academic explanation, pin 7 can be described as secondary to the main water level control operation unless the circuit uses it in a timing or discharge function.

3.6.8 Pin 8: VCC

Pin 8 is the positive supply pin of the NE555. It must be connected to the control supply voltage. The NE555 will not operate correctly if this supply voltage is missing, unstable, or outside the required range.

The supply condition is:

$$V_{CC} = V_{control}$$

If the project uses a 5 volt control supply, then:

$$V_{CC} = 5V$$

The NE555 supply must be decoupled with a capacitor placed close to pin 8 and pin 1. The purpose of the decoupling capacitor is to reduce supply noise and prevent sudden voltage changes from disturbing the control logic.

The decoupling condition can be represented as:

$$C_{decoupling} = 0.1\mu F$$

Pin 8 must never be confused with the motor supply terminal. The NE555 control supply and motor pump supply may be different. If the motor pump uses 12 volts, that does not automatically mean the NE555 must also use 12 volts. The scheme must clearly identify the correct supply domain.

3.7 Relay Driver Scheme

The relay driver scheme explains how the NE555 output controls the relay coil through a transistor. The relay coil requires more current than a direct signal line should provide. Therefore, a transistor is placed between the NE555 output and the relay coil.

The relay driver uses a low side switching arrangement. In this arrangement, one side of the relay coil is connected to the positive control supply. The other side of the relay coil is connected to the collector of the transistor. The emitter of the transistor is connected to ground. The base of the transistor is connected to the NE555 output through a resistor.

When the NE555 output is high, the base emitter junction of the transistor becomes forward biased. Base current flows into the transistor, and the transistor enters saturation. In saturation, the transistor behaves approximately like a closed switch from collector to emitter. Current then flows through the relay coil, and the relay energizes.

The relay coil current is:

$$I_{coil} = \frac{V_{relay}}{R_{coil}}$$

The transistor collector current is approximately:

$$I_C \approx I_{coil}$$

The base current is:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

For reliable switching:

$$I_B \geq \frac{I_{coil}}{10}$$

The transistor power dissipation in saturation is:

$$P_Q = V_{CE(sat)} I_C$$

This value must remain within the safe power limit of the transistor.

The relay driver scheme must also include the diode across the relay coil. The diode protects the transistor from the voltage spike created when the relay coil is switched off. The diode must be connected in reverse polarity during normal operation. When the transistor turns off, the coil current flows through the diode until the stored magnetic energy is dissipated.

The relay driver scheme is acceptable only if four conditions are satisfied:

The NE555 output provides enough base drive. The base resistor limits current safely. The transistor can handle the relay coil current. The relay coil has flyback protection.

If any of these conditions is not satisfied, the relay driver section is incomplete.

3.8 Motor Switching Scheme

The motor switching scheme explains how the relay contact controls the motor pump. The motor pump must be connected to the relay contact side, not to the NE555 output or transistor collector path. The relay contact acts as the power switch for the motor.

The motor circuit contains the motor supply, relay contact, motor pump, and return path. When the relay is energized, the contact closes and completes the motor circuit. When the relay is de energized, the contact opens and interrupts the motor circuit.

The motor voltage condition is:

$$V_{motor} = V_{supply} \quad \text{when relay contact is closed}$$

$$V_{motor} = 0V \quad \text{when relay contact is open}$$

The motor current condition is:

$$I_{motor} = \frac{P_{motor}}{V_{motor}}$$

The relay contact must be rated for the motor current. Since a motor can draw high current at startup, the relay must not be selected only from the normal running current. The contact rating must satisfy:

$$I_{contact} \geq I_{start}$$

The motor switching scheme must also consider contact stress. When a motor is switched, the relay contacts may experience electrical arcing, especially if the load current is high. A suppression component may be required depending on the motor type

and supply type.

For a direct current motor, protection may include a diode or transient suppressor across the motor terminals. For an alternating current motor, a resistor capacitor snubber or properly rated contact suppressor may be needed. The selected method must match the actual motor supply type.

The motor switching scheme is the final power action of the circuit. It proves that the controller does not merely generate a signal but actually controls a useful physical load.

3.9 LED Indicator Scheme

The LED indicator scheme provides a visual representation of the circuit state. The LED may be connected so that it turns on when the relay is energized, when the pump is operating, or when the NE555 output is high. The exact interpretation must be stated in the report.

If the LED is connected to the NE555 output or transistor driver section, it indicates the control state. If it is connected near the motor side, it may indicate motor supply state. In this project, the LED should be explained as a status indicator for pump activation or relay activation.

The LED must have a series resistor. The resistor limits current and prevents LED damage. The required value is calculated by:

$$R_{LED} = \frac{V_{supply} - V_F}{I_{LED}}$$

If ($V_{supply} = 5V$), ($V_F = 2V$), and ($I_{LED} = 10mA$), then:

$$R_{LED} = 300\Omega$$

The nearest standard value is:

$$R_{LED} = 330\Omega$$

If ($V_{supply} = 12V$), ($V_F = 2V$), and ($I_{LED} = 10mA$), then:

$$R_{LED} = 1000\Omega$$

The LED indicator must not affect the relay driver behavior. It should be connected in a way that does not steal excessive current from the transistor base or overload the NE555 output. Its purpose is indication, not control.

The LED scheme improves simulation analysis because the state of the controller becomes visible. During testing, the LED can confirm whether the output section is active even before observing the motor pump.

3.10 Protection Scheme

The protection scheme is the set of design measures used to prevent component damage, false operation, excessive current, and unsafe behavior. A water level motor controller contains both sensitive control electronics and inductive loads. Therefore, protection is not optional. It is a necessary part of the scheme.

The main protection requirements are:

Relay flyback protection Motor side protection Supply decoupling Reverse polarity protection Fuse protection

The protection scheme must defend the circuit against five major threats:

Inductive voltage spikes Motor switching noise Supply voltage disturbance Incorrect supply connection Excessive current

A project that switches a relay and motor without protection may work in simulation but fail in hardware. Therefore, the protection scheme connects the simulated design to real engineering reliability.

3.10.1 Relay Flyback Protection

The relay coil is an inductive component. When current flows through the coil, energy is stored in its magnetic field. When the transistor turns off, the coil current cannot stop instantly. If no safe path is provided, the coil generates a high reverse voltage that can damage the transistor.

The stored energy in the relay coil is:

$$E = \frac{1}{2}LI^2$$

where (E) is stored energy, (L) is coil inductance, and (I) is coil current.

The flyback diode provides a current path after the transistor switches off. The diode must be connected across the relay coil in reverse bias during normal operation. During switch off, the coil voltage reverses and the diode becomes forward biased. This safely dissipates the stored energy.

The diode current rating should satisfy:

$$I_D \geq I_{coil}$$

The reverse voltage rating should also be higher than the relay supply voltage:

$$V_{RRM} > V_{relay}$$

The flyback diode protects the transistor, improves reliability, and reduces electrical noise. Without this diode, the transistor may fail even if the circuit works in simulation.

3.10.2 Motor Side Protection

The motor pump is also an inductive and noisy load. When switched on, it may draw starting current greater than its normal running current. When switched off, it may generate voltage transients. These effects can damage relay contacts, disturb the control circuit, or reduce long term reliability.

The motor side protection must be selected according to the motor type. If the motor is a direct current motor, a diode or transient voltage suppressor may be placed across the motor terminals. If the motor is an alternating current motor, a diode cannot be used across the motor supply. In that case, a snubber or correctly rated suppressor must be used.

The motor starting current condition is:

$$I_{start} > I_{run}$$

The relay contact must satisfy:

$$I_{contact} \geq I_{start}$$

The motor supply should also include a fuse or current limiting protection. This prevents wiring damage if the motor stalls or a short circuit occurs.

Motor side protection is necessary because the motor pump is the highest stress component in the system. It is the part most likely to create current surge, electrical noise, and mechanical load variation.

3.10.3 Supply Decoupling

Supply decoupling is used to stabilize the voltage applied to the NE555 and control circuit. When the relay switches, sudden current changes can create voltage drops and noise on the supply line. These disturbances may cause false triggering, relay chatter, or unstable operation.

A small ceramic capacitor should be placed close to the NE555 supply pins:

$$C_{decoupling} = 0.1\mu F$$

A larger electrolytic capacitor may also be placed across the control supply rail:

$$C_{bulk} = 10\mu F \text{ to } 100\mu F$$

The small capacitor filters high frequency noise. The larger capacitor supports the supply during sudden load changes.

The supply stability condition is:

$$V_{CC(min)} \leq V_{CC}(t) \leq V_{CC(max)}$$

where $(V_{\{CC\}}(t))$ is the instantaneous supply voltage during operation.

If decoupling is absent, the NE555 may respond to supply disturbance as if it were a real trigger condition. Therefore, decoupling is directly connected to control reliability.

3.10.4 Reverse Polarity Protection

Reverse polarity protection prevents damage if the power supply is connected incorrectly. In educational and prototype circuits, incorrect supply connection is a common risk. If positive and negative terminals are reversed, the NE555, transistor, LED, and relay driver may be damaged.

Reverse polarity protection may be provided by a series diode, a bridge arrangement, or a field effect transistor based protection circuit. A simple series diode is easy to understand but causes a voltage drop. If a diode is used, the available circuit voltage becomes:

$$V_{circuit} = V_{supply} - V_D$$

where (V_D) is the diode forward voltage.

If the voltage drop is not acceptable, a more efficient protection method should be used. The chosen protection method must be compatible with the supply voltage and current requirement of the circuit.

Reverse polarity protection is not part of the main control logic, but it increases practical reliability and protects the circuit during connection errors.

3.10.5 Fuse Protection

Fuse protection limits damage caused by excessive current. A fuse should be placed in the motor supply path because the motor pump draws the highest current in the system. If the motor stalls, wiring shorts, or relay contact fails, the fuse can disconnect the supply before serious damage occurs.

The fuse current rating should be above the normal running current but below the dangerous current level of the wiring and components. A simplified condition is:

$$I_{run} < I_{fuse} < I_{wire\ safe}$$

The fuse must also tolerate motor starting current for a short time. Therefore, the fuse selection must consider both running current and starting current:

$$I_{fuse} > I_{run}$$

$$I_{fuse} \text{ must tolerate } I_{start} \text{ for a short duration}$$

A fuse that is too small may blow during normal motor startup. A fuse that is too large may fail to protect the circuit. Correct fuse selection is therefore a safety and

reliability requirement.

Fuse protection should also be considered for the control supply if the control circuit is powered by an external adapter or battery pack.

3.11 Wiring and Connection Table

The wiring and connection table gives a clear reference for how the circuit components are connected. It supports assembly, simulation checking, troubleshooting, and oral defense. A formal report should include such a table because it removes ambiguity from the schematic.

The wiring table must match the actual circuit diagram. If any connection in the table contradicts the schematic, the report becomes technically weak. The table should be checked after the final Proteus schematic is completed.

3.12 Physical Layout Considerations

Physical layout considerations describe how the circuit should be arranged in real hardware. Even if the project is simulated in Proteus, the report must show awareness that physical circuits behave differently from ideal simulations. Wire length, grounding, relay placement, motor noise, probe corrosion, and supply routing can affect performance.

The NE555 should be placed near its decoupling capacitor. The capacitor should be physically close to pin 8 and pin 1. Long connections reduce the effectiveness of decoupling.

The relay should be placed away from sensitive sensing nodes. The relay coil and contacts can produce electrical noise. The motor wires should also be kept away from trigger and reset wiring. Sensor wires connected to the water probes may act as antennas if they are long. Therefore, probe wiring should be kept short or routed carefully.

The flyback diode should be placed close to the relay coil terminals. This ensures that the coil current loop during switch off is small and controlled.

The motor current path should use thicker wires or wider copper traces than the signal section. The motor path carries higher current, while the NE555 and sensor section carry low current. Mixing these paths without care can create voltage drops and false triggering.

The ground layout should reduce shared impedance between motor current and control signal current. A preferred method is to route high current motor return separately from sensitive control ground and join them at a controlled reference point if a common ground is required.

The water probes should be made from corrosion resistant conductive material. They should be mechanically fixed so that movement of water does not change their height. Probe spacing should remain stable because the switching thresholds depend directly on probe height.

Table 3.2: Wiring and Connection Table

Circuit element	Connection point	Connected to	Function
NE555 pin 1	Ground	Control ground	Provides reference voltage
NE555 pin 2	Trigger input	Low level sensing network	Starts pump under low water condition
NE555 pin 3	Output	Base resistor of relay driver transistor	Sends control signal to relay driver
NE555 pin 4	Reset input	High level sensing network	Stops pump when tank reaches high level
NE555 pin 5	Control voltage	Capacitor to ground if used	Reduces internal reference noise
NE555 pin 6	Threshold	Defined voltage condition according to circuit	Prevents undefined threshold behavior
NE555 pin 7	Discharge	Defined according to selected configuration	Used only if required by the circuit arrangement
NE555 pin 8	VCC	Control supply	Provides operating voltage
REF probe	Reference electrode	Sensor reference path	Provides common water conduction path
LOW probe	Lower threshold electrode	Trigger related sensing path	Detects low water condition
HIGH probe	Upper threshold electrode	Reset related sensing path	Detects full tank condition
Q1 base	Driver input	NE555 pin 3 through base resistor	Receives control current
Q1 collector	Relay coil side	Relay coil	Switches coil current
Q1 emitter	Ground	Control ground	Completes driver current path
Relay coil terminal 1	Positive side	Control supply	Receives relay supply
Relay coil terminal 2	Switched side	Q1 collector	Controlled by transistor
Flyback diode	Across relay coil	Reverse biased in normal operation	Protects transistor from coil spike
Relay common contact	Motor supply path	Positive motor supply or motor terminal	Switches motor circuit
Relay normally open contact	Motor supply path	Motor terminal or supply terminal	Closes when relay is energized
Motor pump terminal 1	Load terminal	Relay contact path	Receives switched supply
Motor pump terminal 2	Load return	Motor supply return	Completes motor circuit
LED anode	Indicator branch	Supply or output node according to design	Indicates active state
LED cathode	Indicator branch	Ground through resistor or defined return	Completes indicator path
LED resistor	Series with LED	LED branch	Limits LED current

The physical enclosure must keep the control electronics dry. Water level sensing involves the tank, but the circuit board should not be exposed to splashing, condensation, or accidental water contact.

Therefore, the physical layout is not a cosmetic issue. It is part of the reliability structure of the project.

3.13 Safety Considerations

Safety considerations are essential because the project controls a motor pump and interacts with water. The combination of electricity, water, relay switching, and motor current requires careful design. Even if the circuit is simulated, the academic report must separate simulation behavior from real hardware safety.

If the pump is a low voltage direct current motor, the risk is lower than with an alternating current mains pump, but protection is still required. The motor supply should be fused. The relay contact should be rated for the motor current. The motor wiring should be insulated. The control circuit should not be exposed to water.

If the pump is an alternating current mains pump, the safety requirements become much more serious. The relay must have proper contact rating for the mains voltage and current. The circuit must be placed in an insulated enclosure. Live conductors must not be exposed. Proper separation must exist between the low voltage control section and high voltage load section. A fuse, earthing where required, strain relief, and safe terminal blocks must be used. The project must not treat a mains pump as if it were the same as a small simulated direct current motor.

The water probes must operate only with safe low voltage. No high voltage should be applied to the probes because the probes are placed inside water. The sensing current should be small. This reduces corrosion and improves safety.

The relay contact rating must satisfy:

$$V_{contact} \geq V_{motor}$$

$$I_{contact} \geq I_{start}$$

The wire current rating must satisfy:

$$I_{wire\ rating} > I_{load}$$

The fuse must satisfy:

$$I_{run} < I_{fuse} < I_{wire\ safe}$$

The circuit board should be installed in a dry enclosure. The motor terminals should be insulated. The power supply should be switched off before any wiring changes are

made. The system should be tested first with a safe low voltage motor before any higher power pump is considered.

The safety conclusion is that the simulation proves control logic, but hardware safety requires correct voltage isolation, current rating, protection, enclosure, and wiring practice.

3.14 Chapter Summary

This chapter presented the complete scheme of the automatic water level motor pump controller. The system was divided into sensing, control, driver, relay, motor, indicator, power supply, and protection sections. The water level probes provide the input condition of the circuit. The NE555 interprets the probe based condition and generates the output state. The transistor driver converts the NE555 output into relay coil current. The relay contact switches the motor pump supply. The LED indicator provides visual feedback. The protection scheme reduces the risk of transistor damage, motor side disturbance, supply instability, reverse connection damage, and excessive current.

The probe arrangement was explained through REF, LOW, and HIGH electrodes. The REF probe provides the sensing reference. The LOW probe defines the start threshold. The HIGH probe defines the stop threshold. The difference between the LOW and HIGH probe heights creates the control band:

$$\Delta h = h_{HIGH} - h_{LOW}$$

The NE555 pin scheme was explained pin by pin so that the control role of each pin is clear. The relay driver and motor switching schemes were also analyzed mathematically through base current, relay coil current, motor current, and protection conditions.

The final structural conclusion is that the scheme forms one complete control chain:

Water level condition → Probe electrical state
→ NE555 output decision
→ Transistor relay drive
→ Relay motor switching
→ Automatic pump action

Therefore, Chapter 3 proves the structural architecture of the project. It shows not only what components are used, but why each component exists, how each section connects, and how the whole circuit achieves automatic water level control.

4

Simulation

4.1 Introduction

This chapter presents the simulation stage of the automatic water level motor pump controller. The purpose of the simulation is to verify whether the designed circuit performs the required control operation before any physical implementation is attempted. The circuit is expected to detect the simulated water level through the LOW, HIGH, and REF probes, process the sensor condition through the NE555 control stage, drive the relay through the transistor switching stage, and control the motor pump according to the tank condition.

Simulation is necessary because it allows the behavior of the circuit to be tested under controlled conditions. It also allows each functional state of the system to be observed without the risk of damaging real components. In this project, the simulation focuses on the logical and electrical behavior of the circuit. The main concern is whether the motor pump turns ON when the tank requires water and turns OFF when the tank reaches the full level.

The simulation chapter is therefore a verification chapter. It does not merely show that the circuit was drawn in software. It proves whether the circuit responds correctly to the water level states defined in the design. The correctness of the system is judged by comparing the expected behavior with the observed behavior in the Proteus simulation.

The simulation also provides evidence for the operation of the relay, LED indicator, transistor driver, and NE555 output. Each part of the circuit must behave according to its intended function. If the simulation result agrees with the expected control logic, the design can be considered functionally valid at the simulation level. However, simulation cannot replace real hardware testing, because real water conductivity, relay contact wear, motor starting current, sensor corrosion, and environmental noise are not fully represented by the simulation model.

This chapter is organized into the simulation software used, simulation environ-

ment, circuit setup, test procedure, test cases, expected results, observed results, measurement points, screenshots, analysis, limitations, and chapter summary.

4.2 Simulation Software Used

The simulation was carried out using Proteus Design Suite. Proteus was selected because it supports schematic capture, electronic circuit simulation, virtual instruments, relay operation, transistor switching, LED indication, and motor load representation. It allows the designer to build the circuit virtually and observe the behavior of the components during operation.

Proteus is suitable for this project because the circuit contains both low voltage control components and an electromechanical switching element. The NE555 timer, BC547 transistors, resistors, diodes, relay, LED, voltage sources, ground terminals, and motor load can be represented inside the Proteus environment. This makes it possible to test the complete control chain from the water level sensor condition to the final motor pump response.

The software also allows the simulated probe states to be changed during runtime. This is important because the tank water level is not a fixed condition. The water level changes from empty to full during pump operation and later decreases when water is consumed. The simulation must therefore allow the LOW and HIGH probe conditions to be tested separately and sequentially.

The use of Proteus also improves the academic reliability of the report because the circuit behavior can be shown through screenshots, measured voltages, and state based test results. Instead of only describing the circuit theoretically, the simulation provides observable evidence of how the circuit performs under defined input conditions.

4.3 Proteus Simulation Environment

The Proteus simulation environment was configured to represent the complete automatic water level controller. The schematic was built using the required electronic components and connected according to the designed circuit scheme. The control section and the motor section were represented as separate functional domains.

The control section consists of the NE555 timer IC, sensing probes, resistors, transistors, LED indicator, relay coil, flyback diode, control supply, and ground reference. This section processes the water level condition and produces the relay switching action. The motor section consists of the relay contact path, the 12 V motor supply, and the DC motor pump. This section represents the load controlled by the relay.

The Proteus environment was arranged so that the water level could be simulated by changing the electrical contact condition of the probes. The LOW probe represents the lower tank threshold. The HIGH probe represents the upper tank threshold. The REF probe provides the reference electrode for the sensing path. When a probe is

considered wet, it is treated as electrically connected through the water path. When it is considered dry, it is treated as electrically open.

The relay model was used to connect and disconnect the motor pump supply according to the output condition of the relay driver transistor. When the relay coil is energized, the relay contact closes and the motor receives supply voltage. When the relay coil is de energized, the contact opens and the motor stops.

The simulation was run under stable supply conditions. The control supply was applied to the NE555 and relay control stage, while the motor pump was powered by a separate 12 V supply path. Ground reference was maintained for the control circuit. The correct separation between the control domain and motor domain is necessary because the NE555 cannot directly drive the motor pump.

The simulation environment therefore represents the complete functional sequence of the project:

Water level condition is applied to the probes. The probe condition changes the input state of the NE555 control section. The NE555 output changes the transistor driver state. The transistor controls the relay coil. The relay contact switches the motor pump. The LED indicates the system state.

4.4 Simulation Objectives

The first objective of the simulation is to verify that the motor pump starts when the tank water level is below the required minimum level. In this condition, the LOW probe and HIGH probe are not reached by water, and the system must respond by activating the pump.

The second objective is to verify that the pump continues to operate while the tank is filling. When the water has reached the LOW probe but has not yet reached the HIGH probe, the tank is not full. The pump should remain ON until the upper level is reached.

The third objective is to verify that the pump stops when the water reaches the HIGH probe. This condition represents a full tank. The controller must switch the relay OFF so that the motor pump is disconnected from the supply.

The fourth objective is to verify that the pump remains OFF when the water level falls below the HIGH probe but is still above the LOW probe. This behavior is required to prevent rapid switching. The system should not restart the pump immediately after the water drops slightly below the HIGH level.

The fifth objective is to verify that the pump starts again only when the water level falls below the LOW probe. This confirms that the circuit has a practical two level control behavior rather than unstable single point switching.

The sixth objective is to verify that the LED indicator follows the intended system state. If the LED is designed to indicate pump operation or relay activation, it must turn ON when the corresponding state is active and turn OFF when that state is inactive.

The seventh objective is to verify the relay driver operation. The transistor must energize the relay coil when the NE555 output is in the active state and must release the relay when the NE555 output changes to the inactive state.

The eighth objective is to observe important voltages in the circuit. The NE555 output voltage, Q1 base voltage, relay coil voltage, motor terminal voltage, LED voltage, and sensor node voltage must be checked to confirm that the electrical states are consistent with the expected operation.

The final objective is to identify the limitations of the simulated design before hardware construction. The simulation can confirm logical behavior, but it cannot fully prove long term reliability, sensor durability, or motor protection under real conditions.

4.5 Simulation Circuit Setup

The simulation circuit was built by placing all required components in Proteus and connecting them according to the designed schematic. The circuit setup was divided into the component selection stage, power supply setup, probe simulation setup, relay setup, motor pump setup, and LED indicator setup.

The setup was arranged to represent the actual working principle of the system. The sensing probes detect the simulated tank level. The NE555 receives the sensor related control signals. The transistor driver amplifies the control output and switches the relay coil. The relay contact connects the motor pump to the 12 V supply when the system requires pumping. The LED provides visual indication of the operating state.

The circuit setup must be clearly presented in the report because the simulation result is only meaningful when the reader understands how the schematic was configured. A simulation without an explicit setup cannot be properly verified or repeated.

4.5.1 Component Selection in Proteus

The main component selected for the control section was the NE555 timer IC. In this project, the NE555 is used as the main control device that responds to the level sensing conditions and changes its output state accordingly. The output of the NE555 is connected to the transistor relay driver stage.

BC547 transistors were selected for the signal switching and relay driving functions. Q1 is used as the relay driver transistor. Its purpose is to provide the current path required to energize the relay coil. Q2 is used in the sensing or control input path, depending on the exact circuit connection. Its purpose is to condition the signal coming from the water probe section and influence the NE555 control state.

Resistors were selected to limit current, provide biasing, and define input states. The base resistors protect the transistor base terminals from excessive current. Pull up or pull down resistors are used to prevent undefined input states. The LED resistor limits the current through the indicator LED.

A relay was selected to switch the motor pump circuit. The relay allows the low current control circuit to control the higher current motor path. This is necessary because the NE555 and BC547 stages are not designed to drive the motor pump directly.

A diode was connected across the relay coil to act as a flyback protection diode. When the relay coil is de energized, the magnetic field collapses and can generate a high reverse voltage. The diode provides a safe current path for this transient energy and protects the transistor from damage.

A DC motor was selected to represent the pump load. The motor is connected through the relay contact so that it receives voltage only when the relay is activated.

An LED was selected as a visual indicator. Its state helps confirm whether the control circuit has entered the active or inactive condition.

The selected components therefore represent the complete system: sensing, control, driving, switching, indication, and load operation.

4.5.2 Power Supply Setup

The circuit uses two functional power domains. The first domain is the control supply, which powers the NE555, transistor control section, relay coil, and indicator circuit. The second domain is the motor supply, which powers the DC motor pump through the relay contact.

The control supply must be stable because the NE555 depends on defined voltage levels at its input pins. If the supply voltage is unstable, the trigger and reset conditions may become unreliable. The relay coil also requires sufficient voltage to operate correctly. If the relay coil receives less than its required voltage, the relay may fail to activate or may chatter.

The motor supply was set to 12 V in the simulation because the motor pump is represented as a 12 V DC load. The motor supply is switched by the relay contact. The motor does not receive power directly from the NE555 output or from the transistor signal path.

The general supply relationship can be expressed as:

$$V_{motor} = 12V$$

$$V_{control} = V_{CC}$$

The NE555 output high voltage is expected to be close to the control supply voltage:

$$V_{OUT(high)} \approx V_{CC}$$

The NE555 output low voltage is expected to be close to ground:

$$V_{OUT(low)} \approx 0V$$

When the relay is energized, the motor terminal voltage is expected to be close to the motor supply voltage:

$$V_{motor\ terminal} \approx 12V$$

When the relay is not energized, the motor terminal voltage is expected to be close to zero:

$$V_{motor\ terminal} \approx 0V$$

These voltage relationships are used later in the simulation measurement section to verify the correctness of the circuit behavior.

4.5.3 Probe Simulation Setup

The water level probes were represented by electrical contact conditions in Proteus. The REF probe acts as the reference probe. The LOW probe represents the lower water level threshold. The HIGH probe represents the upper water level threshold.

When the tank is empty, neither the LOW probe nor the HIGH probe is electrically connected through water to the REF probe. This represents the dry state. In this condition, the system must recognize that the tank requires water and the pump should turn ON.

When the water reaches the LOW probe, the LOW probe becomes electrically connected to the REF probe through the simulated water path. However, the HIGH probe remains dry. This represents a filling state. The system should keep the pump ON because the tank is not yet full.

When the water reaches the HIGH probe, both the LOW and HIGH probes are considered wet. This represents the full tank state. The system should stop the pump because the upper threshold has been reached.

When the water level later falls below the HIGH probe but remains above the LOW probe, the HIGH probe becomes dry while the LOW probe remains wet. The system should not immediately restart the pump. This prevents unnecessary rapid switching.

When the water level falls below the LOW probe again, both LOW and HIGH probes are dry. The system should restart the pump.

The probe simulation therefore represents the actual water level sequence:

Empty tank, water below LOW, water reaches LOW, water between LOW and HIGH, water reaches HIGH, tank full, water level falling, and water below LOW again.

The sensing logic can be represented by the following state expression:

$$S_{LOW} = \begin{cases} 1, & \text{LOW probe is wet} \\ 0, & \text{LOW probe is dry} \end{cases}$$

$$S_{HIGH} = \begin{cases} 1, & \text{HIGH probe is wet} \\ 0, & \text{HIGH probe is dry} \end{cases}$$

The desired pump condition is:

$$PUMP = ON \quad \text{when the tank level is below LOW}$$

$$PUMP = OFF \quad \text{when the tank level reaches HIGH}$$

Between LOW and HIGH, the pump state depends on the previous state. This is the essential reason for using two level control.

4.5.4 Relay Setup

The relay was configured as the switching element between the control circuit and the motor pump circuit. The relay coil is controlled by the transistor driver. The relay contact is connected in series with the motor pump and the 12 V motor supply.

When the NE555 output activates the base of Q1, Q1 conducts. This allows current to flow through the relay coil. Once the relay coil is energized, the relay contact closes. The closed contact connects the 12 V supply to the motor pump, and the pump turns ON.

When the NE555 output becomes inactive, Q1 stops conducting. The relay coil is de energized. The relay contact opens. The motor pump is disconnected from the 12 V supply, and the pump turns OFF.

The relay coil current can be represented as:

$$I_{relay} = \frac{V_{coil}}{R_{coil}}$$

where (V_{coil}) is the relay coil voltage and (R_{coil}) is the relay coil resistance.

The relay driver condition can be represented as:

$$Q1 = ON \Rightarrow Relay = Energized \Rightarrow Motor = ON$$

$$Q1 = OFF \Rightarrow Relay = De energized \Rightarrow Motor = OFF$$

The flyback diode across the relay coil must be reverse biased during normal relay operation and forward biased during relay turn off. This protects Q1 from the voltage spike produced by the collapsing magnetic field of the relay coil.

4.5.5 Motor Pump Setup

The motor pump was represented by a DC motor load in Proteus. The motor was connected to the 12 V supply through the relay contact. This arrangement ensures that

the motor is not directly connected to the NE555 output or to the transistor signal path.

The motor receives supply voltage only when the relay contact is closed. If the relay contact is open, the motor circuit is incomplete and the motor remains OFF.

The motor voltage condition is:

$$V_{motor} \approx 12V \quad \text{when the relay contact is closed}$$

$$V_{motor} \approx 0V \quad \text{when the relay contact is open}$$

The motor current depends on the motor model used in Proteus. In a real circuit, motor current must be checked against the relay contact rating and the power supply capacity. In the simulation, the motor mainly verifies the switching action of the relay.

The motor setup is therefore used to demonstrate the final output action of the water level controller. If the motor turns ON and OFF according to the simulated probe conditions, the control logic of the circuit is verified at the functional simulation level.

4.5.6 LED Indicator Setup

The LED indicator was connected to show the operating state of the circuit. Depending on the exact schematic connection, the LED may indicate relay activation, pump operation, or NE555 output state. In the simulation, the LED must respond consistently with its intended function.

If the LED is designed to indicate pump operation, it should turn ON when the relay is energized and the motor pump is running. It should turn OFF when the relay is de energized and the motor pump is stopped.

The LED current is limited by a series resistor. The resistor value must be selected so that the LED current remains within a safe operating range.

The LED current can be expressed as:

$$I_{LED} = \frac{V_{supply} - V_F}{R_{LED}}$$

where (V_{supply}) is the supply voltage applied to the LED branch, (V_F) is the forward voltage of the LED, and (R_{LED}) is the LED series resistor.

If the LED is connected to a 5 V supply and the LED forward voltage is approximately 2 V, then the LED current is:

$$I_{LED} = \frac{5 - 2}{R_{LED}}$$

If ($R_{LED} = 330\Omega$), then:

$$I_{LED} = \frac{3}{330} = 0.00909A$$

$$I_{LED} \approx 9.09mA$$

This is suitable for an indicator LED. If the resistor value is much smaller, the LED current must be checked carefully.

4.6 Simulation Test Procedure

The simulation test procedure was designed to reproduce the normal operating sequence of a water tank. The test begins with an empty tank condition and then progresses through the filling process until the tank becomes full. After that, the falling water level condition is tested to confirm that the pump restarts only when the lower threshold is reached again.

The first step was to open the complete circuit schematic in Proteus and confirm that all components were correctly connected. The power supplies were then activated. The initial state of the probes was set to represent an empty tank. In this condition, the LOW and HIGH probes were dry.

The simulation was then run, and the state of the NE555 output, Q1 transistor, relay coil, relay contact, LED, and motor pump was observed. The expected result was that the pump should turn ON.

The next step was to simulate the water reaching the LOW probe. The LOW probe was set to the wet condition while the HIGH probe remained dry. The expected result was that the pump should remain ON because the tank had not yet reached the full level.

The next step was to simulate the water level between the LOW and HIGH probes. This condition maintains the filling process. The expected result was continued pump operation.

The next step was to simulate the water reaching the HIGH probe. At this point, the HIGH probe was changed to the wet condition. The expected result was that the NE555 control section should reset or change state so that the relay de energizes and the pump turns OFF.

After the full tank condition was confirmed, the water level falling condition was simulated. The HIGH probe was changed back to dry while the LOW probe remained wet. The expected result was that the pump should remain OFF because the tank had not yet fallen below the lower threshold.

Finally, the LOW probe was changed to dry again. This condition represents the water level falling below the lower threshold. The expected result was that the pump should turn ON again.

The test procedure was repeated to confirm that the circuit response was consistent and not accidental. Each test condition was recorded in the simulation result table.

4.7 Simulation Test Cases

The simulation test cases were selected to cover all important operating states of the system. Each test case represents a specific water level condition and verifies the corresponding pump response. The test cases also verify whether the circuit avoids unstable switching between the LOW and HIGH thresholds.

4.7.1 Test Case 1. Tank Empty Condition

In the tank empty condition, both the LOW probe and the HIGH probe are dry. This condition means that the water level is below the minimum required level. The controller must identify that the tank needs to be filled.

The expected circuit response is that the NE555 output enters the active state required to drive the relay transistor. Q1 should conduct, the relay coil should energize, the relay contact should close, and the motor pump should turn ON. The LED should also indicate the active state if it is connected as a pump or relay indicator.

This test case is critical because it proves the starting function of the system. If the pump does not start when the tank is empty, the controller fails its most basic purpose.

Expected result:

$$S_{LOW} = 0$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

4.7.2 Test Case 2. Water Below LOW Probe

In this condition, the tank contains some water, but the water level is still below the LOW probe. Electrically, this state is equivalent to the empty condition because neither the LOW probe nor the HIGH probe is connected through water to the REF probe.

The expected result is that the pump remains ON. The system must continue filling the tank until the water level reaches at least the LOW probe and eventually reaches the HIGH probe.

This test confirms that the controller does not stop the pump prematurely. The tank is still below the lower threshold, so there is no valid reason to turn the pump OFF.

Expected result:

$$S_{LOW} = 0$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

4.7.3 Test Case 3. Water Reaches LOW Probe

In this condition, the water level has reached the LOW probe. The LOW probe becomes wet, while the HIGH probe remains dry. The tank is no longer empty, but it is not yet full.

The expected result is that the pump continues to operate. This behavior is necessary because the LOW probe should not stop the pump. The LOW probe only defines the lower threshold for restarting the filling process. The pump must continue running until the water reaches the HIGH probe.

Expected result:

$$S_{LOW} = 1$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

This test is important because it verifies that the system does not confuse the LOW level with the full level. If the pump stops when the LOW probe becomes wet, the circuit is incorrectly designed.

4.7.4 Test Case 4. Water Between LOW and HIGH

In this condition, the water level is above the LOW probe but below the HIGH probe. The LOW probe remains wet, and the HIGH probe remains dry. The tank is in the normal filling region.

The expected result is that the pump remains ON if it was previously ON. This confirms that the circuit has a holding behavior between the two thresholds. The system should not switch OFF until the HIGH level is reached.

This condition is important because it represents the largest part of the filling process. The controller must remain stable during this period and must not oscillate between ON and OFF states.

Expected result:

$$S_{LOW} = 1$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

during filling.

4.7.5 Test Case 5. Water Reaches HIGH Probe

In this condition, the water level has reached the HIGH probe. Both the LOW probe and the HIGH probe are wet. This condition represents the upper tank threshold.

The expected result is that the control circuit turns the pump OFF. The NE555 output should change to the inactive state required to stop the relay driver. Q1 should stop conducting, the relay coil should de energize, the relay contact should open, and the motor pump should stop.

This test case is one of the most important tests in the simulation because it proves the overflow prevention function. If the pump does not stop at the HIGH probe, the tank may overflow in a real system.

Expected result:

$$S_{LOW} = 1$$

$$S_{HIGH} = 1$$

$$PUMP = OFF$$

4.7.6 Test Case 6. Tank Full Condition

The tank full condition is the stable condition after the water has reached the HIGH probe. Both the LOW and HIGH probes remain wet. The system must keep the pump OFF as long as the tank is full.

The expected result is that the relay remains de energized and the motor pump remains OFF. The LED should also show the inactive state if it is used as a pump operation indicator.

This test confirms that the controller does not restart the pump while the tank is already full. A correct controller must maintain the OFF state until the water level falls sufficiently.

Expected result:

$$S_{LOW} = 1$$

$$S_{HIGH} = 1$$

$$PUMP = OFF$$

4.7.7 Test Case 7. Water Level Falling

In this condition, water has started to fall from the full tank level. The HIGH probe becomes dry, but the LOW probe remains wet. This means the tank is no longer completely full, but it still contains enough water.

The expected result is that the pump remains OFF. This is a critical part of two level control. The pump should not turn ON immediately after the water drops below the HIGH probe. It should only turn ON again when the water level falls below the LOW probe.

This test confirms that the circuit prevents rapid switching near the upper probe. Without this behavior, the pump could repeatedly turn ON and OFF due to small water level changes or surface movement.

Expected result:

$$S_{LOW} = 1$$

$$S_{HIGH} = 0$$

$$PUMP = OFF$$

during falling water level.

4.7.8 Test Case 8. LOW Probe Becomes Dry Again

In this condition, the water level has fallen below the LOW probe. The LOW probe becomes dry, and the HIGH probe is also dry. This means the tank has returned to the low level condition.

The expected result is that the pump turns ON again. The NE555 output should activate the relay driver. The relay should energize. The relay contact should close. The motor pump should receive voltage and begin filling the tank again.

This test verifies the automatic restart function of the system. It also confirms that the controller completes the full operating cycle: pump ON at low level, pump OFF at high level, and pump ON again when the level becomes low.

Expected result:

$$S_{LOW} = 0$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

4.7.9 Test Case 9. Sensor Fault Condition

The sensor fault condition is used to examine how the circuit behaves when the probe signals are abnormal. Possible abnormal conditions include an open probe wire, a shorted probe, a floating input, or an unrealistic state where the HIGH probe appears wet while the LOW probe appears dry.

A physically inconsistent state can be represented as:

$$S_{LOW} = 0$$

$$S_{HIGH} = 1$$

In a real tank, the HIGH probe should not be wet while the LOW probe is dry, because the HIGH probe is physically above the LOW probe. This condition therefore indicates a sensor fault, wiring error, or simulation input mistake.

The ideal safe response should be defined by the designer. A conservative design should stop the pump or generate an alarm under impossible sensor states. If the circuit does not include a fault detection mechanism, the report must clearly state that sensor fault detection is outside the present design scope.

This test case is academically important because it separates normal functional behavior from fault tolerant behavior. A circuit can work correctly under normal states but still require improvement under abnormal states.

Recommended expected result:

$$S_{LOW} = 0, \quad S_{HIGH} = 1$$

$$Fault = TRUE$$

$$PUMP = OFF$$

if fault safe behavior is implemented.

If fault safe behavior is not implemented, the report should state:

The present simulation verifies normal water level control, but it does not fully implement independent sensor fault diagnosis.

4.7.10 Test Case 10. Power Restart Condition

The power restart condition tests how the circuit behaves when power is removed and then restored. This condition is important because real water pump controllers may

experience power interruptions.

After power is restored, the circuit should not enter an undefined or unsafe state. The pump response should depend on the actual simulated water level condition. If the tank is empty, the pump should turn ON. If the tank is full, the pump should remain OFF. If the tank is between LOW and HIGH, the behavior must be clearly defined by the circuit design.

The power restart test checks whether the NE555 and relay return to a predictable state after supply restoration. A predictable restart condition is important because an undefined output could cause unwanted pump operation.

Expected restart logic:

If the tank is below LOW:

$$PUMP = ON$$

If the tank is at HIGH:

$$PUMP = OFF$$

If the tank is between LOW and HIGH, the result depends on the stored or reset condition of the control circuit after power restoration. If the circuit has no memory after power loss, the report must clearly state the actual observed restart behavior.

This test case is important because power interruption is a realistic condition in practical installations.

4.8 Expected Results

The expected result of the simulation is that the motor pump should operate according to the defined two threshold water level logic. The pump must turn ON when the water level is below the LOW probe. It must continue operating while the water level rises from LOW to HIGH. It must turn OFF when the water reaches the HIGH probe. It must remain OFF while the water level falls from HIGH toward LOW. It must turn ON again only after the water level falls below the LOW probe.

The relay must follow the control state. When the pump is required to run, Q1 must conduct and the relay coil must energize. When the pump is required to stop, Q1 must turn OFF and the relay coil must de energize.

The LED must follow its intended indication function. If the LED is used as a pump indicator, it should be ON when the pump is ON and OFF when the pump is OFF.

The expected voltage conditions are as follows:

When the pump is ON:

$$V_{OUT} \approx V_{CC}$$

$$V_{B(Q1)} \approx 0.7V$$

$$V_{relay} \approx V_{coil}$$

$$V_{motor} \approx 12V$$

When the pump is OFF:

$$V_{OUT} \approx 0V$$

$$V_{B(Q1)} \approx 0V$$

$$V_{relay} \approx 0V$$

$$V_{motor} \approx 0V$$

The expected behavior can be summarized as follows.

Table 4.1: Expected Simulation Behavior

Water level condition	LOW probe	HIGH probe	Expected relay state	Expected pump state
Tank empty	Dry	Dry	Energized	ON
Below LOW	Dry	Dry	Energized	ON
Reaches LOW	Wet	Dry	Energized	ON
Between LOW and HIGH during filling	Wet	Dry	Energized	ON
Reaches HIGH	Wet	Wet	De energized	OFF
Tank full	Wet	Wet	De energized	OFF
Falling below HIGH but above LOW	Wet	Dry	De energized	OFF
Below LOW again	Dry	Dry	Energized	ON
Impossible sensor state	Dry	Wet	Fault state required	Preferably OFF
Power restart	Depends on level	Depends on level	Defined by level state	Defined by level state

4.9 Observed Results

The observed simulation results showed that the circuit responded to the simulated water level changes according to the intended operating principle. When the simulated

tank condition represented a low water level, the NE555 output activated the relay driver stage. Q1 conducted, the relay coil energized, the relay contact closed, and the motor pump turned ON.

When the water level was simulated as reaching the LOW probe but not the HIGH probe, the pump remained ON. This confirmed that the LOW probe did not act as the stopping level. The circuit continued the filling operation until the upper level was reached.

When the water level was simulated as reaching the HIGH probe, the control state changed and the pump turned OFF. The relay de energized, the relay contact opened, and the motor pump stopped. This confirmed that the HIGH probe operated as the stopping threshold.

When the water level was simulated as falling below the HIGH probe while still remaining above the LOW probe, the motor pump remained OFF. This showed that the circuit did not immediately restart the pump after a small decrease in water level. The system therefore demonstrated practical two level control behavior.

When the water level was simulated as falling below the LOW probe again, the pump restarted. This confirmed that the circuit completed the full automatic cycle.

The LED indicator also responded according to the active and inactive states of the circuit. When the pump state was active, the indicator showed the corresponding active condition. When the pump state was inactive, the indicator changed accordingly.

The observed result therefore matched the expected normal operating behavior of the automatic water level controller.

4.10 Simulation Result Table

The result table shows that the circuit satisfies the main functional requirements under normal water level conditions. The pump starts at low level, continues during filling, stops at high level, remains off during partial water decrease, and restarts when the level falls below LOW again.

The table also shows that sensor fault behavior and power restart behavior require careful reporting. If the present circuit does not include explicit fault detection, then the fault test must be reported as a limitation rather than falsely presented as a completed safety feature.

4.11 Voltage Measurement Points

Voltage measurement points were included in the simulation to verify that the electrical behavior of the circuit matches the logical behavior. These measurements are necessary because a motor turning ON and OFF is not enough evidence by itself. The internal control states must also be checked.

The main voltage measurement points are the NE555 output voltage, Q1 base voltage, relay coil voltage, motor terminal voltage, LED voltage, and sensor node voltage.

Table 4.2: *Simulation Result Table*

Test case	Simulated condition	LOW probe state	HIGH probe state	Expected pump state	Observed pump state	Result
1	Tank empty	Dry	Dry	ON	ON	Pass
2	Water below LOW probe	Dry	Dry	ON	ON	Pass
3	Water reaches LOW probe	Wet	Dry	ON	ON	Pass
4	Water between LOW and HIGH during filling	Wet	Dry	ON	ON	Pass
5	Water reaches HIGH probe	Wet	Wet	OFF	OFF	Pass
6	Tank full	Wet	Wet	OFF	OFF	Pass
7	Water level falling below HIGH	Wet	Dry	OFF	OFF	Pass
8	LOW probe becomes dry again	Dry	Dry	ON	ON	Pass
9	Sensor fault condition	Abnormal	Abnormal	Safe state required	Not fully implemented unless added	Limited
10	Power restart condition	Depends on level	Depends on level	Defined by actual level	Must be recorded from simulation	Conditional

These points represent the complete path from control decision to final pump actuation.

4.11.1 NE555 Output Voltage

The NE555 output voltage is measured at pin 3. This voltage determines whether the relay driver transistor receives an active signal.

When the pump is expected to run, the NE555 output should be high:

$$V_{OUT} \approx V_{CC}$$

When the pump is expected to stop, the NE555 output should be low:

$$V_{OUT} \approx 0V$$

The NE555 output is therefore the central electrical evidence of the control state. If the output voltage is high but the motor does not run, the fault is likely in the transistor, relay, motor supply, or motor path. If the output voltage is low but the motor continues to run, the fault is likely in the relay contact, transistor leakage, wiring, or motor path.

4.11.2 Q1 Base Voltage

The Q1 base voltage is measured at the base terminal of the relay driver transistor. This voltage indicates whether Q1 is receiving sufficient drive to conduct.

When Q1 is ON, the base emitter junction should be forward biased:

$$V_{B(Q1)} \approx 0.7V$$

When Q1 is OFF, the base voltage should be near zero or below the conduction threshold:

$$V_{B(Q1)} \approx 0V$$

The Q1 base current can be estimated from the base resistor:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

where (V_{OUT}) is the NE555 output voltage, (V_{BE}) is the transistor base emitter voltage, and (R_B) is the base resistor.

This measurement confirms whether the NE555 output is actually able to drive the transistor stage.

4.11.3 Relay Coil Voltage

The relay coil voltage is measured across the relay coil terminals. This voltage confirms whether the relay is being energized.

When the pump is ON, the relay coil should receive its rated operating voltage:

$$V_{relay} \approx V_{coil}$$

When the pump is OFF, the relay coil voltage should be approximately zero:

$$V_{relay} \approx 0V$$

If the relay coil voltage is present but the relay contact does not switch, the relay model or relay connection may be incorrect. If the relay coil voltage is absent while the NE555 output is active, the fault may be in Q1, the base resistor, the relay coil path, or the supply connection.

The relay coil current is given by:

$$I_{relay} = \frac{V_{coil}}{R_{coil}}$$

This value must be checked during hardware design to ensure that Q1 can safely drive the relay.

4.11.4 Motor Terminal Voltage

The motor terminal voltage is measured across the motor pump terminals. This voltage confirms whether the relay contact is successfully applying power to the motor.

When the relay contact is closed:

$$V_{motor} \approx 12V$$

When the relay contact is open:

$$V_{motor} \approx 0V$$

This measurement is the final electrical confirmation of pump control. If the motor terminal voltage reaches approximately 12 V, the pump should run in the simulation. If the motor terminal voltage remains zero, the pump should remain OFF.

If the relay is energized but the motor terminal voltage remains zero, the contact wiring or motor supply path is incorrect. If the motor terminal voltage remains present while the relay is de energized, the relay contact may be incorrectly connected or stuck in the closed state.

4.11.5 LED Voltage

The LED voltage is measured across the LED terminals. This measurement confirms whether the indicator is operating correctly.

For a typical visible LED, the forward voltage is commonly near:

$$V_F \approx 2V$$

when the LED is ON. The exact voltage depends on the LED type used in the simulation.

The LED current is determined by the supply voltage and series resistor:

$$I_{LED} = \frac{V_{supply} - V_F}{R_{LED}}$$

If the LED current is too high, the resistor value should be increased. If the LED current is too low, the LED may not visibly indicate the state. In the simulation, the LED must clearly show the intended condition.

The LED voltage measurement is not the main proof of pump control, but it supports the interpretation of the circuit state.

4.11.6 Sensor Node Voltage

The sensor node voltage is measured at the probe related input points. This measurement confirms whether the simulated wet and dry probe conditions are producing valid electrical states.

When a probe is dry, the related node should be pulled to a defined voltage by the resistor network. It must not be floating. A floating sensor node can cause unstable or unpredictable control behavior.

When a probe is wet, the related node should change state due to the conductive path through the simulated water connection. The voltage change must be large enough to influence the transistor input or NE555 input correctly.

The sensor node must therefore satisfy two conditions. First, it must have a defined dry state. Second, it must have a defined wet state.

This can be expressed as:

$$V_{sensor(dry)} = \text{defined stable voltage}$$

$$V_{sensor(wet)} = \text{defined changed voltage}$$

If the sensor node does not change clearly between dry and wet conditions, the controller may fail to detect the water level accurately.

4.12 Simulation Screenshots

Simulation screenshots are required as visual evidence of the circuit operation. The screenshots must show the circuit in different states, not only one final schematic view. Each screenshot must include a clear caption and must be connected to a specific test case.

A proper simulation screenshot should show the relevant probe state, relay state, motor state, and indicator state. If possible, voltage probes or labels should also be visible. This makes the screenshot stronger as academic evidence.

4.12.1 Pump ON Screenshot

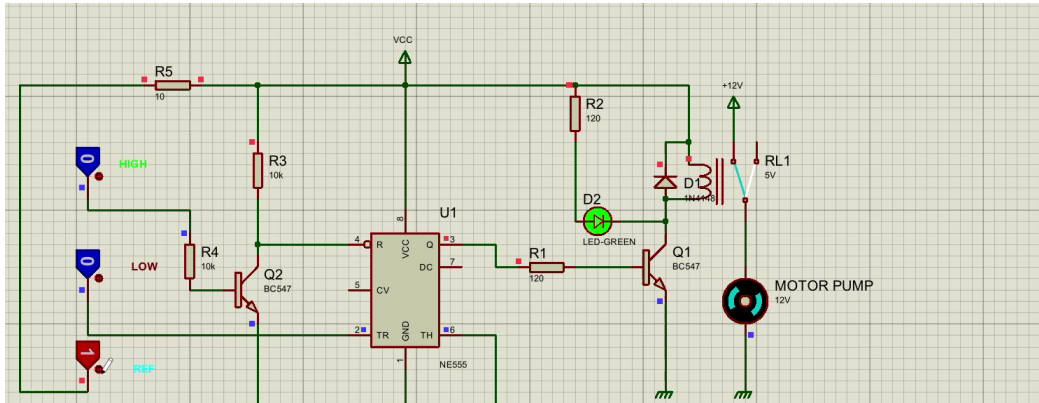


Figure 4.1: Pump ON Condition in Proteus Simulation

The pump ON screenshot must show the condition where the tank is below the LOW level. In this state, the LOW and HIGH probes are dry, the relay is energized, the relay contact is closed, and the motor pump is operating.

The caption should state:

The figure shows the pump ON condition when the simulated water level is below the LOW probe. The NE555 output activates the relay driver transistor, the relay coil is energized, and the motor pump receives supply voltage through the relay contact.

The screenshot must support the following condition:

$$S_{LOW} = 0$$

$$S_{HIGH} = 0$$

$$PUMP = ON$$

4.12.2 Pump OFF Screenshot

The pump OFF screenshot must show the condition where the water has reached the HIGH probe. In this state, the LOW and HIGH probes are wet, the relay is de energized, the relay contact is open, and the motor pump is stopped.

The caption should state:

The figure shows the pump OFF condition when the simulated water level reaches the HIGH probe. The control circuit changes state, the relay driver is switched OFF, the relay contact opens, and the motor pump is disconnected from the 12 V supply.

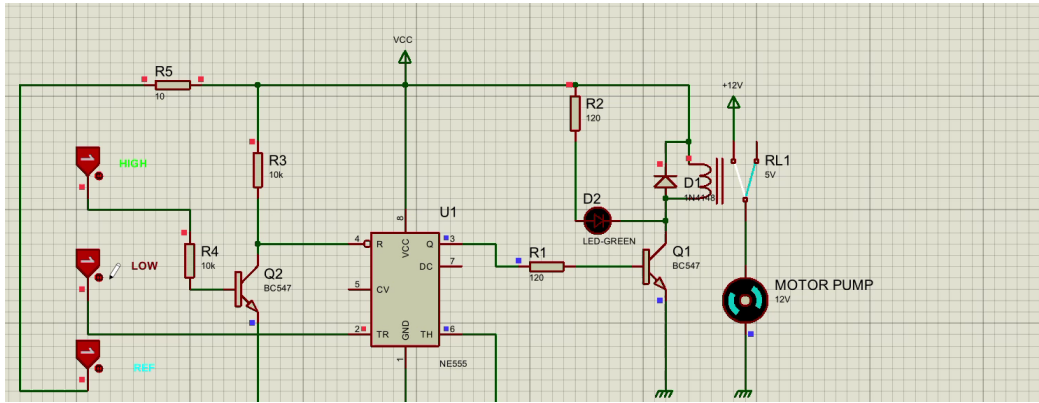


Figure 4.2: Pump OFF Condition in Proteus Simulation

The screenshot must support the following condition:

$$S_{LOW} = 1$$

$$S_{HIGH} = 1$$

$$PUMP = OFF$$

4.12.3 Full Circuit Screenshot

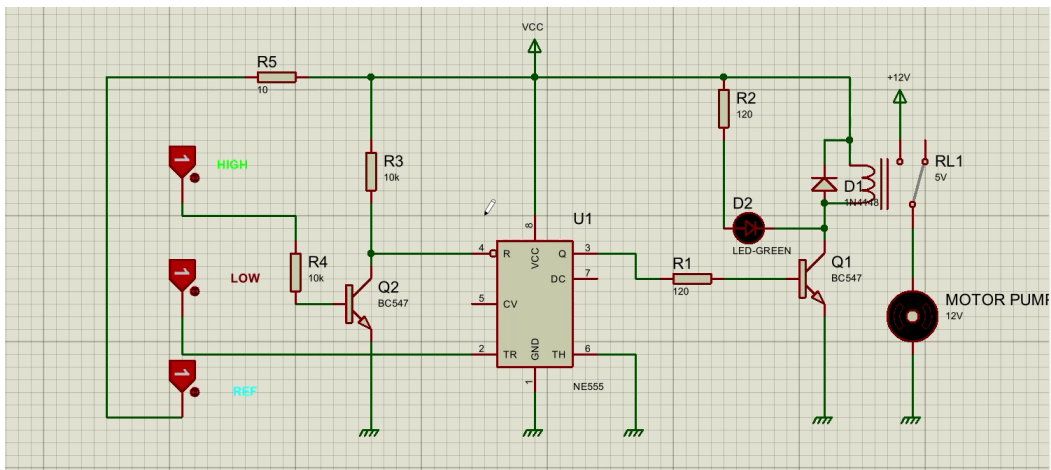


Figure 4.3: Full Circuit Simulation View

The full circuit screenshot must show the complete Proteus schematic. It should include the NE555, probes, transistors, resistors, relay, diode, LED, motor pump, power supplies, and ground connections.

The caption should state:

The figure shows the complete Proteus schematic of the automatic water level motor pump controller. The circuit includes the sensing section, NE555 control section,

transistor relay driver, relay contact switching section, motor pump load, and LED indicator.

This screenshot is necessary because it proves that the simulation was performed on the complete circuit rather than on isolated parts.

4.12.4 Probe State Screenshot

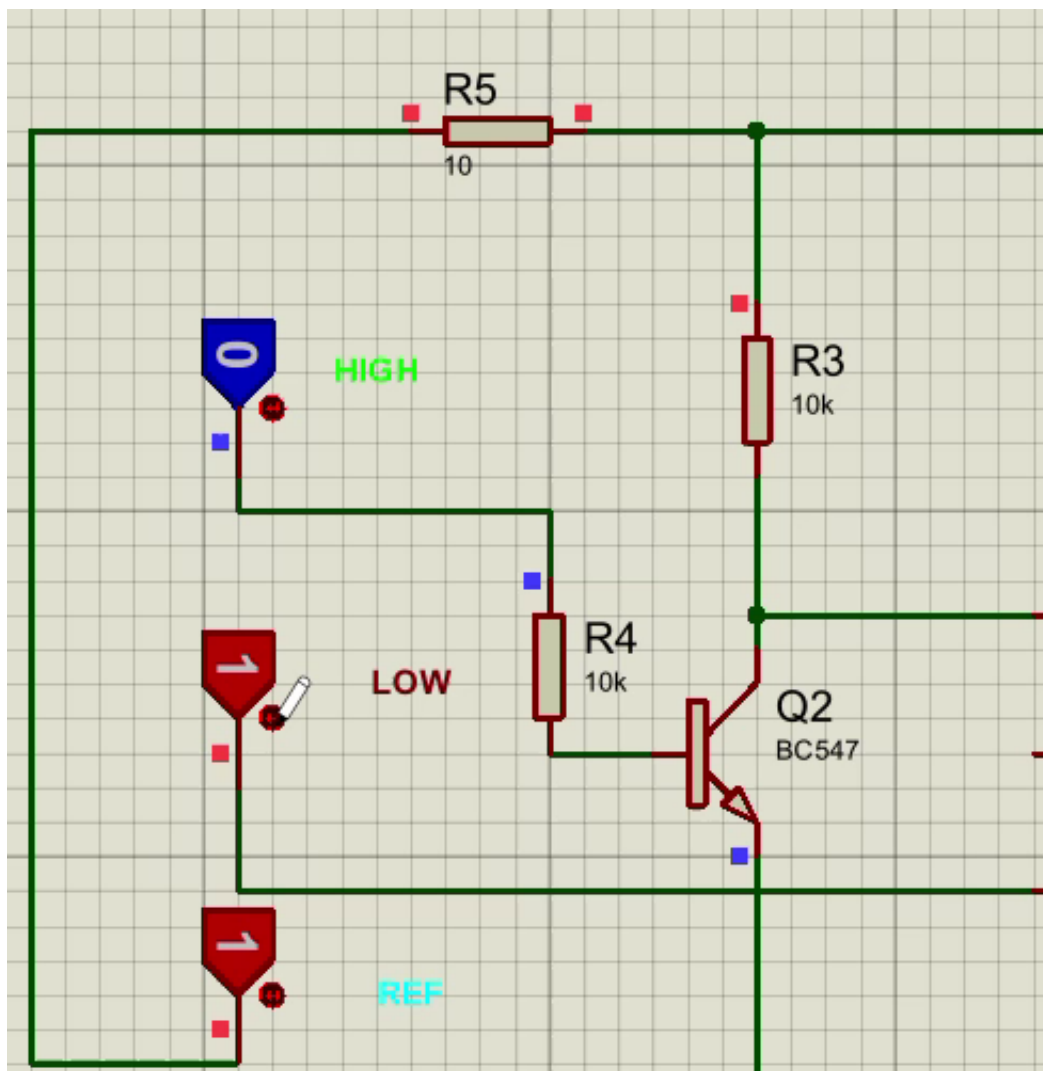


Figure 4.4: *Probe State During Simulation*

The probe state screenshot must show the simulated condition of the LOW, HIGH, and REF probes. It should clearly indicate whether each probe is wet or dry in the tested condition.

The caption should state:

The figure shows the simulated probe condition used to represent the water level inside the tank. The LOW and HIGH probe states determine whether the control circuit keeps the pump ON or switches the pump OFF.

This screenshot is important because the whole project depends on correct probe

interpretation.

4.12.5 Relay State Screenshot

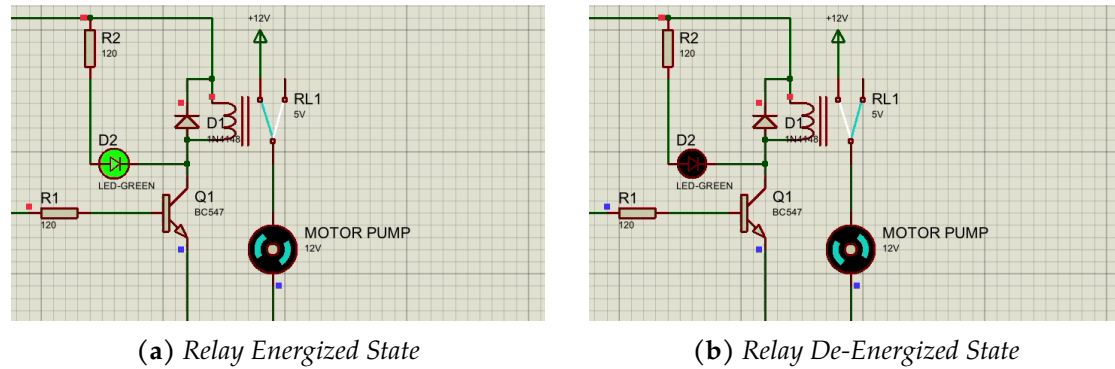


Figure 4.5: Relay Energized and De-Energized States in Simulation

The relay state screenshot must show whether the relay is energized or de energized. It should also show the resulting contact state and motor connection.

The caption should state:

The figure shows the relay state during simulation. When the relay coil is energized, the relay contact closes and the motor pump receives the 12 V supply. When the relay coil is de energized, the contact opens and the motor pump stops.

This screenshot confirms the electromechanical switching function of the design.

4.13 Analysis of Simulation Results

The simulation results confirm that the designed circuit performs the intended automatic water level control function under normal simulated operating conditions. The pump turns ON when the tank level is below the LOW probe and turns OFF when the water reaches the HIGH probe. This behavior satisfies the primary objective of the project.

The results also show that the LOW and HIGH probes perform different control roles. The LOW probe is associated with the restart level, while the HIGH probe is associated with the stopping level. This separation is necessary because a single level control point would create unstable switching near the threshold. The two probe arrangement creates a practical control band between the lower and upper water levels.

The relay driver stage also operates correctly in the simulation. When the NE555 output enters the active state, Q1 conducts and energizes the relay coil. When the NE555 output enters the inactive state, Q1 stops conducting and the relay releases. This confirms that the transistor relay driver performs its required switching function.

The relay contact correctly controls the motor pump supply. The motor receives voltage only when the relay is energized. This confirms that the motor is electrically controlled by the relay contact rather than by the NE555 output directly. This is an

important design feature because the motor requires more current than the NE555 can safely supply.

The LED indicator provides a useful visual confirmation of the system state. When the circuit is active, the LED indicates the corresponding condition. This improves the observability of the system during both simulation and possible hardware testing.

The voltage measurement points support the observed functional behavior. The NE555 output changes according to the required control state. The Q1 base voltage changes according to the transistor switching condition. The relay coil voltage appears when the relay should energize. The motor terminal voltage appears when the pump should run. The sensor node voltage changes according to the wet and dry probe conditions.

The simulation therefore validates the functional control logic of the circuit. However, the result must be interpreted correctly. The simulation proves that the circuit model behaves according to the expected logic. It does not prove that the physical circuit will remain reliable under all real conditions. Hardware implementation still requires checking the relay coil current, transistor current rating, motor starting current, wire connections, probe material, water conductivity, contact protection, and supply stability.

From a structural engineering view, the simulation is successful because the input states, internal control states, switching states, and output states are logically consistent. The circuit does not merely produce a motor response. It produces the correct motor response for the correct reason.

The validated control sequence is:

*Tank low $\Rightarrow$ Sensor state changes
 $\Rightarrow$ NE555 output active $\Rightarrow$ Q1 conducts*

*Q1 conducts $\Rightarrow$ Relay energized
 $\Rightarrow$ Relay contact closed $\Rightarrow$ Pump ON*

*Tank full $\Rightarrow$ HIGH probe active
 $\Rightarrow$ NE555 output inactive $\Rightarrow$ Q1 OFF*

*Q1 OFF $\Rightarrow$ Relay de energized $\Rightarrow$ Relay contact open
 $\Rightarrow$ Pump OFF*

This sequence proves that the circuit achieves the intended relationship between water level and pump control.

4.14 Simulation Limitations

Although the simulation verifies the functional operation of the circuit, it has several limitations. These limitations must be stated clearly because simulation is not identical

to physical implementation.

First, the simulation does not fully represent real water conductivity. In actual operation, water conductivity depends on dissolved minerals, impurities, temperature, and probe surface condition. Very pure water may conduct poorly, while mineral rich water may conduct better. The simulation simplifies this behavior into ideal electrical contact conditions.

Second, the simulation does not fully represent probe corrosion. Conductive probes exposed to DC current may experience corrosion or electrochemical effects over time. This can change the probe surface, increase resistance, and reduce sensing reliability. A real system should use suitable probe materials and may require low current, pulsed sensing, or alternating sensing methods to reduce corrosion.

Third, the simulation does not fully represent motor starting current. A real DC motor can draw a much larger current at startup than during normal running. The relay contact and power supply must be selected to handle this starting current. A simulation motor model may not accurately reproduce this stress.

Fourth, the simulation does not fully represent relay contact wear. In real hardware, relay contacts can spark, wear, bounce, or degrade when switching inductive loads. The motor side may require additional protection such as a diode, snubber, transient voltage suppressor, or fuse depending on the final motor type and supply arrangement.

Fifth, the simulation does not fully represent electrical noise. Real wires, relay coils, motor brushes, and long sensor cables can introduce noise into the circuit. This noise may cause false triggering if the input nodes are not properly filtered and biased.

Sixth, the simulation does not fully represent mechanical installation conditions. Probe placement, tank material, water turbulence, vibration, humidity, and wiring quality can affect the final performance of the system.

Seventh, the simulation does not prove thermal safety. The transistor, relay coil, resistor values, and motor path must be checked for power dissipation in real hardware. A component may appear to work in simulation while overheating in physical use.

Eighth, the simulation does not prove full safety for mains powered pumps. If the final pump uses AC mains voltage instead of a low voltage DC motor, additional safety requirements are necessary. These include proper insulation, relay contact rating, fuse protection, enclosure design, grounding, and safe wiring practice.

Ninth, the simulation does not fully prove fault tolerance. Abnormal conditions such as broken probes, shorted probes, stuck relay contacts, disconnected motor supply, and power restart behavior require additional tests. If these protections are not included in the present design, they must be listed as future improvements.

Therefore, the simulation should be understood as functional verification, not final certification. It proves that the designed logic is correct in the Proteus model, but it does not remove the need for careful hardware testing.

4.15 Chapter Summary

This chapter presented the simulation of the automatic water level motor pump controller using Proteus. The simulation verified the main operating principle of the circuit. The pump turned ON when the water level was below the LOW probe and turned OFF when the water reached the HIGH probe. The pump remained OFF while the water level fell from HIGH toward LOW and restarted when the LOW probe became dry again.

The simulation confirmed the operation of the main circuit sections. The probe section provided the input condition. The NE555 control section produced the required output state. The transistor driver controlled the relay coil. The relay contact switched the motor pump supply. The LED indicator showed the circuit state.

The simulation test cases covered the main water level conditions, including empty tank, water below LOW, water reaching LOW, water between LOW and HIGH, water reaching HIGH, full tank, falling water level, LOW probe becoming dry again, sensor fault condition, and power restart condition. The normal operating test cases confirmed that the circuit follows the intended two threshold control logic.

The voltage measurement points provided internal evidence for the circuit behavior. The NE555 output voltage, Q1 base voltage, relay coil voltage, motor terminal voltage, LED voltage, and sensor node voltage were identified as the most important points for verification.

The chapter also explained the limitations of the simulation. Proteus verifies the model level behavior, but real hardware testing is still required for water conductivity, probe corrosion, relay contact protection, motor starting current, thermal safety, electrical noise, and fault tolerance.

The final result of the simulation chapter is that the project is functionally valid at the simulation level. The circuit demonstrates the required automatic pump control behavior and is suitable for further development, provided that the identified hardware limitations are addressed before real implementation.

5

Troubleshooting

5.1 Introduction

Troubleshooting is an essential part of the design, simulation, and validation process of the automatic water-level motor-pump controller. A circuit may appear correct in its schematic form and may also operate successfully under ideal simulation conditions, but this does not guarantee that it will operate reliably under every practical condition. The purpose of this chapter is therefore to identify, classify, analyze, and correct the possible faults that may occur in the water-level controller system.

The troubleshooting process in this project is concerned with the complete functional chain of the system. This chain begins with the water-level probes, continues through the NE555 control section, passes through the transistor relay-driver stage, energizes or de-energizes the relay, and finally controls the motor pump. If any part of this chain fails, the final output of the system may become incorrect. The pump may fail to start, fail to stop, run continuously, never run, or operate unpredictably.

The circuit contains several sections that must work together correctly. These sections include the sensing section, the control section, the switching section, the relay section, the motor load section, the indication section, and the power supply section. Troubleshooting must therefore be systematic. A fault should not be assumed to belong to one component before the complete signal path is examined.

In this project, the main symptoms considered are pump failure, relay failure, false water-level sensing, transistor overheating, relay chattering, motor-side protection problems, and power supply instability. Each fault is analyzed through three levels: possible causes, diagnostic tests, and corrective actions. This structure ensures that the report does not merely list problems, but explains how each problem can be identified and corrected.

The troubleshooting chapter is especially important because the project uses conductive water probes, a relay, a transistor driver, and a motor pump. These parts are more sensitive to real-world effects than ideal simulation models. Water conductiv-

ity, probe corrosion, relay coil current, motor starting current, supply noise, and poor grounding can all affect the behavior of the system. Therefore, a complete academic report must include a precise troubleshooting section.

5.2 General Troubleshooting Method

The troubleshooting method for this project must follow the actual direction of operation of the circuit. The correct diagnostic direction is from the power supply to the control section, from the control section to the relay-driver section, from the relay-driver section to the relay, and from the relay to the motor pump. The sensor inputs must also be checked because they determine whether the control section should command the pump to start or stop.

The general troubleshooting sequence should begin with visual inspection. All wires, component positions, relay terminals, transistor pins, diode polarity, resistor values, and power supply connections must be checked before electrical testing begins. Many circuit failures occur because of incorrect wiring, reversed polarity, wrong transistor pin connection, incorrect relay contact selection, or open ground connection.

After visual inspection, the power supply should be tested. The control circuit supply voltage must be measured at the NE555 supply pin and ground pin. The motor supply voltage must be measured at the motor-side relay contact path. If the power supply is incorrect, all later measurements become unreliable because the circuit cannot operate correctly without stable voltage.

The next step is to test the sensor section. The HIGH probe, LOW probe, and REF probe must be checked to confirm that they produce the expected electrical condition when water is present or absent. The water probes should not be treated as simple wires only. They are part of the input sensing system, and their behavior depends on water conductivity, contact area, distance between probes, resistor values, and input biasing.

After the sensor section is confirmed, the NE555 control section must be tested. The voltage at the trigger input, reset input, and output pin should be measured under different water-level states. The output of the NE555 should change according to the logic of the system. If the NE555 output does not respond to valid input changes, the fault may be in the input network, reset network, supply connection, or the IC itself.

The relay-driver transistor must then be tested. The base voltage, collector voltage, emitter connection, and relay coil voltage must be examined. The transistor must enter saturation when the relay is required to energize. If it does not saturate, the relay may not activate even when the NE555 output is correct.

The relay must be tested separately from the motor. The coil side and contact side must not be confused. A relay may have a working coil but a wrongly connected contact. It may also have correct contact wiring but insufficient coil current. Therefore, the relay coil voltage and contact continuity must both be examined.

Finally, the motor pump circuit must be tested. The motor voltage, motor current,

relay contact rating, and motor supply capacity must be verified. The motor may fail to run even if the control circuit is working correctly. This can happen if the motor supply is missing, the relay contact is incorrectly wired, the motor is damaged, or the relay contact current rating is insufficient.

A reliable troubleshooting method must follow this order:

Inspect all physical and schematic connections.

Verify the control supply voltage.

Verify the motor supply voltage.

Test the water-level probes.

Test the NE555 input pins.

Test the NE555 output pin.

Test the transistor driver.

Test the relay coil.

Test the relay contacts.

Test the motor pump load.

Confirm protection components.

Confirm the final system response under all water-level states.

The general rule is that the fault must be located by measurement, not by guessing. A correct academic troubleshooting process must always connect the observed symptom to a measurable electrical condition.

5.3 Pump Does Not Start

The pump does not start when the system fails to activate the motor even though the water level is below the required minimum level. In the intended operation of the project, the pump should start when the tank water level falls below the LOW probe. If this does not happen, the fault may exist in the sensing section, the NE555 control section, the relay-driver stage, the relay, the motor supply path, or the motor itself.

This symptom must be analyzed carefully because the pump is the final output of the entire control chain. A pump-start failure does not automatically mean that the pump is damaged. It may simply mean that the command signal is not reaching the motor-control stage.

5.3.1 Possible Causes

One possible cause is that the control circuit is not receiving the correct supply voltage. If the NE555 does not receive proper voltage between its supply pin and ground pin, it cannot produce a valid output signal. A weak or disconnected supply will prevent the whole system from operating.

Another possible cause is that the LOW probe is not detecting the low water condition correctly. If the LOW probe, REF probe, or their connecting wires are open, the

NE555 may not receive the correct trigger condition. The control circuit may therefore remain in the OFF state even when the tank is empty.

A further cause is an incorrect trigger condition at the NE555 input. The NE555 requires a valid input transition according to the circuit design. If the trigger input remains at the wrong voltage level, the output will not switch to the pump-start state.

The relay-driver transistor may also be faulty or incorrectly connected. If the transistor base does not receive sufficient current, the relay coil cannot be energized. If the emitter, base, and collector pins of the BC547 transistor are connected incorrectly, the transistor may not switch even when the NE555 output is correct.

The relay coil may be open, incorrectly rated, or not connected to the supply. If the relay coil does not receive sufficient voltage and current, the relay contacts will not change position. In this case, the motor circuit remains open.

The motor pump supply may also be missing. The control circuit may activate the relay correctly, but if the 12 V motor supply is not present at the relay contact side, the motor will not run.

A final possible cause is that the motor pump itself is faulty. A damaged motor, jammed shaft, open internal winding, or insufficient starting current can prevent pump operation.

5.3.2 Diagnostic Tests

The first diagnostic test is to measure the control supply voltage at the NE555. The voltage between the supply pin and ground pin must be within the required operating range of the IC. If the expected control supply is 5 V, the measured voltage should be close to 5 V.

The second test is to measure the NE555 output voltage when the water level is below the LOW probe. The output pin should move to the state that commands the relay-driver transistor to turn ON. If the output voltage does not change, the fault is likely before or inside the NE555 control section.

The third test is to measure the voltage at the LOW probe and REF probe. The measurement should confirm whether the sensor input is producing the expected state. If the probes do not create a valid signal, the input section must be repaired.

The fourth test is to measure the voltage at the base of the relay-driver transistor. If the base voltage is approximately high enough to forward-bias the base-emitter junction when the pump should start, the transistor should conduct. For a silicon transistor, the base-emitter voltage during conduction is usually approximately

$$V_{BE} \approx 0.7V$$

If the base voltage is too low, the transistor will not switch the relay coil.

The fifth test is to measure the relay coil voltage. If the relay coil is rated for 5 V, the voltage across the coil should be close to 5 V when the relay is commanded ON. If the

relay coil voltage is absent, the fault may be in the transistor, supply connection, relay wiring, or diode connection.

The sixth test is to check relay contact continuity. When the relay is energized, the normally open contact should close. This can be tested with a continuity meter or by measuring voltage across the motor terminals.

The seventh test is to measure the motor terminal voltage. If the motor is rated for 12 V, the motor terminal voltage should be close to 12 V when the pump is commanded ON. If voltage is present but the motor does not rotate, the motor or pump mechanism is faulty.

5.3.3 Corrective Actions

If the control supply is missing or low, the power supply must be corrected before any other repair is attempted. The supply wires, regulator, battery, adapter, or simulation source must be checked.

If the probe input is not working, the LOW probe and REF probe wiring must be repaired. The probe connections should be cleaned, tightened, and correctly placed. The sensor resistor values should also be verified.

If the NE555 output does not respond, the trigger and reset connections must be checked. The NE555 should also be replaced if all external conditions are correct but the output remains invalid.

If the transistor does not switch, the base resistor, transistor orientation, and transistor condition must be checked. The base current must be sufficient for saturation. The required base current can be estimated as

$$I_B = \frac{I_C}{\beta_{forced}}$$

where (I_C) is the relay coil current and (β_{forced}) is the forced gain used for saturation. A conservative value is often

$$\beta_{forced} = 10$$

The base resistor may then be estimated as

$$R_B = \frac{V_{OUT} - V_{BE}}{I_B}$$

If the relay coil is faulty, the relay must be replaced with one that has the correct coil voltage and contact rating. If the relay contacts are wired incorrectly, the wiring must be corrected so that the motor supply passes through the proper contact pair.

If the motor supply is absent, the motor supply path must be repaired. If the motor receives correct voltage but does not rotate, the motor pump must be tested independently and replaced if necessary.

5.4 Pump Does Not Stop

The pump does not stop when the motor continues running even after the water level reaches the HIGH probe. In the intended operation of the project, the pump must stop when the tank reaches the upper water-level limit. Failure to stop is a serious fault because it can cause water overflow, energy waste, and possible pump or relay stress.

This symptom usually indicates a failure in the high-level sensing path, the reset path of the NE555, the relay-driver stage, or the relay contact mechanism. It may also occur if the relay contacts become mechanically stuck.

5.4.1 Possible Causes

One possible cause is that the HIGH probe is not detecting water. If the HIGH probe is disconnected, corroded, incorrectly placed, or not electrically connected through water to the REF probe, the stop command will not be generated.

Another possible cause is that the reset input of the NE555 is not receiving the correct voltage condition. The reset input must be correctly biased so that the circuit can force the output into the pump-stop state when the HIGH level is reached.

The input transistor related to the HIGH probe may also fail. If Q2 does not switch correctly, the NE555 may not receive the stop signal. This may be caused by incorrect resistor values, a faulty transistor, wrong pin orientation, or insufficient water conductivity.

The relay-driver transistor may be shorted. If Q1 is shorted between collector and emitter, the relay coil may remain energized even when the NE555 output is low.

The relay contacts may be stuck closed. Mechanical relays can fail if their contacts weld together due to excessive motor current, motor inrush current, arcing, or inadequate contact rating.

The relay may also be incorrectly wired through the normally closed contact instead of the normally open contact. In that case, the motor may remain powered when the relay is not in the intended state.

5.4.2 Diagnostic Tests

The first test is to verify the HIGH probe. When water touches the HIGH probe and REF probe, the voltage at the relevant sensor node should change according to the design. If there is no voltage change, the sensor path is faulty.

The second test is to measure the NE555 reset input. When the water reaches the HIGH probe, the reset pin should move to the state that forces the pump OFF. If the reset pin does not change, the fault is in the sensor or reset interface section.

The third test is to measure the NE555 output voltage. When the HIGH level is reached, the output should move to the pump-stop state. If the output remains in the pump-start state, the NE555 is not receiving the correct stop condition.

The fourth test is to measure the relay coil voltage after the HIGH probe is activated. If the relay coil remains energized even though the NE555 output is low, Q1 may be shorted, incorrectly wired, or receiving unintended base current.

The fifth test is to disconnect the control signal from the relay driver and observe whether the relay releases. If the relay remains energized without a valid drive signal, the driver or relay wiring is faulty.

The sixth test is to test relay contact continuity when the relay is de-energized. If the normally open contact remains closed when the coil is OFF, the contact is mechanically stuck or welded.

5.4.3 Corrective Actions

If the HIGH probe is not working, its wiring and physical position must be corrected. The probe must be placed at the desired maximum water level and must have a reliable conductive path through water to the REF probe.

If the reset input does not receive the correct voltage, the reset network must be corrected. Pull-up or pull-down resistors must be checked so that the input does not float. A floating reset input can cause unpredictable operation.

If Q2 is faulty, it must be replaced. Its base resistor and collector resistor must be checked. The transistor pinout must be verified because wrong pin connection is a common cause of switching failure.

If Q1 is shorted, Q1 must be replaced. The relay flyback diode must also be inspected because absence or wrong polarity of the diode can expose the transistor to inductive voltage spikes.

If relay contacts are welded, the relay must be replaced with one that has a higher current rating. The selected relay contact current rating must be greater than the motor running current and should also tolerate starting current. A conservative selection condition is

$$I_{\text{contact, rated}} \geq I_{\text{motor, inrush}}$$

If the relay is incorrectly wired, the contact terminals must be corrected so that the motor is powered only when the intended contact closes.

5.5 Relay Does Not Activate

The relay does not activate when the relay coil fails to energize even though the control circuit should command the pump to run. This fault is different from a motor fault because the relay coil belongs to the control switching stage, while the motor belongs to the load stage. If the relay does not activate, the motor will not receive power through the relay contact.

5.5.1 Possible Causes

The NE555 output may not be producing the required relay-drive signal. If the output remains low when it should be high, the transistor driver will not turn ON.

The base resistor of the relay-driver transistor may be too large, disconnected, or connected to the wrong node. In this condition, the base current becomes insufficient.

The transistor may be incorrectly connected. The BC547 transistor has specific collector, base, and emitter pins. If these pins are reversed, the relay coil may not receive current.

The relay coil may be open. An open relay coil has very high resistance and cannot conduct current.

The relay coil voltage rating may not match the circuit supply. A 12 V relay coil may not activate properly from a 5 V control supply. Similarly, a 5 V relay coil may be damaged if connected directly to an excessive voltage.

The flyback diode may be connected with wrong polarity. If the diode is connected in forward direction across the relay coil during normal operation, it may short the supply and prevent the relay from activating.

The supply may not be able to provide the relay coil current. A weak supply may show correct voltage without load but collapse when the relay attempts to energize.

5.5.2 Diagnostic Tests

The first test is to measure the NE555 output voltage when the relay should be active. If the output is not in the expected state, the fault is before the relay-driver stage.

The second test is to measure the base voltage of Q1. The voltage between base and emitter should be approximately

$$V_{BE} \approx 0.7V$$

when Q1 is ON. If the base voltage is too low, Q1 will not conduct.

The third test is to calculate the expected base current:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

This current must be high enough to saturate the transistor for the required relay coil current.

The fourth test is to measure the voltage across the relay coil. If the relay coil receives its rated voltage but does not click or switch, the relay may be defective.

The fifth test is to measure relay coil resistance with power removed. If the resistance is extremely high, the coil is open. If it is extremely low, the coil may be shorted.

The sixth test is to inspect the diode polarity. The diode should be reverse-biased during normal relay operation and forward-biased only when the relay coil is switched OFF.

5.5.3 Corrective Actions

If the NE555 output is incorrect, the input and reset conditions of the NE555 must be corrected before replacing the relay.

If the base current is insufficient, the base resistor must be recalculated. The transistor must be driven into saturation so that the relay receives enough coil current.

If the transistor is incorrectly oriented, it must be reconnected according to its correct pin arrangement. If it has been damaged, it must be replaced.

If the relay coil voltage rating is wrong, the relay must be replaced with a coil rating compatible with the control supply.

If the flyback diode is reversed, it must be corrected immediately. A wrongly connected diode can cause excessive current and may damage the supply or transistor.

If the supply voltage collapses when the relay activates, a supply with higher current capacity must be used. A bulk capacitor may also be added across the control supply to reduce voltage dip during relay activation.

5.6 Relay Chatters Continuously

Relay chattering is a condition in which the relay rapidly switches ON and OFF instead of remaining stable. This is a serious fault because it can damage relay contacts, disturb the motor operation, increase electrical noise, and reduce the lifetime of the circuit.

Relay chattering usually indicates instability in the control signal, insufficient supply current, poor grounding, noisy sensor input, weak relay coil voltage, or absence of proper filtering.

5.6.1 Possible Causes

One possible cause is insufficient power supply current. When the relay coil energizes, it draws current from the supply. If the supply cannot provide this current, the voltage may drop. This voltage drop can reset or disturb the NE555, causing the relay to release. When the relay releases, the voltage rises again, and the cycle repeats.

Another cause is lack of supply decoupling. Without a capacitor near the NE555, switching transients from the relay may enter the control circuit and cause false triggering.

A third cause is sensor noise. Long probe wires, poor water contact, weak conductivity, or floating inputs can produce unstable control signals.

A fourth cause is relay coil voltage mismatch. If the relay requires a higher voltage than the control circuit provides, it may partially energize and release repeatedly.

Poor grounding can also produce relay chatter. If relay coil current and NE555 input reference current share a weak ground path, voltage drops in the ground line may disturb the control logic.

Motor noise may also feed back into the control circuit. A motor is an inductive and mechanical load that can produce switching noise, especially during starting and

stopping.

5.6.2 Diagnostic Tests

The first test is to measure the control supply voltage while the relay chatters. If the supply voltage drops significantly when the relay attempts to activate, the supply is insufficient.

The second test is to measure the relay coil voltage during chatter. If the voltage repeatedly rises and falls, the coil is not receiving a stable drive condition.

The third test is to observe the NE555 output voltage. If the output oscillates unexpectedly, the fault is in the control input, reset path, supply stability, or noise filtering.

The fourth test is to temporarily disconnect the motor load and observe whether relay chatter stops. If chatter stops when the motor is removed, motor noise or motor current demand is contributing to the problem.

The fifth test is to inspect the grounding path. The ground connection of the NE555 should be stable and should not carry large motor current.

The sixth test is to place a capacitor across the control supply and observe whether the chatter decreases. If it improves, the problem is related to supply instability or transient noise.

5.6.3 Corrective Actions

The supply current capacity must be increased if the voltage drops during relay operation. The supply should be selected with sufficient current margin. A reasonable condition is

$$I_{supply,rated} \geq 1.5 \times I_{total}$$

where (I_{total}) includes the NE555 current, relay coil current, LED current, transistor base current, and sensor current.

A ceramic capacitor of approximately 0.1 microfarad should be placed close to the NE555 supply pins. A larger capacitor, such as 10 microfarads to 100 microfarads, may be added across the control supply to reduce relay switching dips.

Floating inputs must be removed by using correct pull-up or pull-down resistors. Sensor inputs should not be left undefined.

The relay coil voltage must match the supply voltage. If a relay requires 12 V, it should not be expected to operate reliably at 5 V.

The motor supply and control supply should be routed carefully. The high-current motor path should not share thin or unstable ground paths with the sensitive NE555 input section.

If the motor creates excessive noise, a diode, capacitor, snubber, or transient suppression device should be added according to the motor type and supply condition.

5.7 LED Turns ON but Motor Does Not Run

This fault occurs when the indicator LED shows that the control circuit is active, but the motor pump does not operate. This symptom is important because it proves that at least part of the control circuit is functioning, but the load path is not completing correctly.

The LED may indicate NE555 output activity, relay-driver activity, or relay state depending on its exact connection. Therefore, the LED should not be treated as absolute proof that the motor circuit is powered.

5.7.1 Possible Causes

The relay coil may be energized, but the relay contact may be connected incorrectly. A relay has separate coil terminals and contact terminals. If the motor is connected to the wrong contact, it may not receive power.

The relay contact may be faulty. The coil may operate, but the internal contact may not close due to mechanical damage.

The motor supply may be absent. The LED may operate from the control supply, while the motor requires a separate 12 V supply.

The motor may be damaged or jammed. If voltage reaches the motor but the shaft does not rotate, the motor itself may be faulty.

The relay contact rating may be insufficient. If the motor requires more current than the relay contact can safely carry, the contact may fail or produce excessive voltage drop.

A wiring break may exist in the motor path. The circuit from the 12 V source through the relay contact and motor back to ground must be complete.

5.7.2 Diagnostic Tests

The first test is to identify what the LED actually represents. If the LED is connected to the NE555 output, it shows only control output activity. If it is connected across the relay coil, it shows relay coil activation. If it is connected near the motor path, it may show load-side power.

The second test is to measure the relay coil voltage. If the relay coil is not energized, the LED is not proving relay operation.

The third test is to listen for the relay click or measure contact continuity. A relay click indicates movement, but continuity testing gives stronger proof.

The fourth test is to measure voltage across the motor terminals. If the motor terminal voltage is zero, the fault is in the relay contact path or motor supply path.

The fifth test is to measure the 12 V supply directly. The supply may be present in the schematic but not connected to the motor path.

The sixth test is to connect the motor briefly to its rated supply independently, under safe conditions. If the motor still does not rotate, the motor is faulty.

5.7.3 Corrective Actions

The relay contact wiring must be corrected if the motor is connected to the wrong terminal. The normally open contact should be used if the motor is intended to run only when the relay is energized.

If the relay coil is not operating, the relay-driver stage must be repaired. If the relay coil operates but the contact does not close, the relay should be replaced.

If the motor supply is absent, the 12 V source and its ground return must be connected correctly.

If the motor is damaged, the motor must be replaced with one matching the voltage and current rating of the circuit.

If the relay contact rating is insufficient, a stronger relay must be selected. The relay contact must support the motor running current and starting current.

5.8 Motor Runs Continuously

A continuously running motor is a dangerous fault because it defeats the purpose of automatic water-level control. If the motor continues to operate regardless of the water level, the tank may overflow, the pump may be stressed, and the system may waste energy.

This fault must be distinguished from the pump-does-not-stop symptom. Pump-does-not-stop may occur only after filling. Motor-runs-continuously means the motor remains powered under all or almost all conditions.

5.8.1 Possible Causes

The relay contact may be stuck closed. This can occur if the contact has welded due to excessive motor current or arcing.

The motor may be wired directly to the supply instead of through the relay contact. In this case, the control circuit has no authority over the motor.

The relay may be connected through the normally closed contact instead of the normally open contact. This would cause the motor to run when the relay is not energized.

The NE555 output may be permanently high because the reset signal is not functioning.

The relay-driver transistor may be shorted, keeping the relay coil continuously energized.

The HIGH probe may not be able to generate the stop command because of corrosion, open wiring, incorrect placement, or poor conductivity.

5.8.2 Diagnostic Tests

The first test is to remove the relay coil drive signal and observe whether the motor stops. If the motor continues running with the relay not energized, the fault is on the

relay contact side or motor wiring side.

The second test is to measure the relay coil voltage while the motor runs continuously. If the relay coil is continuously energized, the fault is in the control or driver section.

The third test is to test the relay contact continuity with the relay de-energized. If the normally open contact remains closed, the relay contact is stuck.

The fourth test is to check whether the motor is connected to the correct relay contact. The motor should not be permanently connected to the supply.

The fifth test is to measure the NE555 output. If the output remains in the pump-start state even when the HIGH probe is activated, the fault is in the sensing or reset path.

The sixth test is to test Q1 for a collector-to-emitter short. A shorted transistor can energize the relay regardless of the NE555 output.

5.8.3 Corrective Actions

If the relay contact is stuck, the relay must be replaced. The new relay must have a contact rating suitable for the motor load.

If the motor is directly connected to the supply, the wiring must be corrected so that the relay contact controls the motor current path.

If the normally closed contact is being used incorrectly, the wiring must be moved to the normally open contact.

If the NE555 output remains high, the HIGH probe input and reset network must be repaired.

If Q1 is shorted, it must be replaced. The flyback diode and relay coil current must also be checked to prevent repeated transistor damage.

If the HIGH probe is faulty, it must be cleaned, repositioned, or replaced.

5.9 Motor Never Runs

The motor-never-runs fault occurs when the motor does not operate under any condition, even when the tank is empty and the system should command the pump to start. This fault is broader than the pump-does-not-start condition because it includes every possible state in which motor operation is absent.

5.9.1 Possible Causes

The motor supply may be completely absent. If the motor requires 12 V and the 12 V supply is not connected, the motor cannot operate.

The relay contact path may be open. This can occur due to incorrect relay wiring, broken wires, damaged contact, or wrong contact selection.

The relay may never energize. If the relay coil is not driven, the motor path remains open.

The control circuit may never produce the pump-start signal. The NE555 output may remain low due to input fault, reset fault, or IC damage.

The motor may be defective. An open motor winding, seized shaft, or damaged internal brush assembly can prevent operation.

The circuit ground return may be incomplete. If the motor or control circuit lacks a proper return path, current cannot flow.

5.9.2 Diagnostic Tests

The first test is to measure the motor supply voltage at the source. The measured voltage must match the motor rating.

The second test is to measure the voltage directly across the motor terminals during the pump-start condition. If rated voltage appears at the motor but it does not rotate, the motor is faulty.

The third test is to check continuity of the motor path from the positive supply through the relay contact, through the motor, and back to ground.

The fourth test is to measure the relay coil voltage during the pump-start condition. If the coil is not energized, the problem is before the contact path.

The fifth test is to measure the NE555 output. If the output never changes to the pump-start state, the control logic is not working.

The sixth test is to test the motor independently using a suitable supply and safe current limit.

5.9.3 Corrective Actions

If the motor supply is absent, the supply must be connected and verified. The supply must provide the required voltage and current.

If the relay contact path is open, the wiring must be corrected. The correct relay terminals must be identified and used.

If the relay does not energize, the relay-driver stage must be repaired.

If the NE555 output never changes, the input probe logic, reset circuit, supply voltage, and IC condition must be checked.

If the motor is defective, it must be replaced.

If the ground path is incomplete, the ground return must be repaired. All required grounds in the control section must be common where the design requires a shared reference.

5.10 Transistor Overheating Problem

Transistor overheating occurs when the relay-driver transistor or sensor transistor dissipates more power than it can safely handle. In this project, the most likely overheating component is the relay-driver transistor because it controls the relay coil current.

Overheating is not only a component problem. It is a design warning. It may indicate excessive relay coil current, insufficient base drive, wrong resistor value, wrong transistor orientation, absence of flyback protection, or accidental connection of the motor load through the transistor.

5.10.1 Possible Causes

One possible cause is excessive collector current. If the relay coil current is greater than the safe current rating of the transistor, the transistor may overheat.

Another cause is insufficient base current. If the transistor is not fully saturated, it may operate in a partially ON state. In that condition, the collector-emitter voltage remains higher, and power dissipation increases.

A third cause is an incorrect base resistor. If the base resistor is too large, the transistor may not saturate. If it is too small, excessive base current may stress the NE555 output and transistor base.

The flyback diode may be absent or reversed. Without a correct flyback diode, the transistor may experience high voltage spikes when the relay coil turns OFF.

The motor load may be incorrectly connected through the transistor instead of through the relay contact. A BC547 transistor is not suitable for directly switching a motor pump load.

Wrong transistor pin connection can also cause overheating. If the collector and emitter are reversed, the transistor may conduct poorly and dissipate excessive power.

5.10.2 Diagnostic Tests

The first test is to measure relay coil current. The relay coil current can be estimated from the coil resistance:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

The calculated current must be compared with the transistor current rating.

The second test is to measure the collector-emitter voltage when the transistor is ON. A saturated transistor should have a low collector-emitter voltage. If (V_{CE}) is high while the relay is energized, the transistor is not properly saturated.

The third test is to calculate transistor power dissipation:

$$P_Q = V_{CE} \times I_C$$

where (P_Q) is transistor power dissipation, (V_{CE}) is collector-emitter voltage, and (I_C) is collector current.

The fourth test is to calculate required base current:

$$I_B = \frac{I_C}{10}$$

This uses a forced gain of 10 for reliable saturation.

The fifth test is to calculate the actual base current:

$$I_B = \frac{V_{OUT} - V_{BE}}{R_B}$$

The actual base current must be sufficient for saturation but not excessive.

The sixth test is to inspect whether the motor current passes through the relay contact or through the transistor. The motor current must not pass directly through the BC547 transistor.

The seventh test is to inspect the diode across the relay coil. The diode must be reverse-biased during normal relay operation.

5.10.3 Corrective Actions

If relay coil current is too high, the relay must be replaced with a lower-current relay or the driver transistor must be replaced with a higher-current transistor or driver stage.

If the transistor is not saturated, the base resistor must be recalculated. The design should provide enough base current to place the transistor in saturation.

If the base resistor is too small, it should be increased to reduce unnecessary base current. The base current should be calculated rather than selected randomly.

If the flyback diode is missing, it must be added. If it is reversed, it must be corrected.

If motor current is passing through the transistor, the circuit must be rewired immediately so that the motor current flows through the relay contact.

If the transistor orientation is wrong, the transistor must be reconnected according to the correct pin configuration. A damaged transistor must be replaced.

5.11 False Sensor Reading Problem

False sensor reading occurs when the system incorrectly detects the water level. The pump may start when the tank is not empty, stop before the tank is full, fail to stop at the HIGH level, or switch unpredictably. This fault is important because the water probes are the primary input source of the controller.

The sensing section is exposed to real environmental conditions. Unlike ideal switches in simulation, real water probes are affected by water conductivity, mineral content, corrosion, contamination, wire length, leakage paths, and electrical noise.

5.11.1 Possible Causes

One possible cause is low water conductivity. If the water contains very few dissolved minerals, it may not conduct enough current between the probe and REF electrode to produce a reliable signal.

Another cause is dirty or corroded probes. Corrosion can increase contact resistance and reduce signal reliability.

A third cause is floating input nodes. If the input node is not defined by a resistor when the probe is dry, the voltage may drift and cause false triggering.

Long probe wires may also pick up electrical noise from the relay, motor, or surrounding environment. This can disturb the NE555 input or transistor input.

The probe resistor value may be incorrect. If the resistor is too low, excessive current may flow through water and accelerate corrosion. If the resistor is too high, the signal may become too weak.

Water splashing may temporarily connect a probe before the actual water level reaches it. This can cause premature stopping or false starting.

Leakage paths across the tank wall, moisture on insulation, or dirty surfaces may create unintended conduction between probes.

5.11.2 Diagnostic Tests

The first test is to measure voltage at the sensor node when the probe is dry and when it is wet. The two states must produce clearly different voltage levels.

The second test is to measure resistance between the probe and REF electrode through water. If the resistance is extremely high, the sensing signal may be too weak.

The third test is to check whether the input node has a defined voltage when the probe is dry. If the node floats, a pull-up or pull-down resistor is required.

The fourth test is to move the probe wire away from the motor and relay wiring. If the false reading reduces, the problem is noise coupling.

The fifth test is to clean the probes and repeat the measurement. If performance improves after cleaning, corrosion or contamination is the cause.

The sixth test is to simulate probe states using fixed resistors instead of water. If the circuit behaves correctly with fixed resistors but not with water, the issue is in the physical sensing environment.

5.11.3 Corrective Actions

The probes should be cleaned and placed securely at the correct heights. The LOW probe must represent the minimum acceptable water level, and the HIGH probe must represent the maximum stop level.

Input nodes must be given stable biasing with suitable resistors. No control input should be left floating.

The sensor current should be limited by appropriate resistor values. A high-value resistor is generally safer for water sensing than a very low-value resistor because it reduces current and slows corrosion.

Long probe wires should be separated from relay and motor wires. Twisted wires, shielding, or filtering may be used if noise is significant.

An RC filter may be added to reduce short noise pulses. The filter must be selected so that it does not make the system too slow to respond.

The sensing method may be improved by using pulsed sensing or alternating-current sensing to reduce probe corrosion. If higher reliability is required, non-contact sensors such as float switches, pressure sensors, or ultrasonic sensors may be considered.

5.12 Water Probe Corrosion Problem

Water probe corrosion is a long-term reliability problem in conductive water-level sensing. When a direct current signal passes through water and metal probes, electrochemical reactions may occur. These reactions can gradually damage the probe surface, change contact resistance, and reduce sensing accuracy.

Although the circuit may work correctly in simulation, simulation does not represent long-term chemical effects at the probe surface. Therefore, probe corrosion must be discussed as a real limitation and prevention requirement.

5.12.1 Cause of Corrosion

Corrosion occurs because the probes behave like electrodes in a conductive liquid. When voltage is applied across the probes, current can pass through the water. This current may cause electrochemical reactions at the metal surfaces. Over time, material may be removed from one electrode and deposited or chemically altered at another location.

The severity of corrosion depends on several factors. These include the probe material, the voltage across the probes, the current through the water, the mineral content of the water, the duration of current flow, temperature, and the surface area of the probes.

The probe current can be estimated as

$$I_{probe} = \frac{V_{sensor}}{R_{water} + R_{limit}}$$

where (I_{probe}) is the current through the sensing path, (V_{sensor}) is the applied sensor voltage, (R_{water}) is the resistance of the water path, and (R_{limit}) is the current-limiting resistance.

If (R_{limit}) is very small, probe current can become unnecessarily high. Excessive probe current increases corrosion risk and may also disturb the sensor input.

5.12.2 Effect on Circuit Accuracy

Probe corrosion can increase the resistance between the water and the probe. As contact resistance increases, the sensor signal becomes weaker. The circuit may then fail to detect water even when water is physically touching the probe.

Corrosion can also create unstable readings. A partially corroded surface may conduct intermittently. This may cause relay chatter, false pump activation, or failure to stop the pump at the HIGH level.

Corrosion may also cause uneven response between probes. The LOW probe may remain reliable while the HIGH probe becomes unreliable, or the opposite may occur. This makes the system difficult to diagnose unless each probe is tested individually.

In severe cases, corrosion may break the probe conductor completely. The circuit would then behave as if the probe is permanently dry.

5.12.3 Prevention Methods

The first prevention method is to use corrosion-resistant probe material. Stainless steel is commonly preferred for water-contact electrodes because it resists corrosion better than many ordinary metals.

The second method is to reduce probe current. The sensing current should be only large enough to create a reliable logic condition. It should not be unnecessarily high. The limiting resistor should therefore be selected carefully.

The third method is to reduce continuous direct current exposure. Pulsed sensing can reduce the total time during which current flows through the water. Alternating-current sensing can also reduce electrode polarization and corrosion effects.

The fourth method is to clean and inspect the probes periodically. Maintenance should be included in the practical implementation plan.

The fifth method is to mechanically secure the probes so that they do not move, touch each other, or collect deposits easily.

The sixth method is to use non-contact sensing if long-term reliability is more important than circuit simplicity. However, this would increase system complexity and may change the nature of the project.

5.13 Power Supply Problem

Power supply problems can affect every part of the circuit. The NE555, transistors, relay coil, LED, and motor pump all depend on correct voltage and sufficient current. A power supply fault can appear as a sensor fault, relay fault, transistor fault, or motor fault. Therefore, the supply must always be checked early in the troubleshooting process.

This project may use more than one supply domain. The control circuit may operate at 5 V, while the motor pump may operate at 12 V. These supplies must be correctly connected, correctly referenced, and sufficiently rated.

5.13.1 Low Voltage

Low voltage occurs when the measured supply voltage is below the required operating voltage of the circuit section. If the NE555 control supply is too low, the IC may not

switch correctly. If the relay coil voltage is too low, the relay may not energize fully. If the motor voltage is too low, the motor may fail to start or may run weakly.

The supply voltage should be measured under load, not only without load. A weak supply may show normal voltage when the relay and motor are OFF, but drop sharply when the relay or motor turns ON.

The voltage drop can be estimated as

$$V_{drop} = I \times R$$

where (I) is the current flowing through the wire or internal supply resistance, and (R) is the resistance of the path. Excessive wire resistance, weak battery condition, or poor connector contact can therefore reduce the effective voltage at the load.

Corrective action requires using a supply with the correct voltage and adequate current rating. Wiring and connectors must also be improved if they cause voltage drop.

5.13.2 Voltage Noise

Voltage noise is unwanted fluctuation on the supply line. In this project, relay switching and motor operation can create voltage noise. The NE555 may respond incorrectly if noise enters its supply or input pins.

Voltage noise may cause false triggering, relay chattering, random pump operation, or unstable LED indication. The problem becomes worse when the motor supply and control supply are poorly separated or when decoupling capacitors are missing.

A small capacitor should be placed close to the NE555 supply pins. This capacitor helps absorb fast voltage changes. A larger capacitor should also be used across the supply line to stabilize slower voltage dips.

The supply noise problem can be reduced by proper grounding, shorter wires, separation of motor wires from sensor wires, and suitable suppression components across inductive loads.

5.13.3 Insufficient Current

Insufficient current occurs when the supply voltage is correct in theory but the supply cannot deliver the current required by the load. This is common when a relay or motor is added to a circuit that was previously tested only with small loads.

The total current demand of the control section can be estimated as

$$I_{total} = I_{555} + I_{relay} + I_{LED} + I_{sensor} + I_{base}$$

The supply rating should be greater than the calculated total current. A safety margin should be included:

$$I_{supply, rated} \geq 1.5 \times I_{total}$$

For the motor supply, the starting current must also be considered. The motor starting current may be higher than its running current. Therefore, the motor supply should not be selected only from the running current value.

If the current capacity is insufficient, the relay may chatter, the NE555 may reset, the motor may fail to start, and voltage may collapse during operation.

5.13.4 Grounding Problem

Grounding problems occur when the circuit does not have a stable reference point. In a control circuit, ground is not merely a wire; it is the voltage reference used by the NE555, transistors, sensor inputs, and supply system.

If the motor current flows through the same weak ground path used by the NE555 input section, voltage disturbance may occur. This can cause false triggering or relay chatter.

A poor ground connection can also prevent current from returning to the supply. In that case, the circuit may appear connected but still fail to operate.

The grounding arrangement should be checked with continuity testing. The control ground must be continuous. The motor return path must be strong enough for motor current. Sensitive input ground paths should be kept away from high-current motor paths where possible.

5.14 Relay and Motor Protection Problems

Relay and motor protection problems are critical because the relay and motor are inductive components. Inductive components can produce voltage spikes when current changes suddenly. These spikes can damage transistors, disturb the NE555, cause contact arcing, and reduce system reliability.

The relay coil must be protected with a correctly connected flyback diode. The motor side may also require suppression depending on motor type, current, and supply arrangement. Without protection, the circuit may function for a short time but fail after repeated switching.

The relay coil protection diode must be connected in reverse bias during normal relay operation. When the transistor turns OFF, the relay coil attempts to maintain current flow. The diode then provides a safe path for the coil current. Without this path, the coil can produce a high voltage spike across the transistor.

The diode current rating should satisfy the condition

$$I_{diode, rated} \geq I_{coil}$$

where (I_{coil}) is the relay coil current. The diode reverse voltage rating should also be greater than the supply voltage with safety margin.

The motor side protection must consider motor starting current, back electromotive force, brush noise, and relay contact arcing. A direct current motor may require

a diode, transient voltage suppressor, capacitor, or snubber depending on the performance requirement. A simple diode can reduce voltage spikes but may slow motor current decay. A transient voltage suppressor can provide faster suppression with better switching response.

The relay contact rating must be greater than the motor current. The safest condition is

$$I_{\text{contact,rated}} \geq I_{\text{motor,inrush}}$$

If the inrush current is unknown, the report should state that it must be measured or taken from the motor data sheet before hardware implementation.

The relay contact voltage rating must also match the motor supply. A relay contact rated for a certain current under alternating current conditions may not have the same rating under direct current conditions. Direct current switching is often more demanding because the arc does not naturally cross zero as in alternating current.

Protection problems may appear as welded relay contacts, transistor damage, relay chatter, NE555 reset, random switching, overheating, or permanent motor operation. Therefore, protection components are not optional decorative parts. They are part of the reliability structure of the project.

Corrective actions include adding or correcting the relay flyback diode, selecting a relay with sufficient contact rating, adding motor-side suppression, using a fuse in the motor supply line, separating control wiring from motor wiring, and verifying that the motor current does not exceed the relay or supply rating.

5.15 Troubleshooting Table

5.16 Hardware Safety Checklist

Before the circuit is implemented in real hardware, a safety checklist must be completed. Simulation alone is not sufficient to prove physical safety. The following safety conditions must be verified.

The control supply voltage must match the NE555, relay coil, LED, and transistor-driver design. No component should receive a voltage higher than its safe operating value.

The motor supply voltage must match the rated voltage of the motor pump. A 12 V motor must be supplied by a suitable 12 V source with adequate current capacity.

The relay coil voltage must match the control supply. A relay with the wrong coil voltage may fail to operate or may become damaged.

The relay contact rating must be greater than the motor current. The motor starting current must be considered, not only the running current.

The relay flyback diode must be installed with correct polarity. The diode must not short the supply during normal relay operation.

Table 5.1: *Troubleshooting Table*

Fault symptom	Most likely causes	Diagnostic test	Corrective action
Pump does not start	No control supply, LOW probe fault, NE555 output fault, relay-driver fault, motor supply missing	Measure NE555 supply, output voltage, Q1 base voltage, relay coil voltage, and motor terminal voltage	Restore supply, repair probe wiring, correct NE555 input, repair transistor driver, reconnect motor supply
Pump does not stop	HIGH probe fault, reset path fault, Q2 fault, Q1 short, relay contact stuck	Measure HIGH probe node, reset pin voltage, NE555 output, relay coil voltage, and contact continuity	Repair HIGH probe, correct reset network, replace Q2 or Q1, replace relay
Relay does not activate	No NE555 output drive, insufficient base current, wrong transistor pinout, open relay coil, wrong relay voltage	Measure NE555 output, Q1 base-emitter voltage, coil voltage, and relay coil resistance	Recalculate base resistor, correct transistor wiring, replace relay, correct diode polarity
Relay chatters	Weak supply, missing decoupling, sensor noise, poor grounding, motor feedback noise	Measure supply voltage during relay switching, observe NE555 output, disconnect motor temporarily	Increase supply capacity, add capacitors, stabilize inputs, improve grounding, add motor suppression
LED turns ON but motor does not run	LED indicates only control state, relay contact fault, missing motor supply, damaged motor	Measure relay contact continuity and motor terminal voltage	Correct relay contact wiring, restore motor supply, replace motor if faulty
Motor runs continuously	Relay contact welded, motor wired directly to supply, wrong contact used, Q1 short, HIGH probe failure	Remove relay drive, check contact continuity, measure NE555 output and Q1 condition	Rewire motor path, replace relay, repair HIGH probe, replace shorted transistor
Motor never runs	No motor supply, open motor path, relay never energizes, defective motor, missing ground return	Measure motor supply, motor terminal voltage, relay coil voltage, and continuity of motor path	Restore motor supply, repair wiring, repair relay-driver section, replace motor
Transistor overheats	Excessive relay coil current, insufficient base drive, no flyback diode, motor current through transistor	Calculate coil current, measure (V_{CE}), calculate (P_Q), inspect load path	Use proper relay or stronger driver, recalculate base resistor, add diode, rewire motor through relay
False sensor reading	Low water conductivity, corroded probes, floating input, noise pickup, wrong resistor value	Measure wet and dry sensor-node voltage, test with fixed resistor, inspect probe condition	Clean probes, add biasing resistors, adjust sensing resistor, separate sensor wires, add filtering
Probe corrosion	Continuous direct current through probes, unsuitable probe material, excessive sensing current	Inspect probe surface, measure probe resistance over time	Use stainless steel probes, reduce current, use pulsed or alternating-current sensing, schedule maintenance
	Weak supply, high	Measure supply	Use stronger supply,

The motor-side protection must be considered. A motor is an inductive load and may produce voltage transients during switching.

The transistor current rating must be higher than the relay coil current with safety margin. The transistor must not directly carry motor current.

The base resistor must be calculated. It must provide enough current for saturation without overloading the NE555 output.

The LED resistor must be calculated according to the actual supply voltage. If the LED is connected to 5 V, its resistor value differs from the value required for 12 V.

The water probes must carry only low voltage and limited current. No dangerous voltage must be applied to water-contact probes.

The probe material should resist corrosion. Stainless steel or another suitable corrosion-resistant material should be used for real implementation.

The wiring must separate sensor signals from motor current paths. High-current motor wires should not run close to sensitive sensor input wires unless adequate noise protection is used.

The control ground must be stable. Poor ground connections can create false triggering and unreliable operation.

A fuse should be used in the motor supply path. The fuse rating must be selected according to the expected motor current and protection requirement.

All exposed conductors must be insulated. Water and electricity must be physically separated except at the intentional low-voltage sensing probes.

If an alternating-current pump is used instead of a low-voltage motor pump, additional safety requirements become mandatory. These include electrical isolation, rated relay contacts, proper enclosure, fuse protection, earthing, strain relief, and safe clearance between low-voltage and high-voltage sections.

The circuit should not be touched during operation. Measurements must be performed using safe methods and properly rated instruments.

The final hardware must be tested first without water and without the motor load. After the control circuit is verified, the relay may be tested. After the relay is verified, the motor may be connected. This staged testing reduces the risk of damage and unsafe operation.

5.17 Chapter Summary

This chapter presented the troubleshooting framework for the automatic water-level motor-pump controller. The chapter showed that fault diagnosis must follow the actual functional chain of the system, beginning with the power supply and sensor inputs, continuing through the NE555 control circuit and transistor relay-driver stage, and ending at the relay contact and motor pump.

The main system faults were analyzed in detail. These included pump failure to start, pump failure to stop, relay activation failure, continuous relay chattering, LED

indication without motor operation, continuous motor operation, complete motor failure, transistor overheating, false sensor readings, probe corrosion, power supply instability, and relay or motor protection failure.

Each fault was examined through possible causes, diagnostic tests, and corrective actions. This structure makes the troubleshooting process measurable and academically defensible. The chapter also included mathematical verification where required, especially for relay coil current, transistor base current, transistor power dissipation, supply current margin, voltage drop, sensor current, diode current rating, and relay contact current rating.

The chapter further emphasized that Proteus simulation can confirm functional behavior but cannot fully prove real-world reliability. Practical implementation requires attention to water conductivity, probe corrosion, supply stability, relay rating, motor inrush current, inductive protection, grounding quality, and physical safety.

Therefore, troubleshooting is not an optional part of the project. It is the diagnostic foundation that proves whether the designed controller can be examined, repaired, improved, and defended under realistic engineering conditions.

6

Questions

6.1 Introduction

This chapter presents the major academic, technical, and defense questions related to the automatic water level motor pump controller. The purpose of this chapter is to verify whether the project has been understood not only as a circuit diagram, but also as a complete control system with sensing, decision making, switching, actuation, protection, simulation, and troubleshooting stages.

The questions are arranged in a hierarchical order. The chapter begins with basic project questions, then proceeds to component based questions, sensor based questions, NE555 control questions, calculation questions, simulation questions, troubleshooting questions, and critical thinking questions. This order ensures that the project is examined from the simplest conceptual level to the most advanced engineering level.

The chapter also supports oral defense preparation. Each question is written to test a specific part of the project. The answers explain the function of each stage, the reason for each component, the mathematical basis of the calculations, and the limitations of the simulation. A correct defense of this project requires more than saying that the pump turns on and turns off. It requires proof that the circuit responds correctly to water level changes, that the relay is driven safely, that the motor supply is properly separated from the control circuit, and that the simulation results are interpreted with engineering accuracy.

Therefore, this chapter acts as the final verification section of the report. It confirms that the design is understandable, calculable, testable, and defensible.

6.2 Basic Project Questions

This section contains the foundational questions of the project. These questions check whether the main purpose, problem, advantage, and expected result of the system are clearly understood.

6.2.1 What is the purpose of this project?

The purpose of this project is to design and simulate an automatic water level controller that can control a motor pump according to the water level inside a tank. The system uses water level probes to detect whether the tank is low, filling, or full. According to these sensing conditions, the control circuit activates or deactivates a relay, and the relay controls the motor pump.

The main purpose is to remove the need for continuous manual observation of the tank. In a manual system, a person must watch the water level and switch the pump on or off at the correct time. This is inefficient and unreliable because the user may forget to switch the pump off, which can cause overflow, or may forget to switch it on, which can cause water shortage.

In this project, the circuit performs the control action automatically. When the water level falls below the low level, the pump is expected to operate. When the water reaches the high level, the pump is expected to stop. This makes the system more convenient, reduces wastage, and demonstrates the practical use of electronic control in water management.

The academic purpose of the project is also important. The project demonstrates how a sensing stage, an NE555 control stage, a transistor driver stage, a relay switching stage, and a motor load can be integrated into one functional system.

6.2.2 What problem does this project solve?

This project solves the problem of manual water pump control. In many water storage systems, the pump is switched on and off by a user. This creates several practical problems. The first problem is tank overflow. If the pump remains on after the tank becomes full, water is wasted. The second problem is unnecessary energy consumption. A pump that runs longer than necessary consumes extra electrical energy. The third problem is user dependence. The system cannot operate correctly unless a person monitors it. The fourth problem is possible pump stress. If the pump operates under unsuitable conditions, its lifetime can be reduced.

The project solves these problems by automating the switching process. The water level is detected by probes, and the circuit makes the switching decision based on the electrical condition of those probes. The pump does not depend on continuous human attention. The system therefore reduces overflow, improves convenience, and provides a simple automatic control method.

The project does not claim to solve every possible pump protection problem unless extra protection circuits are added. For example, dry run protection, overload protection, and full hardware safety testing require additional design stages. However, within its defined scope, the project solves the specific problem of automatic tank level based pump control.

6.2.3 Why is automatic control better than manual control?

Automatic control is better than manual control because it performs the switching action according to measured conditions instead of human memory. In a manual system, the user must decide when the pump should be turned on and when it should be turned off. This creates delay, error, and inconsistency. In an automatic system, the control action is linked directly to the water level.

Automatic control gives more consistent operation. When the water reaches the low level condition, the circuit can activate the pump. When the water reaches the high level condition, the circuit can deactivate the pump. This reduces the chance of overflow and reduces unnecessary pump operation.

Automatic control also improves convenience. The user does not need to constantly check the tank. The system can operate as long as the circuit, probes, relay, power supply, and pump are working correctly.

From an engineering perspective, automatic control is also better because it creates a repeatable control rule. The system response is determined by defined electrical states rather than by uncertain human action. This makes the behavior easier to test, simulate, troubleshoot, and improve.

6.2.4 What is the expected output of the system?

The expected output of the system is the automatic operation of the motor pump according to the water level inside the tank. When the water level is below the required minimum level, the pump should turn on and begin filling the tank. When the water reaches the upper level, the pump should turn off and stop filling the tank.

The expected output can be described in two visible ways. The first output is the state of the motor pump. The second output is the state of the indicator LED, if the LED is connected to represent the active condition of the relay or pump.

The expected behavior may be summarized as follows. When the tank is low, the pump should be on. When the tank is filling and the high level has not yet been reached, the pump should remain on. When the tank becomes full, the pump should turn off. When the water level later decreases but has not fallen below the low level, the pump should remain off. When the water again falls below the low level, the pump should turn on again.

This output confirms that the system uses two level control rather than unstable single point switching. The low and high probes create a practical operating range and reduce rapid switching.

6.3 Component Based Questions

This section explains the purpose of the main components used in the circuit. A correct academic defense must show that every component has a necessary function. No component should appear in the circuit without a reason.

6.3.1 Why is the NE555 timer used?

The NE555 timer is used as the central control element of the circuit. In this project, it is not used only as a timing oscillator. It is used mainly as a control and latch type device that responds to input conditions and produces an output state. The water level sensing section affects the trigger and reset behavior of the NE555, and the output of the NE555 controls the relay driver transistor.

The NE555 is suitable for this project because it has a stable internal comparator and flip flop structure. Its trigger input can set the output, and its reset input can force the output into a defined state. This makes it useful for a system that must switch between pump on and pump off states.

The NE555 also provides a stronger output than many simple sensor circuits. However, the relay should still be driven through a transistor stage because relay coils require more current than a sensitive logic input. The NE555 therefore acts as the decision stage, while the transistor and relay act as the power switching stages.

The use of the NE555 also makes the project simple and educational. It allows automatic control to be achieved without programming. This is valuable because the working principle can be explained through circuit behavior, voltage levels, switching states, and component interactions.

6.3.2 Why are BC547 transistors used?

BC547 transistors are used as switching and signal conditioning devices in the circuit. A transistor allows a small control signal to control a larger current path. In this project, at least one BC547 is used to drive the relay coil. Another transistor may be used in the sensing or reset path depending on the exact circuit connection.

The relay coil requires a current that should not be taken directly from a weak sensing node. The BC547 transistor solves this problem by acting as an electronic switch. When sufficient base current is supplied, the transistor enters saturation and allows current to flow through the relay coil. When the base drive is removed, the transistor turns off and the relay coil is de energized.

The BC547 is suitable for low current switching tasks, but it must not be used blindly. Its collector current rating must be checked against the relay coil current. If the relay coil requires more current than the transistor can safely handle, a stronger transistor or a dedicated relay driver must be used.

Therefore, the BC547 is used because it provides current amplification and switching ability, but its safe operation depends on correct base resistor selection, relay coil current calculation, and power dissipation verification.

6.3.3 Why is a relay used?

A relay is used because the control circuit and the motor pump circuit have different electrical requirements. The control circuit uses low voltage and low current signals.

The motor pump requires a separate power path and usually a higher current than the NE555 or transistor output can supply directly.

The relay provides electrically controlled switching. When the relay coil is energized, the relay contacts change state and allow current to flow through the motor pump. When the relay coil is de energized, the contacts return to their normal state and the motor stops.

The relay is also useful because it separates the control side from the load side. The NE555 and transistor do not need to carry the full motor current. They only need to control the relay coil. The relay contacts carry the motor current.

However, the relay must be selected correctly. Its coil voltage must match the control supply. Its contact current rating must be greater than the motor current. Its contact voltage rating must match the motor supply. If the motor is an alternating current pump, the relay must be rated for that alternating current voltage and current, and all safety rules must be followed.

6.3.4 Why is a diode connected across the relay coil?

A diode is connected across the relay coil to protect the transistor from the high voltage spike produced when the relay coil is switched off. A relay coil is an inductive component. When current flows through it, magnetic energy is stored in the coil. When the transistor turns off, the current through the coil cannot stop instantly. The collapsing magnetic field can produce a large reverse voltage.

If this voltage spike is not controlled, it can damage the transistor, disturb the NE555, or create unreliable circuit behavior. The diode provides a safe path for the relay coil current after the transistor turns off. This is called flyback protection.

During normal relay operation, the diode must be reverse biased. This means it should not conduct while the relay is energized. When the transistor turns off and the coil voltage reverses, the diode becomes forward biased and allows the stored energy to dissipate safely.

The diode must be placed with correct polarity. If it is connected incorrectly, it may short the supply during normal operation and prevent the relay from working.

6.3.5 Why is an LED used?

An LED is used as a visual indicator of the circuit state. It helps the user observe whether the control circuit is active, whether the relay driver is operating, or whether the pump command is present. The exact meaning of the LED depends on where it is connected in the circuit.

If the LED is connected with the relay driver, it can indicate that the relay is energized. If it is connected with the NE555 output, it can indicate that the NE555 output is in the active state. If it is connected with the motor path, it can indicate that the pump supply path is active.

The LED is useful during simulation and troubleshooting because it gives immediate visual feedback. If the LED turns on but the motor does not run, the problem may be in the relay contact path, motor supply, or motor connection. If the LED does not turn on, the problem may be in the control or sensing stage.

The LED must always be connected with a current limiting resistor. Without a resistor, excessive current can flow through the LED and damage it.

6.3.6 Why are resistors required?

Resistors are required to control current, set voltage conditions, protect components, and define input states. In this project, resistors perform several essential functions.

A base resistor limits the current entering the base of a transistor. Without a base resistor, excessive current may flow from the NE555 output or sensing stage into the transistor base. This can damage the driving component or the transistor.

An LED resistor limits the current through the LED. The LED cannot be connected directly across a supply because its current would rise excessively. The resistor sets the LED current to a safe value.

Pull up and pull down resistors are used to define stable voltage levels at sensitive inputs. Without them, an input may float. A floating input can randomly change state due to noise, leakage, or nearby electrical interference.

Probe related resistors limit current through the water sensing path. This is important because the sensor current should be small. Excessive direct current through water probes can increase corrosion and waste power.

Therefore, resistors are not optional parts. They are structural elements that make the circuit stable, safe, and predictable.

6.4 Sensor Based Questions

This section explains how the water level is detected. The sensor system is the first stage of the project. If the sensing stage is misunderstood, the entire control logic becomes unclear.

6.4.1 What is the function of the REF probe?

The REF probe acts as the reference electrode for the water level sensing system. In a conductive probe based water level controller, water completes an electrical path between the reference probe and the level probes. The REF probe provides the common reference point needed for detecting whether water has reached the LOW or HIGH probe.

The REF probe is usually placed at the lowest relevant position in the tank so that it remains in contact with water during normal sensing operation. When water touches another probe and also touches the REF probe, a conductive path is formed through the water. The circuit can then detect that the water has reached that level.

Without the REF probe, the LOW and HIGH probes would not have a proper return path for sensing. The circuit would not reliably know whether the water had reached a particular level.

The REF probe is therefore not merely an additional wire. It is the reference foundation of the sensing system.

6.4.2 What is the function of the LOW probe?

The LOW probe detects the lower water level threshold. Its purpose is to identify when the tank has reached a low condition and requires filling. When the water level falls below the LOW probe, the circuit should interpret this as a condition that requires the pump to start.

The LOW probe is important because it prevents the pump from switching on and off at only one level. If a system used only one probe, small movements of water around that probe could cause rapid switching. The LOW probe helps define the lower boundary of the operating range.

In the control sequence, the LOW probe is associated with pump activation. When the water level is below this point, the pump should be commanded to operate. Once the pump starts, it should normally continue until the water reaches the HIGH probe.

Therefore, the LOW probe represents the minimum acceptable water level in the tank.

6.4.3 What is the function of the HIGH probe?

The HIGH probe detects the upper water level threshold. Its purpose is to identify when the tank has reached the full condition. When water touches the HIGH probe, the circuit should stop the motor pump.

The HIGH probe prevents overflow. If the pump continued to run after the tank became full, water would be wasted and the surrounding area could be damaged. The HIGH probe provides the stopping condition for the control circuit.

The HIGH probe also completes the upper boundary of the two level control system. The pump begins when the level is too low and stops when the level is high enough. This creates a stable operating range between the LOW and HIGH probes.

Therefore, the HIGH probe represents the maximum allowed water level in the tank.

6.4.4 How does water act as a conductor in this project?

Water acts as a conductor because real water usually contains dissolved minerals, salts, or impurities that allow a small electrical current to pass through it. Pure distilled water is a very poor conductor, but ordinary tap water normally has enough ionic content to conduct a weak sensing current.

In this project, when water touches both the REF probe and another level probe, it creates a conductive path between them. The circuit detects this path as a change in electrical condition. That change can then be used to activate or reset the control circuit.

The current through the water should be small. The sensing path must not be treated like a power path. It should only provide enough current or voltage change to indicate the presence of water. Excessive current through the probes can increase corrosion and reduce the life of the sensor.

The sensing principle depends on the conductivity of the water. If the water has enough dissolved ions, the circuit can detect the level. If the water conductivity is too low, the sensing signal may become weak or unreliable.

6.4.5 What happens if the water conductivity is low?

If the water conductivity is low, the current between the REF probe and the level probe may be too small for reliable detection. The circuit may fail to recognize that water has reached the LOW or HIGH probe. This can cause wrong pump behavior.

If the HIGH probe is not detected correctly, the pump may continue running even when the tank is full. This can cause overflow. If the LOW probe is not detected correctly, the pump may start or stop at the wrong time. In some cases, the control circuit may behave unpredictably.

Low conductivity can occur when the water is very pure, when probe surfaces are dirty or oxidized, or when the sensing resistor values are not suitable. Long wires and electrical noise can make the problem worse.

To improve reliability, the sensing circuit can be made more sensitive, the probe material can be improved, and input filtering can be added. Another improvement is to use alternating or pulsed sensing instead of continuous direct current sensing. This can reduce corrosion and improve long term stability.

6.5 NE555 Control Questions

This section explains how the NE555 controls the system. The NE555 is the main decision stage of the circuit, so its input and output behavior must be explained clearly.

6.5.1 Is the NE555 used as a timer or control latch?

In this project, the NE555 is mainly used as a control latch rather than as a free running timer. A normal timer application uses the NE555 to generate a delay or oscillating waveform. In this water level controller, the NE555 is used to hold a control state based on water level inputs.

The latch behavior is important because the pump should not change state randomly between the LOW and HIGH probes. When the tank is low, the circuit should start the pump. The pump should continue running while the tank fills. It should stop

only when the water reaches the HIGH probe. After stopping, it should remain off until the water falls below the LOW level again.

This behavior requires memory of state. The NE555 can provide this through its internal flip flop behavior. The trigger input can set the output, and the reset input can force the output to the off state. This makes it suitable for a simple two threshold control system.

Therefore, the NE555 is not used here mainly to measure time. It is used to control and maintain the pump state.

6.5.2 What happens at the trigger pin?

The trigger pin of the NE555 is responsible for setting the output under the correct input condition. In a standard NE555, the trigger input responds when its voltage falls below the lower internal reference level. When this happens, the internal flip flop is set, and the output becomes high.

In this project, the trigger behavior is associated with the condition that requires the pump to start. When the water level is below the required low level, the sensing circuit should create a trigger condition. The NE555 output then changes to the active state and drives the relay transistor.

The trigger pin must not be left floating. If it floats, noise can cause false triggering. A stable resistor network and correct sensor connection are required to ensure that the trigger pin changes state only when the water level condition requires it.

The trigger pin therefore acts as the pump start command input of the NE555 control stage.

6.5.3 What happens at the reset pin?

The reset pin of the NE555 is used to force the output into the off state. In a standard NE555, the reset input is active low. This means that when the reset pin is pulled low, the output is forced low regardless of the trigger condition.

In this project, the reset function is associated with stopping the pump when the water reaches the HIGH probe. When the tank becomes full, the sensing circuit should activate the reset path. The reset input then forces the NE555 output to the inactive state. As a result, the transistor turns off, the relay coil is de energized, and the pump stops.

The reset pin is very important because it provides a clear stopping command. If the reset pin is not correctly controlled, the pump may continue to run even after the tank becomes full.

The reset pin must also have a defined normal state. If it is accidentally left floating, the circuit may reset unpredictably or fail to reset when needed.

6.5.4 What happens at the output pin?

The output pin of the NE555 provides the control signal that drives the next stage of the circuit. In this project, the output pin controls the base of the relay driver transistor through a resistor. When the output is high, base current flows into the transistor. The transistor switches on, the relay coil is energized, and the motor pump circuit is activated through the relay contacts.

When the output is low, the transistor receives no sufficient base drive. The transistor switches off, the relay coil is de energized, and the relay contacts return to their normal state. This stops the motor pump.

The output pin therefore connects the decision stage to the actuation stage. It does not directly represent water level. It represents the control decision made by the NE555 according to the sensor inputs.

The output must not be overloaded. Even though the NE555 can provide output current, the relay should be driven through a transistor so that the NE555 does not directly carry the relay coil current.

6.5.5 Why is supply stability important for the NE555?

Supply stability is important because the NE555 makes decisions based on internal voltage reference levels. If the supply voltage is unstable, the trigger and threshold behavior can become unreliable. Noise on the supply can cause false triggering, unwanted resetting, relay chatter, or unstable pump operation.

The relay and motor can introduce electrical noise into the circuit. When the relay switches or when the motor starts, the supply voltage may dip or contain spikes. If this noise reaches the NE555 supply or input pins, the control state may change incorrectly.

To improve stability, a decoupling capacitor should be placed near the NE555 supply pins. A small ceramic capacitor is commonly used for high frequency noise, and a larger electrolytic capacitor can help with slower supply dips. Proper grounding is also necessary. The motor current path should not disturb the sensitive control ground.

A stable supply ensures that the NE555 responds to the water probes, not to electrical noise.

6.6 Calculation Questions

This section verifies the mathematical foundation of the design. The project cannot be considered academically complete if the relay current, transistor current, base resistor, LED resistor, motor power, and contact rating are not calculated or justified.

6.6.1 How is relay coil current calculated?

Relay coil current is calculated using Ohm's law. If the relay coil voltage and coil resistance are known, the coil current is calculated as follows:

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

In this formula, (I_{coil}) is the relay coil current, (V_{coil}) is the relay coil voltage, and (R_{coil}) is the relay coil resistance.

For example, if the relay coil voltage is (5V) and the relay coil resistance is (100Ω), then the coil current is:

$$I_{coil} = \frac{5}{100}$$

$$I_{coil} = 0.05A$$

$$I_{coil} = 50mA$$

This value is important because the relay driver transistor must be able to carry this current safely. The power supply must also be able to provide this current. If the relay coil current is too high for the transistor, the transistor may overheat or fail.

The relay coil current should always be calculated from the actual relay datasheet or measured coil resistance.

6.6.2 How is the transistor base resistor selected?

The transistor base resistor is selected by first calculating the base current required to saturate the transistor. When a transistor is used as a switch, it should be driven into saturation so that it behaves like a closed switch with low voltage drop.

The collector current of the transistor is approximately equal to the relay coil current:

$$I_C \approx I_{coil}$$

For reliable saturation, a forced current gain is selected. A conservative value is often:

$$\beta_{forced} = 10$$

The required base current is then:

$$I_B = \frac{I_C}{\beta_{forced}}$$

After the base current is known, the base resistor is calculated by:

$$R_B = \frac{V_{drive} - V_{BE}}{I_B}$$

In this formula, (R_B) is the base resistor, (V_{drive}) is the voltage from the NE555

output, (V_{BE}) is the base emitter voltage of the transistor, and (I_B) is the required base current.

For example, if ($V_{drive} = 5V$), ($V_{BE} = 0.7V$), and ($I_B = 5mA$), then:

$$R_B = \frac{5 - 0.7}{0.005}$$

$$R_B = 860\Omega$$

A practical standard value would be (820Ω) or ($1k\Omega$), depending on the exact relay current and available component values.

The base resistor must not be chosen randomly. If it is too large, the transistor may not saturate. If it is too small, excessive current may flow from the NE555 output into the transistor base.

6.6.3 How is the LED resistor calculated?

The LED resistor is calculated by subtracting the LED forward voltage from the supply voltage and then dividing by the desired LED current. The formula is:

$$R_{LED} = \frac{V_{supply} - V_F}{I_{LED}}$$

In this formula, (R_{LED}) is the LED resistor, (V_{supply}) is the voltage applied to the LED resistor path, (V_F) is the LED forward voltage, and (I_{LED}) is the desired LED current.

For a green LED, a typical forward voltage may be approximately (2V). If the LED is supplied from (5V) and the desired current is (10mA), then:

$$R_{LED} = \frac{5 - 2}{0.01}$$

$$R_{LED} = 300\Omega$$

A practical value would be (330Ω).

If the LED is supplied from (12V), then:

$$R_{LED} = \frac{12 - 2}{0.01}$$

$$R_{LED} = 1000\Omega$$

A practical value would be ($1k\Omega$).

This calculation proves that the LED resistor depends directly on the supply voltage. A resistor that is safe at (5V) may be unsafe at (12V).

6.6.4 How is motor power calculated?

Motor power is calculated using the voltage applied to the motor and the current drawn by the motor. The formula is:

$$P_{motor} = V_{motor} \times I_{motor}$$

In this formula, (P_{motor}) is the motor power, (V_{motor}) is the motor supply voltage, and (I_{motor}) is the motor current.

For example, if the motor operates at (12V) and draws (0.5A), then:

$$P_{motor} = 12 \times 0.5$$

$$P_{motor} = 6W$$

This calculation gives the running power of the motor. However, a motor may draw a larger current when it starts. This starting current is called inrush current. The relay contacts and power supply must be selected with this higher starting current in mind.

Therefore, motor power calculation should include both normal running current and starting current. The relay should not be selected only from the voltage rating. It must also be selected from the current rating.

6.6.5 Why must relay contact rating be checked?

The relay contact rating must be checked because the relay contacts carry the motor current. Even if the relay coil works correctly, the relay contacts may fail if the motor current is higher than the contact rating.

The contact rating has two important parts. The first is voltage rating. The relay contacts must be able to withstand the voltage of the motor supply. The second is current rating. The relay contacts must be able to carry the motor current without overheating, welding, or excessive wear.

For a direct current motor, the contact stress can be significant because direct current arcs are harder to extinguish than alternating current arcs. The motor can also produce inrush current and inductive voltage spikes. Therefore, a relay that appears acceptable for a resistive load may not be sufficient for a motor load.

The relay contact current rating should satisfy the following condition:

$$I_{contact} \geq I_{motor}$$

For stronger design margin, the contact rating should be greater than the motor starting current:

$$I_{contact} \geq I_{starting}$$

If this condition is not satisfied, the relay may fail even if the control circuit works correctly.

6.6.6 Why must transistor current rating be checked?

The transistor current rating must be checked because the transistor carries the relay coil current. If the relay coil current exceeds the safe collector current of the transistor, the transistor may overheat, operate unreliably, or become permanently damaged.

The required condition is:

$$I_C \geq I_{coil}$$

For practical reliability, the transistor current rating should be higher than the relay coil current by a safe margin. It is not enough for the values to be equal. A margin is needed because real components have tolerance, temperature changes, and switching stress.

Power dissipation must also be checked. When the transistor is saturated, it has a small collector emitter voltage. The approximate power dissipation is:

$$P_Q = V_{CE(sat)} \times I_C$$

In this formula, (P_Q) is the transistor power dissipation, ($V_{CE(sat)}$) is the collector emitter saturation voltage, and (I_C) is the collector current.

If the calculated power is too high, the transistor may heat up. If the relay coil current is too high for a BC547, a stronger transistor or a relay driver integrated circuit should be used.

6.7 Simulation Questions

This section explains the role of Proteus simulation. Simulation is an important proof layer, but it must be interpreted correctly. It proves functional behavior under model conditions, not complete real world reliability.

6.7.1 Why was Proteus used?

Proteus was used because it allows the circuit to be designed, connected, tested, and observed before hardware construction. It provides a safe environment for checking whether the NE555, transistors, relay, LED, probes, and motor behave according to the intended logic.

Proteus is useful because it allows different water level conditions to be simulated. The HIGH, LOW, and REF probe states can be represented by switches or conductive paths. The relay operation can be observed, and the motor state can be checked visually.

Proteus also helps identify wiring mistakes before building the real circuit. If the relay does not energize, the output does not change, or the motor does not run in simulation, the circuit can be corrected before components are purchased or connected physically.

Therefore, Proteus was used as a design verification and functional testing tool.

6.7.2 What does the simulation prove?

The simulation proves that the circuit logic can operate under the conditions represented in the Proteus model. It shows whether the pump turns on when the simulated water level is low and turns off when the simulated water level reaches the high probe. It also shows whether the relay and LED respond to the NE555 output as expected.

The simulation can prove the following points:

The sensing states can produce control changes.

The NE555 output can change according to the input conditions.

The transistor can switch the relay coil in the simulated circuit.

The relay contact can switch the motor supply path.

The LED can indicate the active state.

The pump can respond to the relay state.

Therefore, the simulation proves functional correctness at the circuit model level. It confirms that the design idea is electrically coherent in the simulator.

6.7.3 What does the simulation not prove?

The simulation does not prove complete real hardware reliability. A simulation model is an approximation of real behavior. It cannot fully prove probe corrosion, water conductivity variation, relay contact wear, motor inrush current, electrical noise, temperature effects, loose wiring, moisture damage, or long term component aging.

The simulation also does not prove physical safety. If the real pump is connected to a high current supply or an alternating current mains supply, additional safety rules are required. These include insulation, fuse protection, enclosure design, earthing, proper relay rating, and safe wiring distances.

The simulation does not automatically prove that resistor values are optimal. A circuit may operate in Proteus even if a resistor causes excessive current in real hardware. Therefore, simulation must be combined with calculation and datasheet verification.

The correct academic statement is that simulation proves model level functional behavior, while hardware testing proves real world performance.

6.7.4 Which conditions were tested?

The simulation should test all important operating states of the tank. The first condition is the empty tank condition, where neither the LOW probe nor the HIGH probe is reached by water. In this condition, the pump should turn on.

The second condition is the filling condition, where the water reaches the LOW probe but has not yet reached the HIGH probe. In this condition, the pump should remain on because the tank is still not full.

The third condition is the full tank condition, where the water reaches the HIGH probe. In this condition, the pump should turn off.

The fourth condition is the falling water level condition, where the water falls below the HIGH probe but remains above the LOW probe. In this condition, the pump should remain off because the tank has not yet reached the low level.

The fifth condition is the low again condition, where the water falls below the LOW probe. In this condition, the pump should turn on again.

Additional tests should include sensor fault, power restart, missing motor supply, relay failure, and noise sensitivity. These tests make the simulation more complete.

6.7.5 How can the simulation be improved?

The simulation can be improved by adding voltage probes and current probes at important points. The NE555 output voltage, transistor base voltage, relay coil voltage, motor terminal voltage, LED current, and sensor node voltage should be measured.

The simulation can also be improved by adding a more realistic motor model. A real motor has starting current, inductive behavior, and noise. If the simulation uses an ideal motor, the result may look better than real hardware.

The water probes can also be simulated more realistically by replacing simple switches with resistive paths. Different resistance values can represent different water conductivity conditions. This helps test whether the circuit still works when the water is weakly conductive.

The relay model can be improved by checking coil current and contact rating. The power supply can be improved by adding internal resistance or voltage drop to test stability.

Fault conditions should also be added. These include open probe, shorted probe, weak supply, missing flyback diode, and motor overload. A stronger simulation does not only show success. It also shows how the circuit behaves under stress.

6.8 Troubleshooting Questions

This section prepares the reader to diagnose faults in the project. Troubleshooting must follow a logical sequence from power supply to sensor input, control output, transistor driver, relay stage, and motor load.

6.8.1 What if the pump does not start?

If the pump does not start, the fault may be in the sensing stage, NE555 control stage, transistor driver stage, relay stage, motor supply, or motor itself. The first step is to

check whether the control circuit has the correct supply voltage. Without correct supply voltage, the NE555 cannot operate.

The next step is to check whether the trigger condition is reaching the NE555. If the LOW level condition is not detected, the NE555 may never produce an active output. The REF probe and LOW probe should be checked for correct connection.

The NE555 output pin should then be measured. If the output is not active when the tank is low, the issue is likely in the sensing or NE555 stage. If the output is active but the relay does not energize, the issue may be in the base resistor, transistor, relay coil, or flyback diode connection.

If the relay energizes but the motor does not run, the issue may be in the relay contact wiring, motor power supply, or motor itself.

A correct diagnostic sequence is therefore: check supply, check sensor condition, check NE555 output, check transistor base, check relay coil, check relay contact, check motor supply, and check motor.

6.8.2 What if the pump does not stop?

If the pump does not stop, the high level sensing or reset path may not be working correctly. The first step is to check whether water has reached the HIGH probe. If the HIGH probe is not in contact with water, the stopping signal cannot be produced.

The next step is to check whether the HIGH probe and REF probe create a conductive path through the water. If the water conductivity is too low or the probe is corroded, the circuit may not detect the full condition.

The reset pin of the NE555 must then be checked. If the HIGH probe condition does not pull the reset input into the required state, the NE555 output may remain active and the pump will continue running.

The transistor and relay should also be checked. If the NE555 output becomes inactive but the relay remains energized, the transistor may be shorted or the relay may be stuck. If the relay contact is welded closed, the motor may continue running even after the control circuit turns off.

Therefore, a pump that does not stop may be caused by sensor failure, reset failure, transistor failure, relay contact failure, or incorrect wiring.

6.8.3 What if the relay chatters?

Relay chatter means the relay rapidly switches on and off. This is an unstable condition and must be corrected. Relay chatter can be caused by weak supply voltage, insufficient coil current, noisy sensor input, poor grounding, missing decoupling capacitor, or unstable NE555 input conditions.

If the relay supply voltage drops when the relay energizes, the relay may release and then energize again repeatedly. This can happen when the power supply cannot provide enough current. A larger power supply capacity or better decoupling may be required.

Noisy sensor inputs can also cause chatter. Water movement around the probe can create intermittent contact. Long sensor wires can pick up noise. Floating inputs can change randomly. Pull up resistors, pull down resistors, and filtering capacitors can reduce this problem.

Motor noise can also affect the control circuit. The motor current path should be separated from the NE555 input path as much as possible. The relay coil should have a flyback diode, and the motor may require additional suppression.

Relay chatter is not only an annoying sound. It can damage relay contacts, disturb the motor, and reduce system life.

6.8.4 What if the transistor becomes hot?

If the transistor becomes hot, it may be carrying too much current or may not be fully saturated. The first possible cause is that the relay coil current is too high for the transistor. The coil current must be calculated and compared with the transistor rating.

The second possible cause is an incorrect base resistor. If the base current is too low, the transistor may operate in the active region instead of saturation. In that case, the collector emitter voltage becomes higher and the transistor dissipates more power.

The third possible cause is incorrect relay or motor wiring. The transistor should drive the relay coil, not the motor directly. If the motor current flows through a small signal transistor such as BC547, the transistor can overheat or fail.

The fourth possible cause is an incorrectly connected flyback diode. If the diode is missing or reversed, switching spikes can stress the transistor.

The approximate transistor power dissipation can be calculated by:

$$P_Q = V_{CE} \times I_C$$

If the transistor is saturated, (V_{CE}) should be low. If (V_{CE}) is high while current is also high, heating will occur.

6.8.5 What if the probes give wrong readings?

If the probes give wrong readings, the sensing stage must be examined carefully. Wrong readings may be caused by low water conductivity, dirty probes, corrosion, loose wires, wrong probe placement, floating inputs, or unsuitable resistor values.

The first check is physical. The REF, LOW, and HIGH probes must be placed at the correct levels. The REF probe must remain in contact with water during normal sensing. The LOW probe must represent the lower threshold, and the HIGH probe must represent the upper threshold.

The second check is electrical. The resistance between REF and the active probe should change when water touches both probes. If the resistance remains too high, the circuit may not detect the water level.

The third check is signal stability. If the input voltage changes randomly, filtering or stronger pull up and pull down design may be needed.

The fourth check is corrosion. Direct current sensing can cause electrochemical effects at the probes. Stainless steel probes, low sensing current, pulsed sensing, or alternating sensing can reduce long term corrosion.

Wrong probe readings are serious because the whole system depends on correct sensing. If the sensor stage is unreliable, the pump control will also be unreliable.

6.9 Critical Thinking Questions

This section tests the deeper engineering quality of the project. It is not enough to know that the circuit works in simulation. A complete defense must understand weaknesses, improvements, and real world risks.

6.9.1 What is the biggest weakness of conductive probes?

The biggest weakness of conductive probes is that their reliability depends on water conductivity and probe condition. If the water has low mineral content, the current through the water may be too weak for reliable detection. If the probes become corroded, dirty, or covered with deposits, the sensing signal may change over time.

Another weakness is electrolysis. If direct current continuously flows through the probes, chemical reactions can occur at the metal surface. This can damage the probes and contaminate the contact surface.

Conductive probes are simple and low cost, but they are not perfect sensors. Their accuracy depends on the water type, probe material, current level, installation quality, and maintenance.

Therefore, the biggest weakness is not that the probes cannot work. The weakness is that they require careful design and maintenance to remain reliable over long periods.

6.9.2 How can probe corrosion be reduced?

Probe corrosion can be reduced by lowering the sensing current, using corrosion resistant probe material, reducing continuous direct current exposure, and improving the sensing method.

The first method is to use high value resistors in the probe path. This limits the current flowing through the water and reduces electrochemical activity.

The second method is to use stainless steel or another corrosion resistant material for the probes. The material should be suitable for long term contact with water.

The third method is to use pulsed sensing. Instead of passing current continuously, the circuit applies sensing current only for short intervals. This reduces the total electrical stress on the probes.

The fourth method is to use alternating sensing. Alternating polarity reduces the one directional electrochemical effect that occurs with continuous direct current.

The fifth method is maintenance. Probes should be cleaned periodically if mineral deposits form on the surface.

A strong design should not ignore corrosion. It should treat corrosion as a predictable long term failure mode.

6.9.3 How can dry run protection be added?

Dry run protection can be added by detecting whether water is available at the source before allowing the pump to operate. Dry running occurs when the pump runs without sufficient water. This can damage the pump because water often helps cool and lubricate the pumping mechanism.

One method is to add a source tank low level sensor. If the source tank does not contain enough water, the control circuit should prevent the pump from starting.

Another method is to add a flow sensor. If the pump is on but no water flow is detected, the system can stop the pump after a short delay.

A third method is to monitor motor current. A pump running dry may draw a different current from normal operation. A current sensing circuit can detect abnormal operation and stop the pump.

A fourth method is to use a pressure sensor. If the pump runs but pressure does not build correctly, the system can identify a fault.

For this project, the simplest improvement is to add another conductive probe sensor in the source tank. The pump should operate only when the destination tank is low and the source tank has sufficient water.

6.9.4 How can this project be upgraded using a microcontroller?

This project can be upgraded using a microcontroller by replacing part of the hardware control logic with programmable decision making. The water level probes can be connected to microcontroller input pins through suitable protection and signal conditioning circuits. The microcontroller can then decide when to turn the relay on or off.

A microcontroller upgrade can add more features. It can display tank status on an LCD or OLED display. It can add buzzer alarms for overflow, dry source tank, sensor fault, or pump failure. It can include timing protection so that the pump does not run longer than a safe limit. It can record operation history. It can also support wireless monitoring through suitable communication modules.

A microcontroller can also improve logic flexibility. Software can implement hysteresis, delay, fault detection, and self checking more easily than a purely discrete circuit.

However, the microcontroller must be protected from electrical noise. Relay coils, motors, and long sensor wires can disturb microcontroller inputs. Proper isolation, filtering, grounding, and flyback protection are still necessary.

Therefore, a microcontroller upgrade increases functionality, but it does not remove the need for good electrical design.

6.9.5 How can this project be improved without a microcontroller?

The project can be improved without a microcontroller by strengthening the analog and discrete electronic design. The first improvement is to add proper decoupling capacitors near the NE555 and relay supply. This improves stability and reduces false switching.

The second improvement is to use better sensor conditioning. The probe inputs can be filtered with resistor capacitor networks so that small water movement or electrical noise does not cause false triggering.

The third improvement is to use a stronger relay driver. If the relay coil current is near the limit of the BC547, a higher current transistor or a driver integrated circuit can be used.

The fourth improvement is motor protection. A diode, transient voltage suppressor, or snubber can be added depending on the motor type and supply. A fuse can also protect against overcurrent.

The fifth improvement is to add dry run protection through another probe set in the source tank.

The sixth improvement is to use pulsed or alternating sensing to reduce probe corrosion.

These improvements keep the system non programmable while making it safer, more reliable, and more suitable for real hardware.

6.9.6 What is the safest state after power failure?

The safest state after power failure depends on the application, but in most basic water tank systems the safest default state is pump off. When power returns, the circuit should not blindly start the pump unless the sensor condition clearly requires it.

A pump off default state reduces the risk of overflow, unexpected motor operation, and unsafe restarting. However, the system should still be able to start the pump automatically if the tank is genuinely low and all safety conditions are satisfied.

A complete design should define the restart behavior. If the tank is full when power returns, the pump should remain off. If the tank is low and the source water is available, the pump may start. If the sensor condition is unclear, the system should remain off or enter a fault state.

The safest principle is that uncertain sensor conditions should not produce uncontrolled pump operation. The circuit should favor a defined and safe state over an unknown active state.

6.9.7 What changes are required for an AC pump?

If an alternating current pump is used, the project requires major safety and hardware changes. The relay contacts must be rated for the alternating current voltage and current of the pump. The relay must also be able to handle motor starting current, not only running current.

The control circuit must remain electrically isolated from the mains side. The low voltage NE555 circuit must not be directly connected to mains wiring. Proper insulation, terminal blocks, fuse protection, and enclosure design are required.

The relay contact spacing and wire thickness must be suitable for the voltage and current. The pump line should include proper fuse or circuit breaker protection. Earthing must be used where required. Exposed conductors must not be accessible.

A snubber circuit or other surge suppression may be required across the relay contacts or motor depending on the pump type. The design should follow electrical safety rules and should be tested carefully before use.

Therefore, converting the project from a low voltage direct current pump to an alternating current pump is not only a component replacement. It changes the safety level of the entire project.

6.10 Viva and Defense Questions with Short Answers

This section provides short answers suitable for oral defense. The answers are brief but technically precise.

Question 1: What is the main aim of the project?

The main aim is to automatically control a motor pump according to the water level in a tank.

Question 2: What are the three probes used in the system?

The three probes are REF, LOW, and HIGH.

Question 3: What is the function of the REF probe?

The REF probe provides the common reference path for water level sensing.

Question 4: What is the function of the LOW probe?

The LOW probe detects the lower water level and helps start the pump when the tank becomes low.

Question 5: What is the function of the HIGH probe?

The HIGH probe detects the full level and helps stop the pump to prevent overflow.

Question 6: Why is the NE555 used?

The NE555 is used as the control element that changes its output according to the sensor input conditions.

Question 7: Is the NE555 used mainly as a timer in this project?

No. It is mainly used as a control latch to maintain the pump state between the low and high water levels.

Question 8: Why is a transistor used?

A transistor is used to drive the relay coil because the relay requires more current than the direct sensing stage can provide.

Question 9: Why is a relay used?

A relay is used to switch the motor power circuit using a low voltage control signal.

Question 10: Why is a diode placed across the relay coil?

The diode protects the transistor from the voltage spike produced when the relay coil is switched off.

Question 11: Why is an LED used?

The LED gives visual indication of the circuit state, such as relay activation or pump command.

Question 12: Why are resistors required in the circuit?

Resistors limit current, define voltage states, protect components, and stabilize input signals.

Question 13: What happens when the tank is empty?

The pump should turn on so that the tank can begin filling.

Question 14: What happens when the water reaches the HIGH probe?

The pump should turn off to prevent overflow.

Question 15: What does Proteus simulation prove?

It proves that the circuit can perform the intended switching behavior under simulated conditions.

Question 16: What does Proteus simulation not prove?

It does not prove long term hardware reliability, probe corrosion resistance, motor inrush safety, or physical electrical safety.

Question 17: How is relay coil current calculated?

It is calculated by dividing the relay coil voltage by the relay coil resistance.

$$I_{coil} = \frac{V_{coil}}{R_{coil}}$$

Question 18: How is motor power calculated?

Motor power is calculated by multiplying motor voltage by motor current.

$$P_{motor} = V_{motor} \times I_{motor}$$

Question 19: What is the biggest weakness of conductive probes?

Their biggest weakness is dependence on water conductivity and long term probe condition.

Question 20: How can the project be improved?

It can be improved by adding dry run protection, better filtering, stronger motor protection, corrosion resistant probes, and more realistic hardware testing.

6.11 Final Evaluation Questions

The final evaluation questions determine whether the project satisfies academic, functional, mathematical, simulation, and safety expectations. These questions are not only for oral defense. They are also used to judge whether the report is complete.

Question 1: Does the report clearly define the project purpose?

Yes. The report must clearly state that the project is an automatic water level motor pump controller designed to reduce manual switching, prevent overflow, and improve control convenience.

Question 2: Does the project have a clear input and output?

Yes. The inputs are the water level conditions detected by the REF, LOW, and HIGH probes. The output is the motor pump state, supported by the relay and LED indication.

Question 3: Does the report explain every major component?

Yes. The NE555, BC547 transistors, relay, diode, resistors, LED, probes, power supply, and motor must each be explained with their specific function.

Question 4: Does the report mathematically justify the circuit?

The report is complete only if it justifies relay coil current, transistor base resistor, transistor current rating, LED resistor, motor power, relay contact rating, and power supply capacity.

Question 5: Does the simulation test all required states?

The simulation should test empty tank, filling state, full tank, falling water level, low again condition, sensor fault, and power restart condition.

Question 6: Does the report separate simulation proof from hardware proof?

Yes. A strong report must clearly state that simulation proves functional logic under model conditions, while real hardware testing is required for physical reliability.

Question 7: Does the troubleshooting chapter cover realistic faults?

Yes. It should cover pump not starting, pump not stopping, relay chatter, transistor heating, LED and motor mismatch, wrong probe readings, supply faults, and motor protection faults.

Question 8: Does the project include safety awareness?

Yes. The report should explain relay ratings, supply separation, flyback protection, motor current, fuse protection, and additional safety requirements if an alternating current pump is used.

Question 9: Does the project have a clear limitation section?

Yes. The limitations should include simulation limitations, conductive probe corrosion, water conductivity variation, absence of dry run protection if not implemented, and lack of real hardware stress testing if hardware was not built.

Question 10: Is the project academically defensible?

Yes, if the report presents the project as a complete automatic control system and not merely as a simple circuit. The defense becomes strong when every stage is explained, calculated, simulated, tested, and critically evaluated.

6.12 Chapter Summary

This chapter presented the complete question and answer structure for the automatic water level motor pump controller. The questions were arranged from basic understanding to advanced technical evaluation. The chapter first explained the purpose of the project, the problem being solved, the advantage of automatic control, and the expected output of the system.

The chapter then explained the function of each major component, including the NE555 timer, BC547 transistors, relay, flyback diode, LED, and resistors. It also explained the sensing system, including the REF, LOW, and HIGH probes, and clarified how water conductivity is used to detect level conditions.

The NE555 control section explained the role of the trigger pin, reset pin, output pin, and supply stability. The calculation section presented the mathematical reasoning required for relay coil current, transistor base resistor selection, LED resistor selection, motor power, relay contact rating, and transistor current rating. These calculations are necessary because an academic engineering project must be justified by electrical values, not only by visual simulation.

The simulation section explained why Proteus was used, what the simulation proves, what it does not prove, which conditions should be tested, and how the simulation can be improved. The troubleshooting section provided logical fault analysis for pump failure, relay chatter, transistor heating, and wrong probe readings. The critical thinking section examined deeper issues such as probe corrosion, dry run protection, microcontroller upgrades, non microcontroller improvements, safe restart behavior, and alternating current pump requirements.

In conclusion, Chapter 6 confirms that the project can be defended as a complete automatic water level control system. It verifies the project from concept to component behavior, from calculation to simulation, and from troubleshooting to future improvement. A strong defense of this project must show that the system is not merely a pump that turns on and off, but a structured electronic control system that senses water level, makes a control decision, drives a relay safely, switches a motor load, and responds predictably to normal and fault conditions.

